Digital logic Design

* Numbering Systems :-

① Decimal (Base 10) → [0-9]

② Binary (Base 2) → [0-1]

③ Octal (Base 8) → [0-7]

④ Hexadecimal (Base 16) → [0-9] + [A(10) B(11) C(12) D(13) E(14) F(15)]

* Decimal representation :-

$*(7230)_{10}$

$7000 \quad 200 \quad 30 \quad 0$

$7\times10^3 + 2\times10^2 + 3\times10^1 + 0\times10^0$

* يمين الفاصلة ($*10^{-x}$)

يسار الفاصلة ($*10^{x}$)

$*(146.53)_{10}$

$1\times10^2 + 4\times10^1 + 6\times10^0 + 5\times10^{-1} + 3\times10^{-2}$

* to convert from any System (Base U) to decimal we multiply the digit by the corrosponding weight and find the summation.

Ex :- Convert the following to decimal :-

① $(1011.11)_2$

$= 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 + 1\times2^{-1} + 1\times2^{-2}$

$= 8 + 2 + 1 + \frac{1}{2} + \frac{1}{4}$

$= \boxed{(11.75)_{10}}$

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② $(127.4)_8$

$= 1\times8^2 + 2\times8^1 + 7\times8^0 + 4\times8^{-1}$

$= 64 + 16 + 7 + \frac{1}{2}$

$= \boxed{(87.5)_{10}}$

③ $(10.B.A)_{16}$

$= 1\times16^2 + 0\times16^1 + 11\times16^0 + 10\times16^{-1}$

$= \boxed{(267.625)_{10}}$

④ $(142)_5$

$= 1\times5^2 + 4\times5^1 + 2\times5^0$

$= 25 + 20 + 2$

$= \boxed{(47)_{10}}$

⑤ $(11010.11)_2$ (16 8 4 2 1 · ½ ¼) $= (26.75)_{10}$

$= 1\times2^4 + 1\times2^3 + 1\times2^1 + 1\times2^{-1} + 1\times2^{-2}$

$= 16 + 8 + 2 + \frac{1}{2} + \frac{1}{4}$

$= \boxed{(26.75)_{10}}$

⑥ $(4021.2)_5$

$= 4\times5^3 + 0\times5^2 + 1\times5^0 + 2\times5^{-1} + 2\times5^1$

$= \boxed{(511.4)_{10}}$

⑦ $(127.4)_8$

$= 1\times8^2 + 2\times8^1 + 7\times8^0 + 4\times8^{-1}$

$= 64 + 16 + 7 + \frac{1}{2}$

$= \boxed{(87.5)_{10}}$

⑧ $(B65F)_{16}$

$= 12\times16^3 + 6\times16^2 + 5\times16^1 + 15\times16^0$

$= \boxed{(46687)_{10}}$

⑨ $(10010)_2$ (16 8 4 2 1) $= 16+2 = (18)_{10}$

$= 1\times2^4 + 1\times2^1$

$= \boxed{(18)_{10}}$

* Notes :- (Binary System) :-

→ A Binary digit is called a bit

8 bits (b) → Byte (B)

2^{10} bit (b) = Kilo (K)

2^{20} bit (b) = Mega (M)

2^{30} bit (b) = Giga (G)

2^{40} bit (b) = Tera (T)

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* Faster way to convert Binary → Decimal :-

① $(101)_2 = (5)_{10}$

② $(1101)_2 = (13)_{10}$

③ $(10010)_2 = (18)_{10}$

④ $(11010)_2 = (26)_{10}$

⑤ $(101011)_2 = (43)_{10}$

* To convert from Decimal to any system with base (r) we divide the decimal by the system I want to convert then the remainder is the answer.

→ For the whole part (الجزء الصحيح) divide by base (r) and keep remainder.

→ For the fraction (الكسر) multiply by base (r).

Ex :- Convert to Bainary :-

① $(41)_{10} = (101001)_2$

41	/2	
20	1	least S.B
10	0	
5	0	
2	1	
1	0	
0	1	Most S.B

② $(0.625)_{10} = (0.101)_2$

Most S.B

0.625 x 2 = 1.25

0.25 x 2 = 0.5

0.5 x 2 = 1.0

0 x 2 = 0 least S.B

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64 32 16 8 4 2 1

65 → 1 0 0 0 0 0 1

64 32 16 8 4 2 1

67 → 1 0 0 0 0 1 1

16 8 4 2 1

17 → 1 0 0 0 1

32 16 8 4 2 1

60 → 1 1 1 1 0 0

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③ $(12.125)_{10} = (1100.001)_2$

12	÷2
6	0
3	0
1	1
0	1

$(1100)_2 = 12$

$0.125 \times 2 = 0.25$

$0.25 \times 2 = 0.5$

$0.5 \times 2 = 1.0$

$0 \times 2 = 0$

$(0.001)_2 = 0.125$

Ex :- Decimal to octal (8) :-

① $(41)_{10} = (51)_8$

41	÷8
5	1
0	5

② $(12.125)_{10} = (14.1)_8$

12	÷8
1	4
0	1

$0.125 \times 8 = 1.0$

$0 \times 8 = 0$

$(0.1)_8$

$(14)_8 = 12$ ✓

Ex :- Decimal to Hex (16) :-

① $(41)_{10} = (29)_{16}$

41	÷16
2	9
0	2

$(29)_{10} = 41$

② $(12.125)_{10} = (\quad)_{16}$

12	÷16

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Notes :- * Each Octal digit = 3 bits ($2^3 = 8$)

* Each Hex digit = 4 bits ($2^4 = 16$)

* Convert from octal to binary → replace each digit by 3 bits.

Hex to binary → replace each digit by 4 bits.

* Convert from binary to Octal → group 3 bits → replace by octal digit

binary to Hex → group 4 bits → replace by Hex digit

8

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

16

Hex	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10 A	1010
11 B	1011
12 C	1100
13 D	1101
14 E	1110
15 F	1111

2 4 2 1

0 = 0 0 0 0

A 10 = 1 0 1 0

2 = 0 0 1 0

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Ex :- Convert :-

① $(01001110.)_2 \rightarrow (4E)_H$ (افتراض 8 4 2 1 8 4 2 1)

4 — 14 — E

② $(001001110.11000100)_2 \rightarrow (26.C4)_H$

2 — 10 — A — 6 — 12 — C — 4

② $(D5.5)_{16} \rightarrow (1101.0101.0101)_2$

1101 — 0101 — 0101

③ $(73.4)_8 \rightarrow (111011.100)_2$

111 — 011 — 100

④ $(001001110)_2 \rightarrow (116)_8$

1 — 1 — 6

⑤

⑥ $(001010100110.110001)_2 \rightarrow (1246.61)_8$

1 2 4 6 . 6 1

* To convert from Hex / octa to octa / Hex, we must convert to Binary first.

Ex: ① $(317)_8 \rightarrow (0CF)_{16}$

000 011 001 111

0 — 12 — C — 15 — F

② $(45.12)_8 \rightarrow (25.28)_{16}$

100 101 . 001 010

2 — 5 . 2 — 8

③ $(136)_{16} \rightarrow (0466)_8$

0001 0011 0110

0 — 4 — 6 — 6

④ $(72.5A)_{16} \rightarrow (162.264)_8$

0111 0010 0101 1010

1 — 6 — 2 . 2 — 6 — 4

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* Mathematical operations :-

• Binary :-

① Addition :-

0 + 0 = 0

0 + 1 = 1

1 + 0 = 1

1 + 1 = 0 (Carry = 1)

32 16 8 4 2 1

45 1 0 1 1 0 1

\+

39 1 0 0 1 1 1 +

= ✓ 84 1 0 1 0 1 0 0

② Subtraction :-

0 - 0 = 0

0 - 1 = 1 (Borrow = 1)

1 - 0 = 1

1 - 1 = 0

32 16 8 4 2 1

45 1 0 1 1 0 1

\-

39 1 0 0 1 1 1 -

= ✓ 6 0 0 0 1 1 0

③ Multiplication :-

1 0 1 1 71

x

1 0 1 x 5

① 1 0 1 1

0 0 0 0 +

1 0 1 1 +

1 1 0 1 1 1 55 ✓

32 16 8 4 2 1

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* Binary Addition :-

x	y	sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Ex :-

$(0101)_2 + (0100)_2 = (1001)_2$

$$\begin{array}{r} 0101 \\ +\ 0100 \\ \hline 1001 \end{array}$$

* Hexadecimal Addition :-

Ex :-

① $(3C + 12)_{16} = (4E)_{16}$

$$\begin{array}{r} 3C \\ +\ 12 \\ \hline 4E \end{array}$$

② $(99 + 37)_{16} = (D0)_{16}$

$$\begin{array}{r} 99 \\ +\ 37 \\ \hline D0 \end{array}$$

$9 + 7 = 16$

$16 - 16 = 0$

③ $(101 + 011)_{16} = (1000)_{16}$

$$\begin{array}{r} 101 \\ +\ 011 \\ \hline 1000 \end{array}$$

④ $BFC + 8AB = (14A7)_{16}$

$$\begin{array}{r} BFC \\ +\ 8AB \\ \hline 14A7 \end{array}$$

$12 + 11 = 23$, $23 - 16 = 7$

$15 + 10 = 26$, $26 - 16 = 10$

$11 + 8 = 20$, $20 - 16 = 4$

* Octal Addition :-

Ex :-

① $(354)_8 + (113)_8$

$4 + 7 = 11$, $11 \geq 8$, $11 - 8 = 3$

$5 + 3 = 9$, $9 \geq 8$, $9 - 8 = 1$

② $(123)_8 + (776)_8 = (1121)_8$

$3 + 6 = 9$, $9 \geq 8$, $9 - 8 = 1$

$2 + 7 = 10$, $10 \geq 8$, $10 - 8 = 2$

$1 + 7 = 9$, $9 \geq 8$, $9 - 8 = 1$

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* Binary Multiplication :-

Ex :-

① $9 \times 2 = 18$

$$\begin{array}{r} 1001 \quad 9 \\ \times \quad 10 \quad 2 \\ \hline 0000 \\ 1001 \quad + \\ \hline 10010 \quad 18 \end{array}$$

② $3 \times 3 = 9$

$$\begin{array}{r} 11 \quad 3 \\ \times \quad 11 \quad 3 \\ \hline 11 \\ 11 \quad + \\ \hline 1001 \quad 9 \end{array}$$

③ $11 \times 3 = 33$

$$\begin{array}{r} 1011 \quad 11 \\ \times \quad 11 \quad 3 \\ \hline 1011 \\ 1011 \\ \hline 100001 \quad 33 \end{array}$$

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* Binary Codes :-

* Binary Coded Decimal (BCD):

- A number with "n" decimal digits will require "4n" bits in BCD.
- The number in Binary, BCD have the same representation if the decimal number has one digit only.

Ex :- (BCD)

① 452 → 0100 0101 0010 (4, 5, 2)

② 9 → Binary: 1001, BCD: 1001

③ 18 → Binary: 10010, BCD: 0001 1000 (1, 8)

* BCD Addition :-

- We can perform arithmatic operation directly when they are stored as BCD
- When the binary sum is equal to or less than (9 → 1001) [without carry]
  ⇒ BCD digit correct ✓
- Otherwise the BCD must be corrected by adding (6 → 0110)

Ex :- (BCD Addition)

① 6 + 2 = 8

6 0110
2 0010 +
8 1000

② 5 + 5 = 10

5 0101
5 0101 +
1010 > 9 correct
0110 +
10000
1 0

③ 7 + 9 = 16

7 0111
9 1001 +
10000 > 9 correct
0110 +
10110
1 6

④ 134 + 149 = 283

0001 0011 0100
1 4 9 +
0001 0100 1001
0010 0111 1101 > 9 correct
0110 +
0010 1000 0011
2 8 3

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⑤ 184 + 576 = 760

1 8 4
0001 1000 0100
5 7 6 +
0101 0111 0110
0110 1111 1010 +6 correct
0110 +
0111 0000 0000
0110
0111 0110 0000
7 6 0

⑥ 73 + 62 = 135

1 7 3
0111 0011
6 2 +
0110 0010
carry 1101 0101 +
0110
10011 0101
1 3 5

* Other Decimal Codes :-

Decimal	2,4,2,1	Excess-3	8,4,-2,-1
0	0000	0011	0000
1	0001	0100	0111
2	1000	0101	0110
3	1001	0110	0101
4	1010	0111	0100
5	1011	1000	1011
6	1100	1001	1010
7	1101	1010	1001
8	1110	1011	1000
9	1111	1100	1111

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* Gray Code :-

- Only one bit changes between consective numbers.

Binary	Gray
000	000
001	001
010	011
011	010
100	110
101	111
110	101
111	100

* How to convert from binary to Gray code :-

① 0 0 0 → 0 0 0

② 0 0 1 → 0 0 1

③ 0 1 0 → 0 1 1

④ 0 1 1 → 0 1 0

⑤ 1 0 0 → 1 1 0

⑥ 1 0 1 → 1 1 1

⑦ 1 1 0 → 1 0 1

⑧ 1 1 1 → 1 0 0

* When we change from binary to Gray code we put the first number the same then we compare with the next number.

Same = 0

Diff = 1

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Ex :- Convert to Gray Code :-

① 1 0 1 0 1 1 → 1 1 1 1 1 0

② 1 0 0 0 1 1 → 1 1 0 0 1 0

* How to convert From Gray to Binary :-

① 0 0 0 → 0 0 0 Decimal 0 (same, same)

② 1 1 0 → 1 0 0 Decimal 4 (diff, same)

③ 1 0 0 → 1 1 1 Decimal 7 (same, same)

④ G = 1 1 0 0 1 0
B = 1 0 0 0 1 1

⑤ G = 1 1 1 1 1 0
B = 1 0 1 0 1 1

* AscII Code :-

- 7 bits are used to represent different characters. (numbers, symbols, letters, special characters).

* Binary numbers operations :-

* Complements :-

- Complements are used to preform the subtraction operation.

A - B = (A) + (-B)

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[1] one's complement (First complement) :-

- Change all 0-bits to 1-bits
  1-bits to 0-bits

Ex :- Find the First complement :-

① $(110101001)_2 \xrightarrow{1's} 001010110$

② $(100010101)_2 \xrightarrow{1's} 011101010$

[2] Two's complement (second complement) :-

- To Find the 2nd complement We have two methods:
  ① Find the 1st complements and then add "1".
  ② Scan binary number From right to left, when finding the First "1" bit, Keep it and then all the next bits.

Ex :- Find the second complement :-

① $(11010100)_2$

* 1st comp: 00101011
  $+\ 1$
  00101100

* 11010100
  00101100

② $(10101101)_2$

* 1st comp: 01010010
  $+\ 1$
  01010011

* 10101101
  01010011

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* Subtraction with Complements :-

1] 2's Complement :-

$M - N = M + (-N)$

- add M to the 2's complement of N.
- $M \geq N \rightarrow$ Sum will produce carry (discard carry)
- $M < N \rightarrow$ No carry $\rightarrow M - N = -$(2's complement) (sum)

Ex:- preform the following subtraction using 2's complement :-

① $(1011)_2 - (0110)_2$ (11 − 6 = 5) $= (1011)_2 + 2\text{'s}(0110)$

$= 1011 + 1010$

$= 0101$ 5

1011
1010 +
1 0101
discard

② $(1101)_2 - (0100)_2$ (13 − 4 = 9) $= (1101)_2 + 2\text{'s}(0100)_2$

$= 1101 + 1100$

$= 1001$ 9

1101
1100 +
1 1001
discard

③ $(0100)_2 - (1101)_2$ (4 − 13 = −9) $= (0100)_2 + 2\text{'s}(1101)$

$= 0100 + 0011$

$= -$ 2's(0111)

$= -(1001) - 9$

0100
0011 +
0111

$A - B$ → $A > B$ (Carry ✗)

$A - B$ → $A < B$ (−2's(sum))

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1 2's complement :-

$M - N = M + (-N)$

- add M to the 2's complement of N
- $M \geq N \rightarrow$ Carry (add carry to sum)
- $M < N \rightarrow$ No carry (Answer = - 2's comp (sum))

Ex :- perform the following subtraction using 2's complement :-

① $10111 - 0110 = 10111 + 2's(0110)$ (11 − 6 = 5)

$= 10111 + 1001$

$= 01015$

carry 1 → 1+

01015

② $1\,1\,01 - 0100 = 1101 + 2's(0100)$ (13 − 4 = 9)

$= 1101 + 1011$

$= 10019$

carry 1 → 1+

10019

③ $0100 - 1101 = 0100 + 2's(1101)$ (4 − 13 = −9)

$= 0100 + 0010$

$= -2's(0110)$

$= -(1001) = -9$

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Ex :- $X = 1010100$ (84), $y = 1000011$ (67)

① 2's complement :-

$X - y = 1010100 - 1000011$ (84 − 67 = 17)

$= 1010100 + 2's(1000011)$

$= 1010100 + 0111101$

$= 0010001$ (17)

1 1 1 1	
1010100	
0111101 +	
1 0010001	discard

$y - X = 1000011 - 1010100$ (67 − 84 = −17)

$= 1000011 + 2's(1010100)$

$= 1000011 + 0101100$

$= -2's(1101111)$

$= -(0010001)$ -17

1000011	
0101100 +	
1101111	No carry

② 1's complement :-

$X - y = 1010100 - 1000011$ (84 − 67 = 17)

$= 1010100 + 1's(1000011)$

$= 1010100 + 0111100$

$= 0010001$ (17)

1 1 1	
1010100	
0111100 +	
1 0010000	
0010001 +	
0010001	

$y - X = 1000011 - 1010100$ (67 − 84 = −17)

$= 1000011 + 2's(1010100)$

$= 1000011 + 0101011$

$= -2's(1101110)$

$= -(0010010)$ -17

1000011	
0101011 +	
1101110	No carry

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* Binary logic :-

- Basic (primitive) logic operation :-

1. AND : this operation is represented by a dot or the absence of an operator.

$Z = X \text{ AND } y$

$Z = X \cdot y$

$Z = Xy$

Def : Truth Table (TT) :-

A table of all the possible combinations of the input variables, the output of these combinations

* # entries of the truth table $= 2^{\#\text{inputs}}$

2 inputs $\rightarrow$ #entries $= 2^2 = 4$

3 inputs $\rightarrow$ #entries $= 2^3 = 8$

Truth Table (AND) :- $Z = Xy$

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

0 $\rightarrow$ False

1 $\rightarrow$ True

inputs $= 2 \rightarrow$ #entries $= 2^2 = 4$

- AND. For the output to be true all inputs must be true.

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[2] OR: The operator is represented by (+)

$Z = X \text{ OR } y$

$Z = X + y$

- For the output to be true one of the inputs must be true.

X + y	Z
0 0	0
0 1	1
1 0	1
1 1	1

[3] NOT: This operation is represented by a prime or over bar

$Z = NOT(x)$

$Z = X'$ #entries (TT) = $2^1 = 2$

$Z = \bar{X}$

X	X'
0	1
1	0

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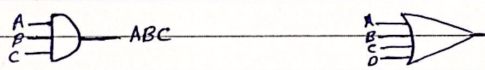
* primitive logic Gates :-

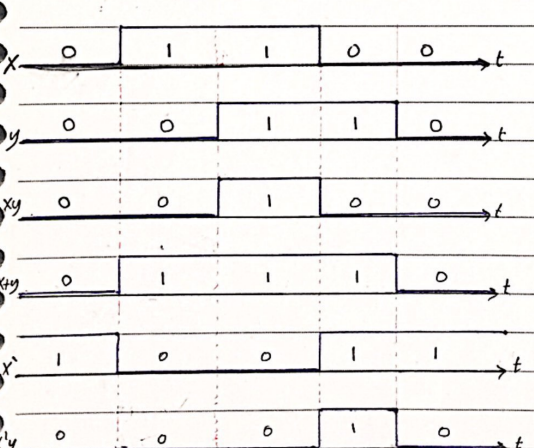
- logic gates are electronic circuits that operate one or more inputs signals to produce an output signals.
- Symbols for primitive digit a logic gates.

2-inputs AND gate 2-inputs OR gate input-NOT gate
Inverter

* Gates with multipleinputs :-



3-inputs AND gate 4-input OR gate



For the inputs signals X, y
Draw the output signal :-

① X.y
② X+y
③ X`
④ X`.y

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* Signed Numbers :-

① 1's complement

② 2's complement

③ Singned magnitude → +3 (+ → 0), -3 (- → 1) — sing bit (MSB)

* Represent (-9) with 8-bits :-

0000 1001

① 1's comp → -9 = 1's (9)

= 1's (0000 1001)

= 1111 0110 $\xrightarrow{1's}$ 0000 1001 -9 ✓

② 2's comp → 2's = 1111 0111 $\xrightarrow{2's}$ 0000 1001 -9 ✓

③ Singed mag → 1000 1001 -9 ✓ (9)

recall: for signed mag + → 0, - → 1

Ex :- ① 2's: 1101001 → 64 32 16 8 4 2 1 : 0010111 = -23

② 1's: 1101001 → 0010110 = -22

③ Singd mag: 1101001 → 1 (-) 101001 (41) = -41

* Boolean Algebra (simplification)

- Inputs (Boolean Variables $\begin{cases} 0 \\ 1 \end{cases}$), (Basic operations (+, ·, ')

* Boolean Algebra Basics :-

① Identity Element :-

- Identity Element for OR (+) → "0" ⟹ $X + 0 = X$

  $0 + 0 = 0$

  $1 + 0 = 1$

- Identity Element for AND (·) → "1"

  ⇓

  $X \cdot 1 = X$

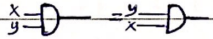
  $0 \cdot 1 = 0$

  $1 \cdot 1 = 1$

② Commutative Law :-

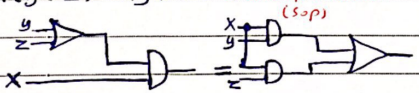
- $X + y = y + X$

- $X \cdot y = y \cdot X$



③ Distributive Law :-

- $X(y + z) = Xy + Xz$ (sum of products) (SOP)



- $X + yz = (X + y) \cdot (X + z)$ (product of sums) (POS)

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④ Complement :-

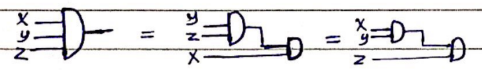
$(X')' = X$

X	X'	(X')'
0	1	0
1	0	1

⑤ Associative Law :-

$X+y+z = X+(y+z) = (X+y)+z$

$Xyz = X(yz) = (Xy)z$



Notes :- $X+1 = 1$ $X \cdot 0 = 0$ $X+X = X$ $X \cdot X = X$

$X+X = X$ $0+0 = 0$ $1+1 = 1$

$X \cdot X = X$ $0 \cdot 0 = 0$ $1 \cdot 1 = 1$

- $Xy + Xyz + Xy$ ($Xy + Xy = Xy$)
- $Xy + Xyz$
- $Xy(1+z)$
- $Xy \cdot 1$
- Xy

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Prove :-

① $Xy + Xy' = X$

$X(y+y')^1$

$X \cdot 1$

X ✓

② $X + Xy = X$

$X(1+y)^1$

$X \cdot 1$

X ✓

③ $(X+y')y = Xy$

$Xy + yy'^0$

$Xy + 0$

Xy ✓

Simplify :-

① $(X+y) \cdot (X+y')$

$XX + Xy' + yX + yy'^0$

$X + Xy' + Xy$

$X(1+y'+y)^1$

$X \cdot 1$

X

② $X(X+y)$

$X \cdot X + Xy$

$X + Xy$

$X(1+y)^1$

$X \cdot 1$

X

③ $X \cdot y' + y$

$(X+y) \cdot (y'+y)^1$

$(X+y) \cdot 1$

$X+y$

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* Demorgan's Theory :-

- used to find the complement of a Boolean Function
- $F \Rightarrow F'$ (Apply Demorgan's)

Replace $\cdot \rightarrow +$

$+ \rightarrow \cdot$

$x \rightarrow x'$

$x' \rightarrow x$

* Apply Demorgan :-

Prove by T.T

$F = x.y \Rightarrow F' = x' + y'$

x y	x · y	(x.y)'	x'	y'	x'+y'
0 0	0	1	1	1	1
0 1	0	1	1	0	1
1 0	0	1	0	1	1
1 1	1	0	0	0	0

↳ $x' + y'$ ✓

* Boolean Function :-

- Boolean Function can be represented as :-

① Algebric expression :-

$F_1 = x + y'z$

$F_2 = x'z + xy'$

② Truth Table.

③ Gate Implementation (Schematic diagram)

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① $F_1 = X + y'z$ — 3 inputs

* Combination $= 2^3 = 8$ [000 → 111]

$F_1 = 1$, when $X = 1$ OR $y'z = 1$ (01)

X	y	z	F_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

② $F_2 = X'z + Xy'$ — 3 inputs — Comb = 8 [000 → 111]

$F_2 = 1$, when, $X'z = 1$ (01) OR $Xy' = 1$ (10)

X	y	z	F_2
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

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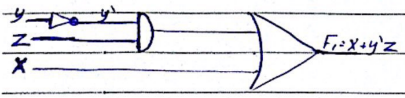
③ $F_3 = wx'z + w'x'yz + xy'z$ 4 inputs combin $= 2^4 = 16$

$F_3 = 1$ when, $wx'z = 1$ OR $w'x'yz = 1$ OR $xy'z = 1$
(1 0 1) (0 0 1 1) (1 0 1)

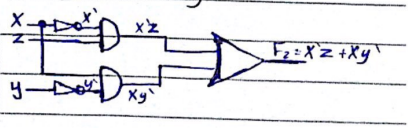
W	X	y	z	F_3
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

* Gate Implementation :-

① $F_1 = X + y'z$



② $F_2 = X'z + Xy'$



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Ex :- Simplify the following :-

① $F = AB + A'C + BC$

$= AB + A'C + BC(A + A')$

$= AB + A'C + ABC + A'BC$

$= AB(1 + C) + A'C(1 + B)$

$= AB + A'C$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

② $F = C + (BC)'$

$= C + B' + C$

$= 1 + B'$

$= 1$

Recall: (Demorgan)

$+ \rightarrow \cdot$

$\cdot \rightarrow +$

$X \rightarrow X'$

$X' \rightarrow X$

③ $F = xyz' + x'yz' + xy'z' + x'y'z' + z$

$= yz'(x + x') + y'z'(x + x') + z$

$= yz' + y'z' + z$

$= z'(y + y') + z$

$= z' + z$

$= 1$

Ex :- Find the complement ← Demorgan

① $F_1 = x'yz' + x'y'z$

$= (x'yz' + x'y'z)'$

$= (x'yz')' \cdot (x'y'z)'$

$= (x + y' + z) \cdot (x + y + z')$

② $F_2 = x(y'z' + yz)$

$= [x(y'z' + yz)]'$

$= x' + (y'z' + yz)'$

$= x' + (y'z')' \cdot (yz)'$

$= x' + (y + z) \cdot (y' + z')$

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* Dual of a Function :-

$+ \rightarrow \cdot$

$\cdot \rightarrow +$

Ex:- Find the Dual :-

① $F_1 = x'yz' + x'y'z$

$F_{1Dual} = (x'yz')_D \cdot (x'y'z)_D$

$= (x'+y+z')_D \cdot (x'+y'+z)_D \xrightarrow[\substack{x \rightarrow x' \\ x' \rightarrow x}]{F_1'} F_1' = (x+y'+z) \cdot (x+y+z')$

② $F_2 = x(y'z' + yz)$

$F_{2D} = x + (y'z' + yz)_D$

$= x + (y'z')_D \cdot (yz)_D$

$= x + (y+z') \cdot (y+z) \xrightarrow{F_2'} F_2' = x' + (y'+z) \cdot (y'+z')$

* Canonical, Standard forms :-

* Minterms, Maxterms :-

- A binary variable in a boolean function in the original form (x) or complemented form (x').

- Consider two binary variables $x, y \Rightarrow$ 4 possible combinations.

  $x \; x' \quad y \; y'$

$x'y'$
$x'y$
xy'
xy

these combinations are called minterms

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x y z	and Minterms	Minterms designation	OR Maxterms	Maxterms designation
0 0 0	$x'y'z'$	m_0 000	$x+y+z$	M_0
0 0 1	$x'y'z$	m_1 001	$x+y+z'$	M_1
0 1 0	$x'yz'$	m_2 010	$x+y'+z$	M_2
0 1 1	$x'yz$	m_3 011	$x+y'+z'$	M_3
1 0 0	$xy'z'$	m_4 100	$x'+y+z$	M_4
1 0 1	$xy'z$	m_5 101	$x'+y+z'$	M_5
1 1 0	xyz'	m_6 110	$x'+y'+z$	M_6
1 1 1	xyz	m_7 111	$x'+y'+z'$	M_7

Note :-

each maxterm in the complement of the corresponding minterm.

$000 \rightarrow \underset{x'y'z'}{m_0} \rightarrow \underset{x+y+z}{M_0}$

$110 \rightarrow \underset{xyz'}{m_6} \rightarrow \underset{x'+y'+z}{M_6}$

Ex :- Represent the functions in canonical forms :-

x y z	F_1	F_2
0 0 0	0	0
0 0 1	(1)	0
0 1 0	0	0
0 1 1	0	1
1 0 0	(1)	0
1 0 1	0	1
1 1 0	0	1
1 1 1	(1)	1

* Sum of minterms (1's)

$F_1 = m_1 + m_4 + m_7$

$= \Sigma(1,4,7) = \Pi(0,2,3,5,6)$

* Product of Maxterms (0's)

$F_1 = M_0 + M_2 + M_3 + M_5 + M_6$

$= \Pi(0,2,3,5,6)$

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* A Boolean Function can be expressed as :-

1. Sum of miterms
2. product of maxterms

] canonical Forms

← From last example :-

$F_2 = \Sigma(3,5,6,7) = \Pi(0,1,2,4)$

Ex :- Represent $F = ABC + A'B'C + BC'$ in canonical forms :-

(1 1 1) (0 0 1) (B=1, C=0)

	A	B	C	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

$F = \Sigma(1,2,6,7)$

$= \Pi(0,3,4,5)$

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* Standard , Canonical Forms :-

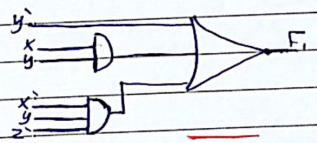
[1] Canonical Forms :-
- Sum of minterms (1's) and-OR
- product of maxterms (0's) OR-and

[2] Standard Forms :-
- Sum of products (sop) (1's) and-OR
- product of sum (pos) (0's) OR-and

2- level gate Implementation

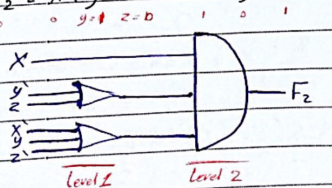
Ex :-

① $F_1 = y' + xy + x'yz'$ (sop) (1's)



x	y	z	F_1
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

② $F_2 = x(y'+z).(x'+y+z')$ (pos) (0's)



x	y	z	F_2	F_2'	F_2
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	0	1	0
1	0	0	1	0	1
1	0	1	0	1	0
1	1	0	0	1	0
1	1	1	1	0	1

0 → x

1 → x'

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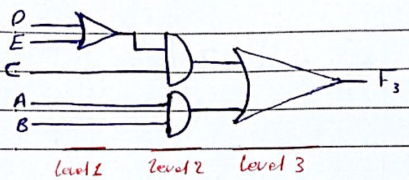
③ $F_3 = AB + C(D+E)$ [non standard form]

$= AB + CD + CE$



Ex:- Convert $F = A + B'C$ to sum of minterms :-

A B C

① Truth table :-

A	B	C	F	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	2
0	1	1	0	3
1	0	0	1	4
1	0	1	1	5
1	1	0	1	6
1	1	1	1	7

$F = \Sigma(1, 4, 5, 6, 7)$

② Boolean Algebra :- $(x + x')$

$F = A + B'C$

$= A + B'C(A + A')$

$= A + AB'C + A'B'C$

$= A(B + B') + AB'C + A'B'C$

$= AB + AB' + m_5 + m_1$

$= AB(C + C') + AB'(C + C') + m_5 + m_1$

$= ABC + ABC' + AB'C + AB'C' + m_5 + m_1$

$= m_7 + m_6 + m_5 + m_4 + m_5 + m_1$

$= \Sigma(1, 4, 5, 6, 7)$

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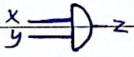
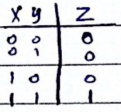
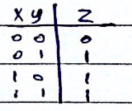
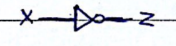
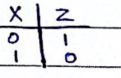
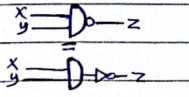
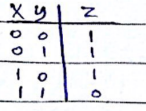
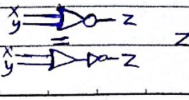
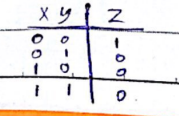
Ex: Express $F = xy + x'z$ as pos ← 0's

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$F = \Pi(0,2,4,5)$

* Digital Logic Gates :-

Name	Symbol	Function	T.T
① AND		$z = x.y$ $z = xy$	
② OR		$z = x+y$	
③ Inverter (Not)		$z = x'$ $z = \bar{x}$	
④ NAND Not AND		$z = (x.y)'$	
⑤ NOR Not OR		$z = (x+y)'$	

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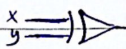
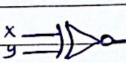
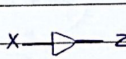
Name	Symbol	Function	T.T
⑥ XOR Exclusive OR		$Z = X \oplus y$ $Z = x'y + xy'$	X y | Z 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 0
⑦ XNOR Exclusive NOR		$Z = (X \oplus y)'$ $Z = x'y' + xy$	X y | Z 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 1
⑧ Buffer		$Z = X$	X | Z 0 | 0 1 | 1

* multiple input XOR

1's odd → 1

1's even → 0

* Karnaugh map (K-map) :-

- The map method is used to find the simplest form of Boolean function.
- The K-map is a map of squares each square represents a minterm
- # Squares = # minterms = 2^n [n = # variables]

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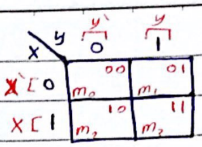
1 2-Variable K-map :-

- # Var = 2 → # squares = $2^2 = 4$
- each square minterm



Steps :-

① Draw Map (x-section, y-section, number squares).

② Fill in the 1's from T.T.

③ Circle largest number of 1's.

→ Bubble size = 4, 2, 1

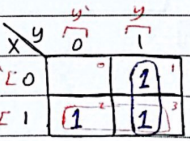
④ Write the function from the map.

Ex :- Simplify :-

$F(x,y) = x'y + xy' + xy$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

$F = \sum(1,2,3)$



$F = x + y$

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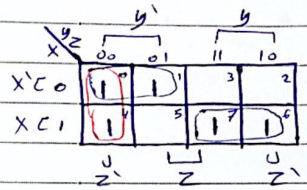
[2] 3- Variable K- map :-

- # Squares $= 2^3 = 8$
- Bubble Size $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$
  $\quad\quad\quad

④ $F = \Pi(2,3,5)$

$= \Sigma(0,1,4,6,7)$



$F = x`y` + xy + y`z`$

$= x`y` + xy + xz`$

⑤ $F = A`C + A`B + AB`C + BC$ ← a. sop, b. pos

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$F = \Sigma(1,2,3,5,7)$

$= \Pi(0,4,6)$

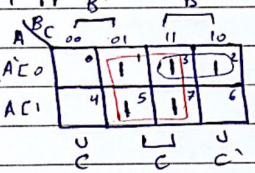
Notes:

Bubble Size	term
8 →	0
4 →	1
2 →	2
1 →	3

$F`$ (sop)

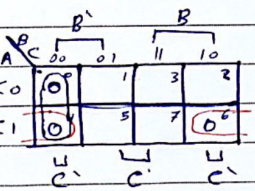
$F_{pos} = (F`_{sop})`$

ⓐ sop :- $A`C$ 0, AC 1



$F = C + A`B$

ⓑ pos :- $A`C$ 0, AC 1



① $F`_{sop} = B`C` + AC`$

② $F_{pos} = (F`_{sop})` = (B`C` + AC`)`$

③ $F_{pos} = (B+C) \cdot (A`+C)$

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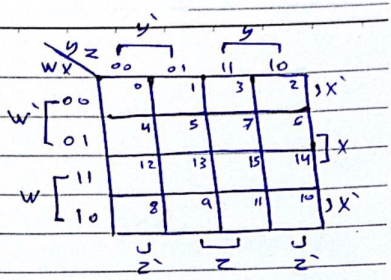
[3] 4-Variable K-map :-

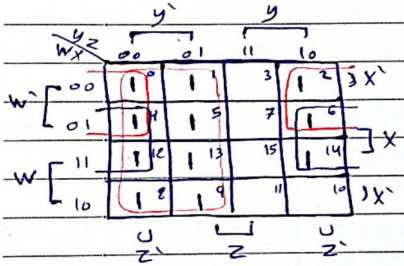
- ✱ Squares = $2^4 = 16$
- Bubble size 16 → 8 → 4 → 2 → 1

✱ terms 0 1 2 3 4



Ex :- Simplify :-

① $F(w,x,y,z) = \Sigma(0,1,2,4,5,6,8,9,12,13,14)$

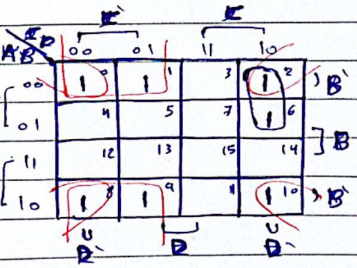


$F = y' + w'z' + xz'$

② $F = A'B'C' + B'CD' + A'BCD' + AB'C'$

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$F = \Sigma(0,1,2,6,8,9,10)$



$F = B'D' + B'C' + A'CD'$

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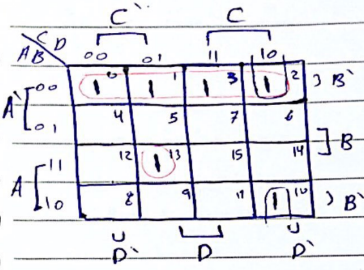
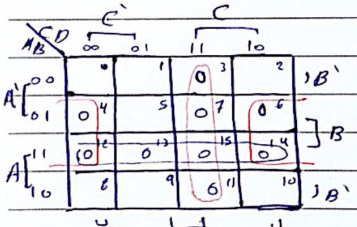
③ $F = \Sigma(0,1,2,3,10,13)$



$F = A`B` + B`CD` + ABC`D$

④ $F(A,B,C,D) = \Sigma(0,1,2,5,6,9,10)$ in pos

$= \Pi(3,4,6,7,11,12,13,14,15)$



$F_{sop}` = CD + BD` + AB$

$F_{pos} = (F_{sop}`)` = (CD + BD` + AB)`$

$F_{pos} = (C`+D`).(B`+D).(A`+B`)$

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* Don't Care Conditions :-

- In some Boolean Function we don't care what is the value of the Function for some minterms.

Ex : Simplify :-

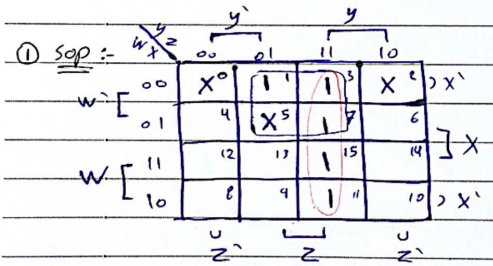
$F(w,x,y,z) = \Sigma(1,3,7,11,15) = \Pi(4,6,8,9,10,12,13,14)$

$d(w,x,y,z) = \Sigma(0,2,5)$

① SOP

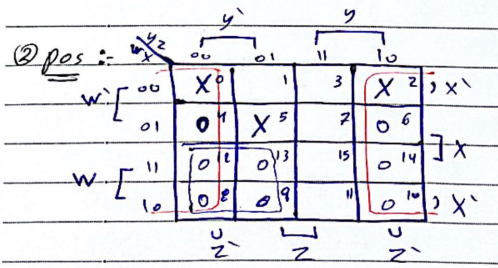
② POS

① SOP :-



$F_{sop} = yz + w'z$

② POS :-



① $F_{sop}' = z' + wy'$

② $F_{pos} = (F_{sop}')' = (z' + wy')'$

③ $F_{pos} = z \cdot (w' + y)$

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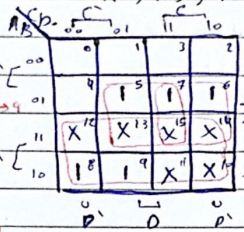
Ex: Design a Circuit that convert from BCD to Excess-3 :

BCD A B C D	Ex-3 E F G H
0 0 0 0 0	0 0 1 1
0 0 0 1 1	0 1 0 0
0 0 1 0 2	0 1 0 1
0 0 1 1 3	0 1 1 0
0 1 0 0 4	0 1 1 1
0 1 0 1 5	1 0 0 0
0 1 1 0 6	1 0 0 1
0 1 1 1 7	1 0 1 0
1 0 0 0 8	1 0 1 1
1 0 0 1 9	1 1 0 0
1 0 1 0 10	X X X X
1 0 1 1 11	X X X X
1 1 0 0 12	X X X X
1 1 0 1 13	X X X X
1 1 1 0 14	X X X X
1 1 1 1 15	X X X X

don't BCD

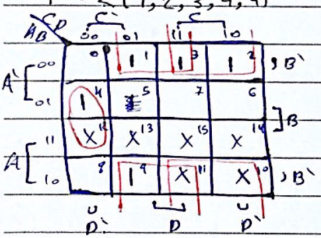
9 → 15

$E = \Sigma(5,6,7,8,9)$. $E(x) = \Sigma(10,11,12,13,14,15)$



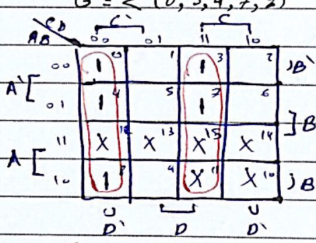
$E = A + BD + BC$

$F = \Sigma(1,2,3,4,9)$



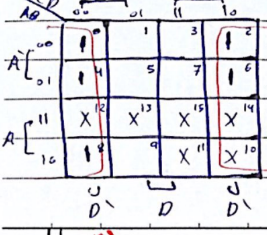
$F = B'C + B'D + BC'D'$

$G = \Sigma(0,3,4,7,8)$

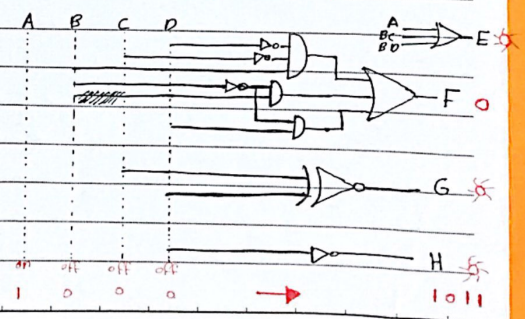


$G = C'D' + CD = (C \oplus D)'$

$H = \Sigma(0,2,4,6,8)$



$H = D'$



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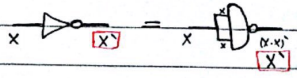
* Universal Gates :-

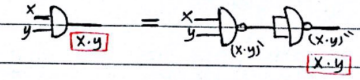
- primitive gates (and, or, not)
- from universal gates (nand / nor) I can implement primitive gates
- any circuit can be implemented using universal gates

* Nand gate implementation :-

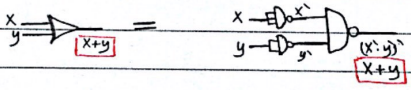
1 not



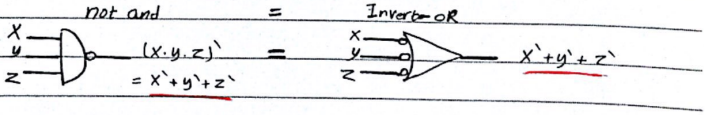
2 and



3 or



Note :-



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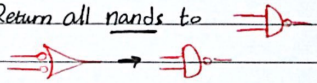
* How to Implement a function using nand gates only :-

① write the function as sop.

② Draw as and - OR

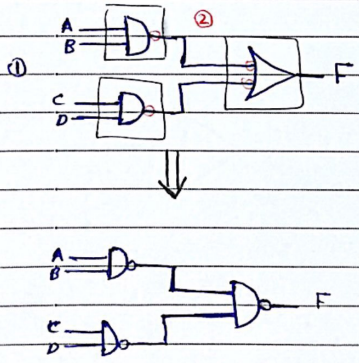
③ Replace and → nand , OR → invert OR

④ Return all nands to



Ex :- Implement the following function using nand gates only :-

① F = AB + CD sop ✓



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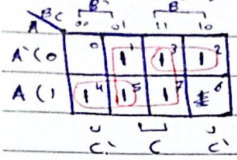
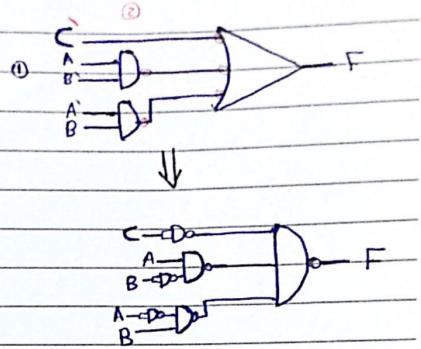
② $F = \Sigma(1,2,3,4,5,7)$



$F = AB` + C + A`B$ sop ✓



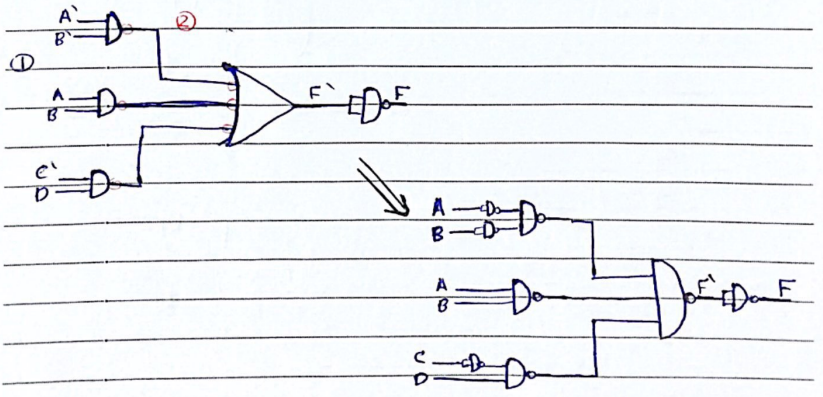
③ $F = (AB` + A`B).(C+D`)$ pos $\xrightarrow[\times]{\text{mult}}$ sop

$F` = (AB` + A`B)` + (C+D`)`$

$= (A`+B).(A+B`) + C`D$

$= A`A + A`B` + AB + BB` + C`D$

$F`_{sop} = \boxed{A`B` + AB + C`D}$ sop ✓



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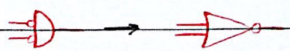
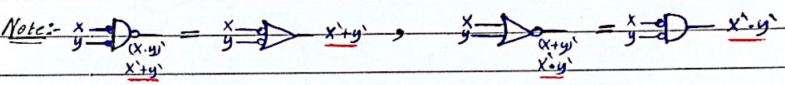
* NOR Implementation :-

① F as pos

② Draw OR/AND

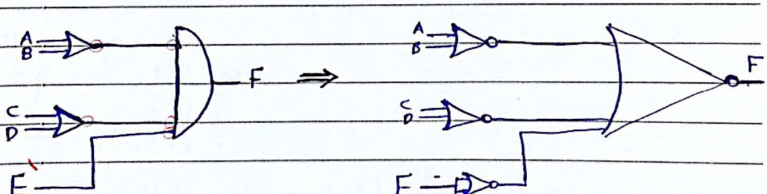
③ Replace

④ 

Note:- 

Ex:- Implement the following functions using NOR gates only:- pos

① F = (A+B)(C+D)E' pos ✓



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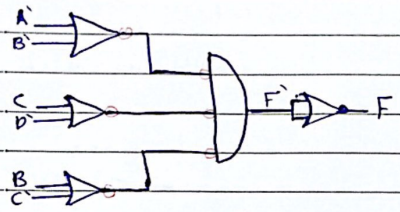
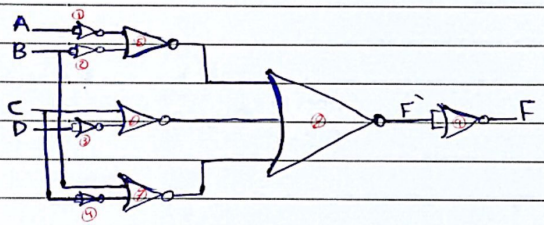
② $F = AB + C'D + B'C$ sop must → pos

$F' = (A'+B').(C+D').(B+C')$ pos ✓



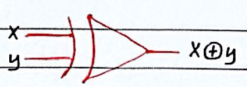


✱ Exclusive OR Function :-

① XOR. Function

$X \oplus y =$ $X'y + Xy'$

X	y	X⊕y
0	0	0
0	1	1
1	0	1
1	1	0



- XOR is an odd Function → # 1's @ input odd → output = 1

even → output = 0

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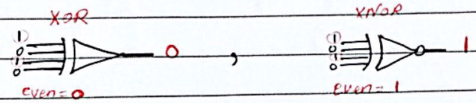
[2] XNOR. Function :-

$(X \oplus y)' = X'y' + Xy$

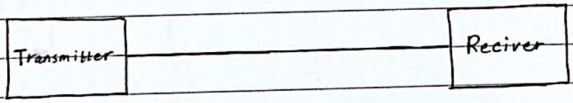
X	y	$(X \oplus y)'$
0	0	1
0	1	0
1	0	0
1	1	1

- XNOR is an even function → # 1's even → output = 1
  odd → output = 0

Ex :- 

* parity check :-

- used for error detection
- Two types of parity ← odd parity, even parity



Ex :- what is the transmitted msg for the following using odd parity :-

① 1101101 → 11011010 = 7
6 ✗ msg P

② 1111101 → 11011010
7 ✓ msg P

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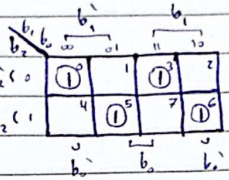
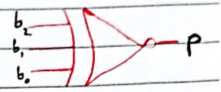
Ex:- Design an odd parity bit generator for a 3-bit word:-

	b_2	b_1	b_0	P
#1's even=1	0	0	0	1
odd=0	0	0	1	0
odd=0	0	1	0	0
even=1	0	1	1	1
odd=0	1	0	0	0
even=1	1	0	1	1
even=1	1	1	0	1
odd=0	1	1	1	0





↳ XNOR Function

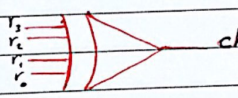
$P = \bar{b_2}\bar{b_1}\bar{b_0} + \bar{b_2}b_1b_0 + b_2\bar{b_1}b_0 + b_2b_1\bar{b_0}$

Ex:- Design an odd parity check circuit for received 4-bit word:-

	r_3	r_2	r_1	r_0	check
#1's even=0	0	0	0	0	0
odd=1	0	0	0	1	1
odd=1	0	0	1	0	1
even=0	0	0	1	1	0
odd=1	0	1	0	0	1
even=0	0	1	0	1	0
even=0	0	1	1	0	0
odd=1	0	1	1	1	1
odd=1	1	0	0	0	1
even=0	1	0	0	1	0
even=0	1	0	1	0	0
odd=1	1	0	1	1	1
even=0	1	1	0	0	0
odd=1	1	1	0	1	1
odd=1	1	1	1	0	1
even=0	1	1	1	1	0

#1's odd → Correct → check = 1

#1's even → Incorrect → check = 0



↳ XOR Function

even → 0

odd → 1

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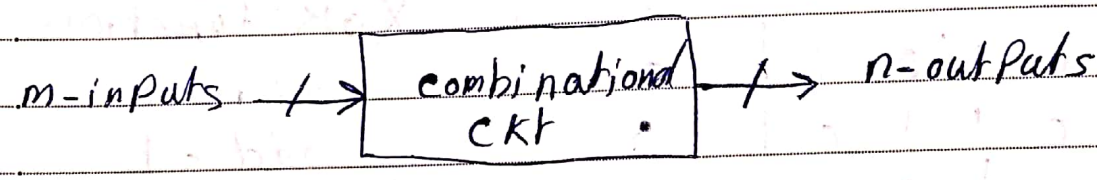
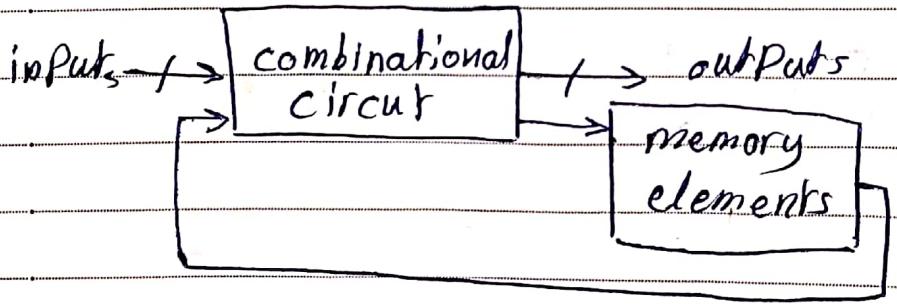
* Combinational circuits :-

→ Two types of logic circuits :-

[1] combinational circuits :-
consists of logic gates whose outputs at any time are determined from only the present inputs



[2] Sequential circuits :-
- employ memory elements in addition to logic gates
- Their outputs are a function of the inputs, the state of the storage elements.

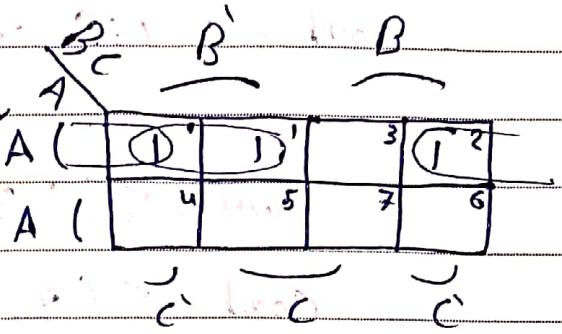


* Design Procedure :-

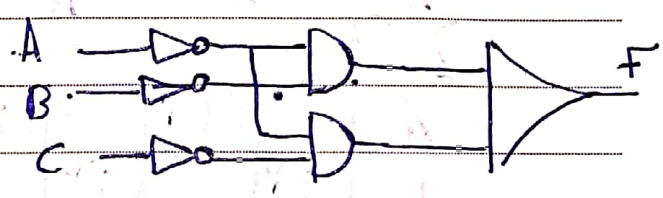
① circuit specification
② Truth table
③ K-maps
④ Boolean function
⑤ logic design (gates)

Ex: Design a combinational circuit which has a 3-bit input, a single output (F) where: F = 1, when input number is less than 3
F = 0, otherwise.

A B C	F
0 0 0	1
0 0 1	1
0 1 0	1
0 1 1	0
1 0 0	0
1 0 1	0
1 1 0	0
1 1 1	0



$F = A'B' + A'C'$



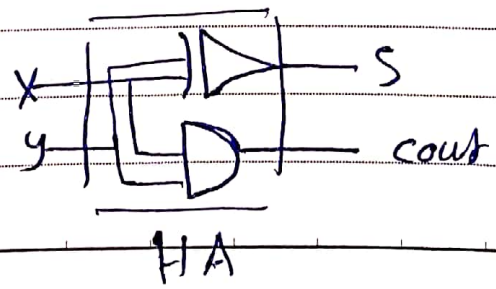
* Binary adders, subtractors :-

1) Half Adder (HA):
- 2 inputs (x, y)
- 2 outputs (sum(s), carry out (cout))
- Sum = x + y (binary addition)
- cout = 1 (if carry occurs)

x y	S	cout
0 0	0	0
0 1	1	0
1 0	1	0
1 1	0	1

$S = X \oplus Y$

$cout = XY$



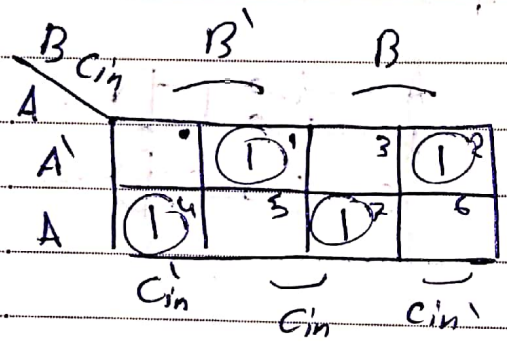
[2] Full Adder (FA) :

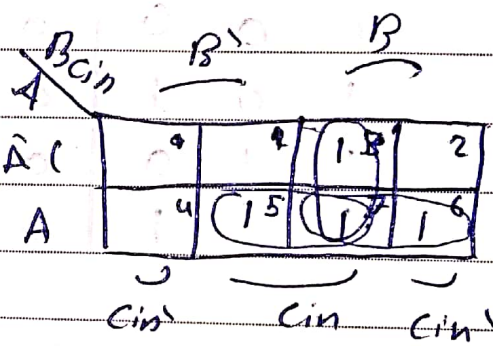
- 3 - inputs (A, B, Cin)
- 2 - outputs (sum(S), carry out (cout))
- sum = A + B + Cin
- cout = 1 (if cout occurs)

A	B	Cin	S	cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$sum = \Sigma(1,2,4,7)$ ←

$cout = \Sigma(3,5,6,7)$





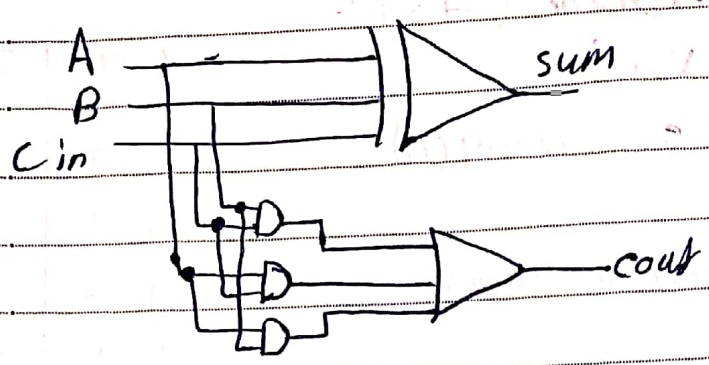
$sum = A\bar{B}\bar{C}in + \bar{A}\bar{B}\bar{C}in + A\bar{B}\bar{C}in + ABCin$

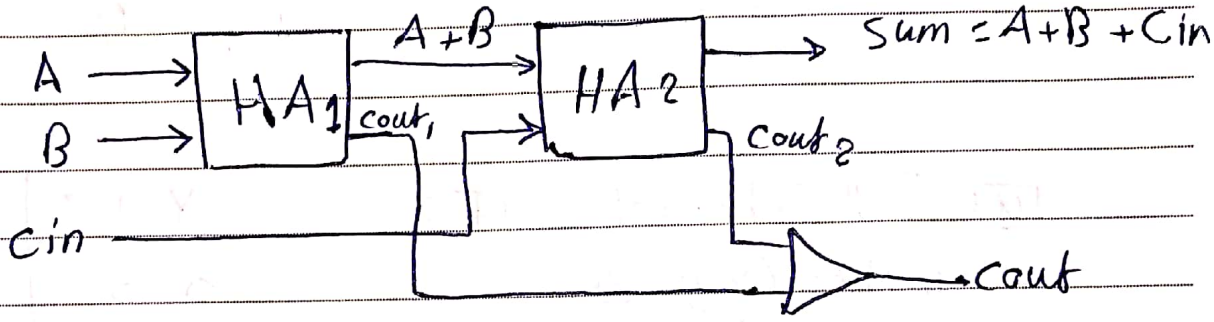
$cout = BCin + ACin + AB$

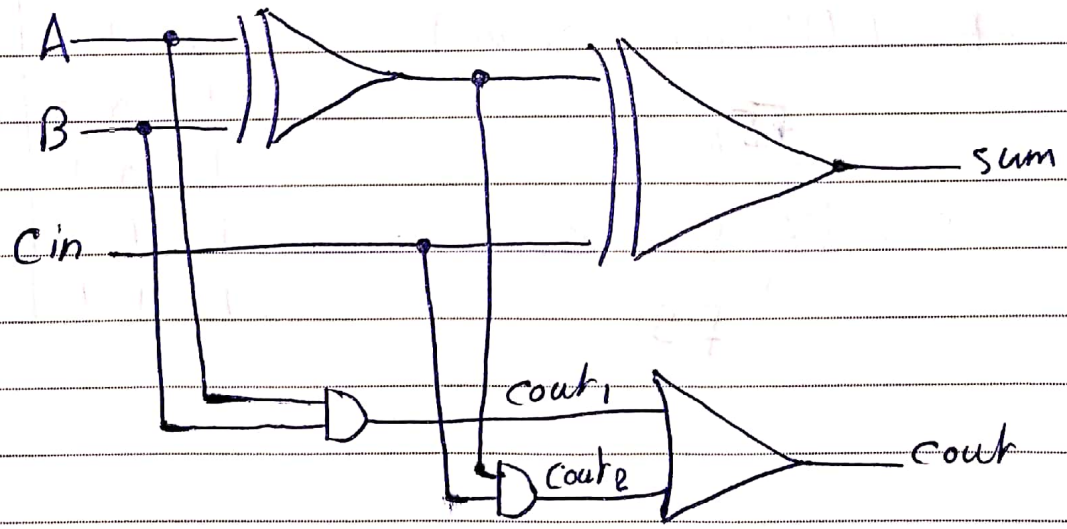
HA

$S = (A + B) + Cin$

HA_2


A
B
Cin
sum
cout

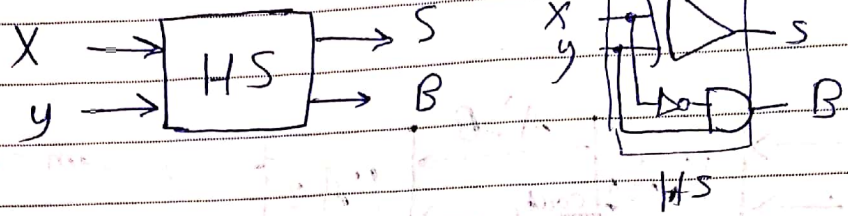

A
B
Cin
HA1
A+B
cout1
HA2
Sum = A+B+Cin
cout2
cout


A
B
Cin
sum
cout1
cout2
cout

[3] Half Subtractor (HS):-

- 2-inputs (X, y)
- 2-outputs (S, Borrow (B))
- S = X - y
- B = 1 (If borrow occurs)

X	y	S	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

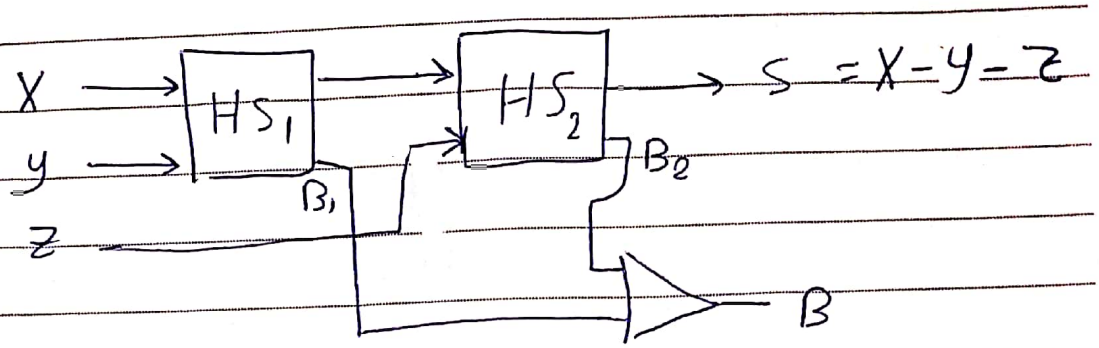


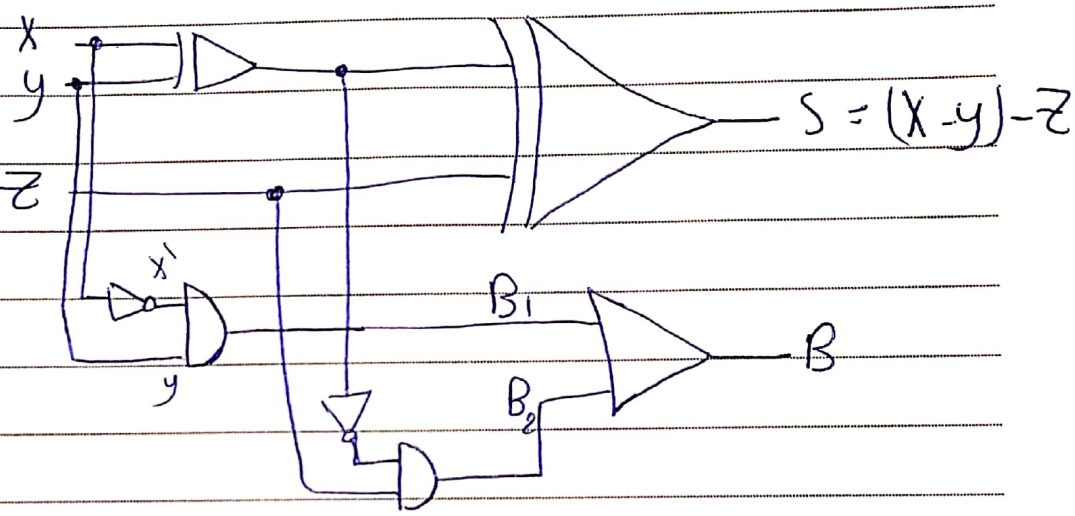
[4] Full Subtractor (FS):-

- 3-inputs (X, y, Z)
- 2-outputs (S, B)
- S = X - y - Z
- B = 1 (If borrow occurs)

$S = \underbrace{(X - y)}_{HS_1} \underbrace{- Z}_{HS_2}$

X	y	Z	S	B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1


X
y
z
HS1
HS2
B1
B2
S = X - y - z
B


X
y
z
x'
y
B1
B2
S = (X - y) - z
B

* 4-bit adder (4bit ripple adder)

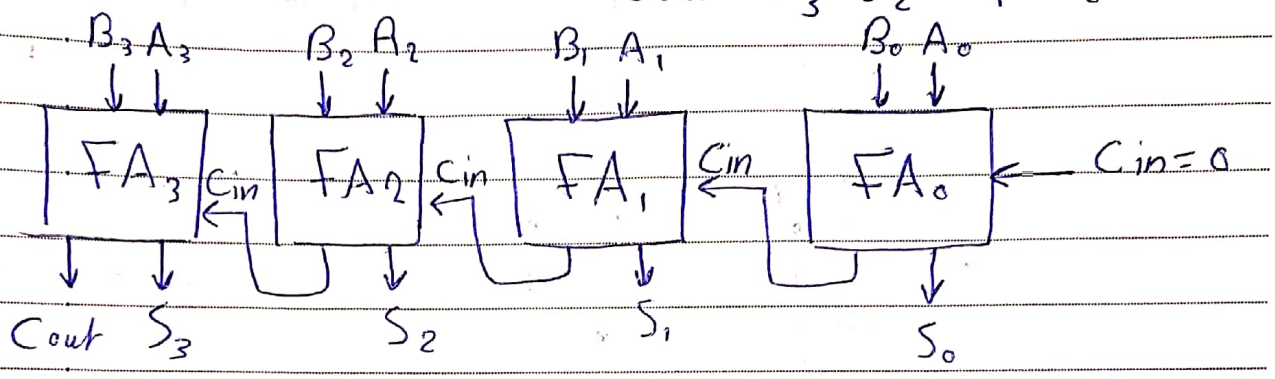
$A = A_3 A_2 A_1 A_0$

$B = B_3 B_2 B_1 B_0$

$S = A + B$

FA FA FA Cout HA

	A_3	A_2	A_1	A_0
	B_3	B_2	B_1	B_0 +
Cout	S_3	S_2	S_1	S_0



* Binary subtractor

$D = A - B$

$= A + (-B)$

$= A + 2's(B)$

$= A + 1's(B) + 1$

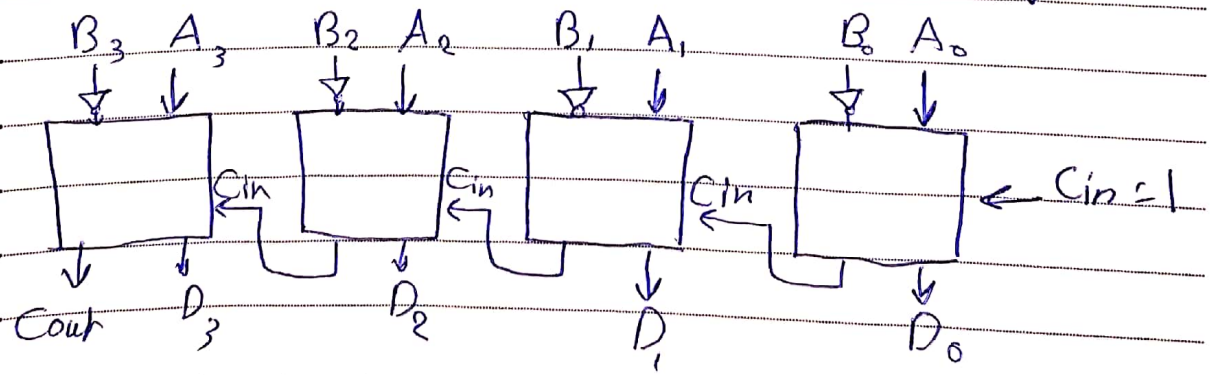
$A = A_3 A_2 A_1 A_0$

$B = B_3 B_2 B_1 B_0$

$D = A - B$

$= A_3 A_2 A_1 A_0 + \bar{B_3}\,\bar{B_2}\,\bar{B_1}\,\bar{B_0} + 1$

	A_3	A_2	A_1	A_0
	$\bar{B_3}$	$\bar{B_2}$	$\bar{B_1}$	$\bar{B_0}$ +
			1	+



* 4 bit Adder subtractor :

- we will use xor

$0 \oplus X = X$

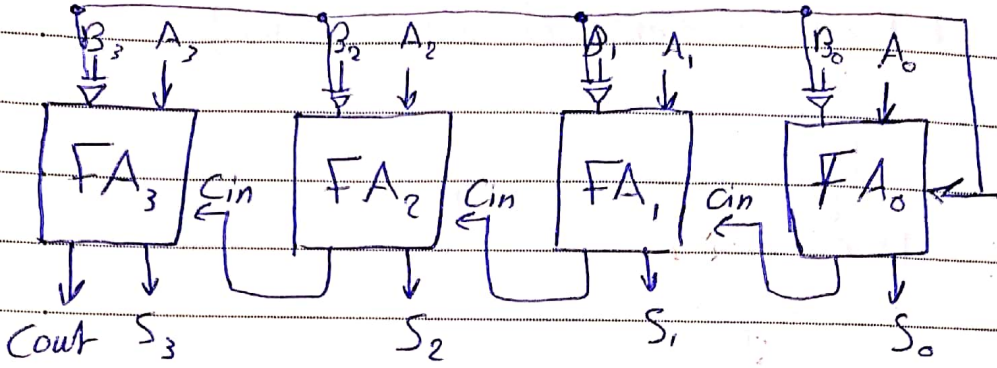
$1 \oplus X = X'$

* $A + B = A_3 A_2 A_1 A_0$
  $B_3 B_2 B_1 B_0$ +

* $A - B = A_3 A_2 A_1 A_0$
  $B_3 B_2 B_1 B_0$ +
  1 +

$C_{in} \rightarrow$ =0 adder

=1 subtractor



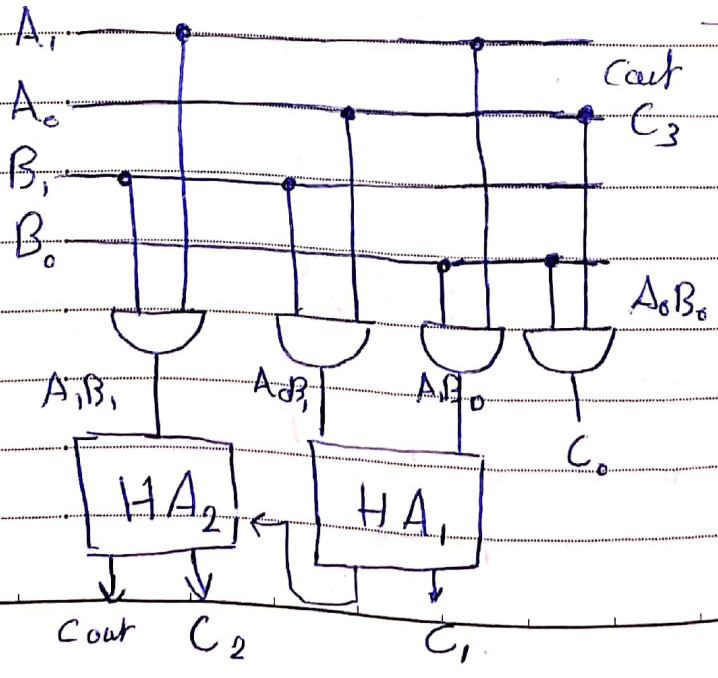
* Binary Multiplier

Ex: Design a 2-bit binary multiplier

$C = A \times B$

$A_1 A_0 \times B_1 B_0$

$A_1 A_0$
$B_1 B_0$ x
A_1B_0 A_0B_0
A_1B_1 A_0B_1 +
Cout C_2 C_1 C_0



* BCD Adder

- A BCD adder adds two BCD digits and Produces sum in BCD

→ Carry → correct digit [+6]

→ not BCD → correct digit [+6]

↓

>9

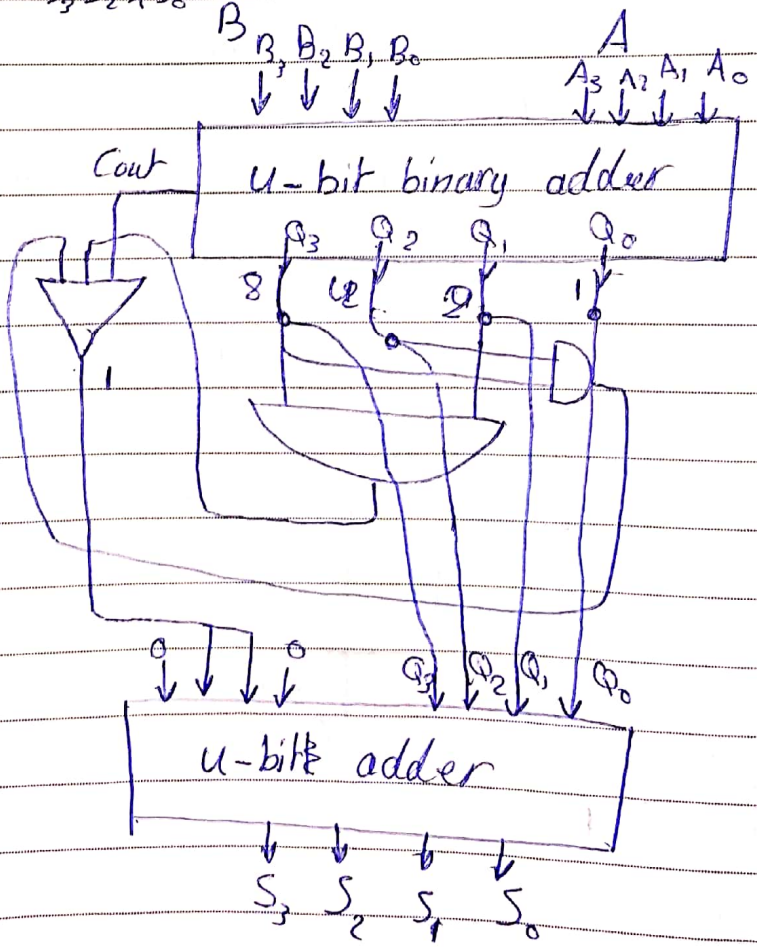
0000 ↓ 1000] BCD ≤ 9

1001

8+2 [1010, 1011]

8+4 [1100, 1101, 1110, 1111] Not BCD > 9

$S_3 S_2 S_1 S_0$



* Magnitude Comparator:
- combinational circuit that compares two numbers, A, B and determines their relative magnitude.
- Inputs: 2 numbers A, B
- outputs: 3, $A>B=1$ if $A>B$
$A<B=1$ if $A<B$
$A=B=1$ if $A=B$

Ex: Design a 2-bit comparator.
$A = A_1 A_0$
$B = B_1 B_0$

$(A \odot B)' = A_1 B_1 + A_0 B_0$

* If $A=B \rightarrow$ =)D- $A=B$

$A_1=1$, $B_1=0$

* If $A>B \rightarrow A_1 \bar{B_1} + A_1 B_1 A_0 \bar{B_0}$
$+ \bar{A_1} \bar{B_1} A_0 \bar{B_0}$

$\bar{A_1} \bar{B_1} A_0 \bar{B_0}$: $A_1=0$, $B_1=0$, $A_0=1$, $B_0=0$

$A_1 B_1 A_0 \bar{B_0}$: $A_1=1$, $B_1=1$, $A_0=1$, $B_0=0$

* If $A<B \rightarrow \bar{A_1} B_1 + A_1 B_1 \bar{A_0} B_0$
$+ \bar{A_1} \bar{B_1} A_0 B_0$

$\bar{A_1} B_1$: $A_1=0$, $B_1=1$

$\bar{A_1} \bar{B_1} A_0 B_0$: $A_1=0$, $B_1=0$, $A_0=0$, $B_0=1$

$A_1 B_1 \bar{A_0} B_0$: $A_1=1$, $B_1=1$, $A_0=0$, $B_0=1$

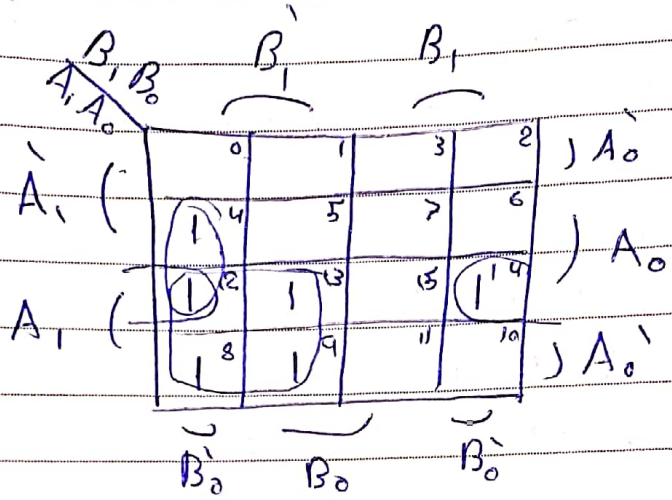
A_1	A_0	B_1	B_0	$A>B$	$A<B$	$A=B$
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	1	0
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	0	1

$(A>B) = \Sigma(4, 8, 9, 12, 13, 14)$

$(A<B) = \Sigma(1, 2, 3, 6, 7, 11)$

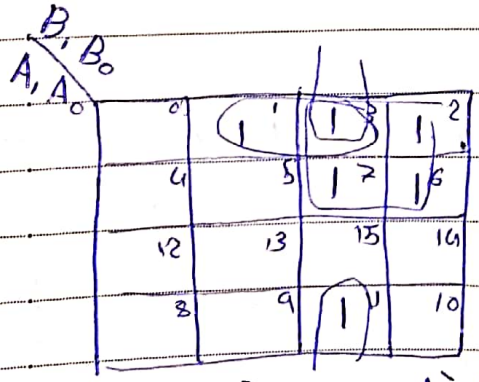
$(A=B) = \Sigma(0, 5, 10, 15)$

$(A>B) = \Sigma(4,8,9,12,13,14)$



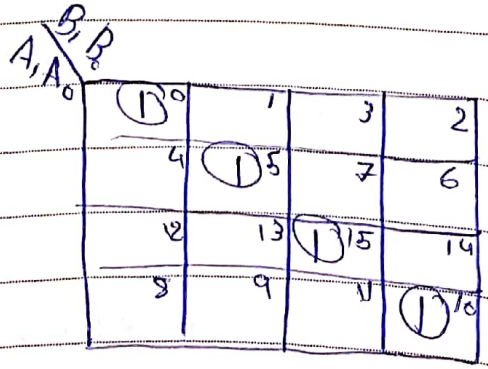
$(A>B) = A_1B_1' + A_1A_0B_0' + A_0B_1'B_0$

$(A<B)$



$(A<B) = A_1'B_1 + A_1'A_0'B_0 + A_0'B_1B_0$

$(A=B)$



$(A=B) = A_1'B_1'A_0'B_0' + A_1'B_1A_0B_0$

$+ A_1B_1A_0B_0 + A_1B_1A_0'B_0'$

* Decoders:-

- A decoders is a combinational circuit that converts binary information from (n) inputs to (2^n) out put lines

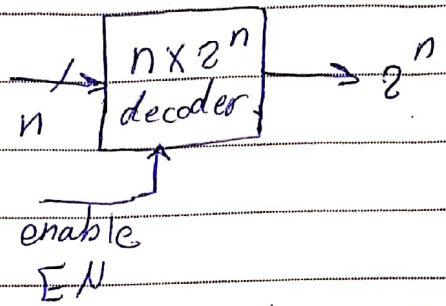
- n to 2^n decoder

2x4

3x8

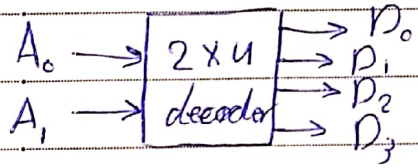
1x2

4x16



- each input of the decoder represents a minterm

Ex: Design a 2x4 decoder



A_1	A_0	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

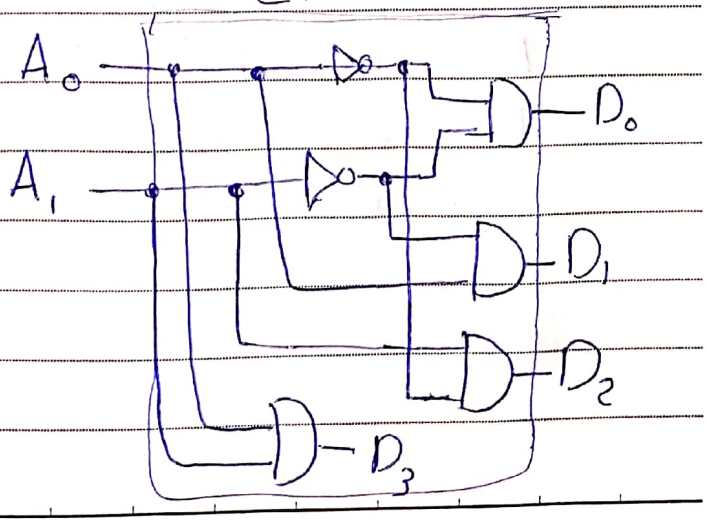
$D_0 = A_1' A_0' = m_0$

$D_1 = A_1' A_0 = m_1$

$D_2 = A_1 A_0' = m_2$

$D_3 = A_1 A_0 = m_3$

2x4 Decoder



Ex: Design a 3x8 decoder Inputs $A_2 A_1 A_0$

$D_0 = m_0 \rightarrow \grave{A}_2 \grave{A}_1 \grave{A}_0$

$D_1 = m_1 \rightarrow \grave{A}_2 \grave{A}_1 A_0$

$D_2 = m_2 \rightarrow \grave{A}_2 A_1 \grave{A}_0$

$D_3 = m_3 \rightarrow \grave{A}_2 A_1 A_0$

$D_4 = m_4 \rightarrow A_2 \grave{A}_1 \grave{A}_0$

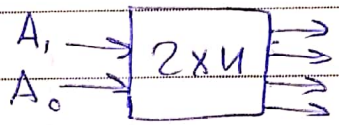
$D_5 = m_5$

$D_6 = m_6$

$D_7 = m_7$

Ex: Design a 2-to-4 Active low decoder

$A_1 A_0$	$D_0 D_1 D_2 D_3$
0 0	0 1 1 1
0 1	1 0 1 1
1 0	1 1 0 1
1 1	1 1 1 0



→ In active low decoder we used nand

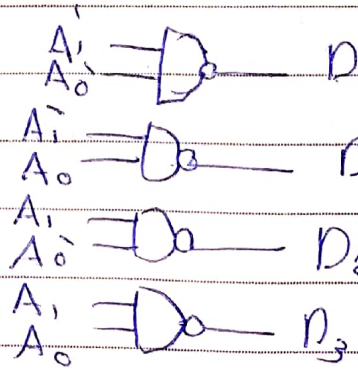
m_0

$D_0 \rightarrow \grave{m_0} \quad (\grave{A_1} \grave{A_0})\grave{}$

$D_1 \rightarrow \grave{m_1} \quad (\grave{A_1} A_0)\grave{}$

$D_2 \rightarrow \grave{m_2} \quad (A_1 \grave{A_0})\grave{}$

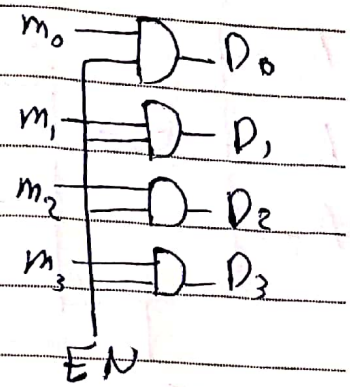
$D_3 \rightarrow \grave{m_3} \quad (A_1 A_0)\grave{}$



Ex: Design 2-to-4 Active high decoder with enable → EN = 1 decoder on
EN = 0 decoder off

$X \cdot 0 = 0$
$X \cdot 1 = X$

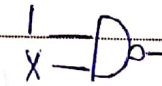
EN	A_1	A_0	D_0	D_1	D_2	D_3
0	X	X	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1



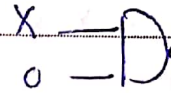
$m_0 = A_1' A_0'$ $m_1 = A_1' A_0$ $m_2 = A_1 A_0'$ $m_3 = A_1 A_0$

Ex: Design a 2x4 Active low decoder with enable EN = 0 → on
EN = 1 → off

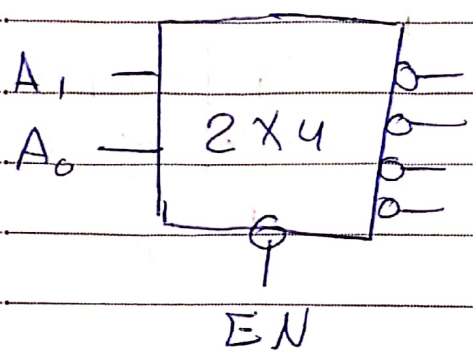
$(1 \cdot X)' = X'$

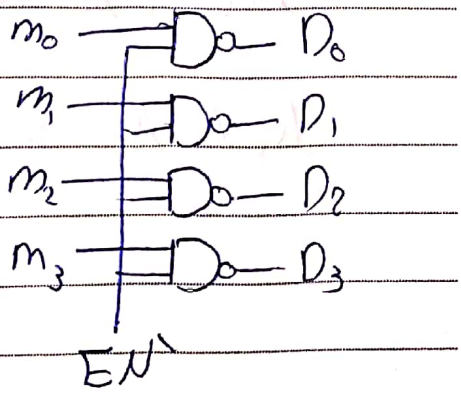


$(X \cdot 0)' = 0' = 1$

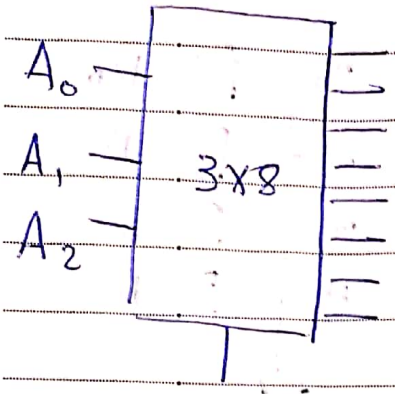


EN	A_1	A_0	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0





Ex: Active High 3 x 8 decoder with enable



EN	A_2	A_1	A_0	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	X	X	X	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1

* Decoder Expansion.
- Building larger decoders using two or more smaller decoders

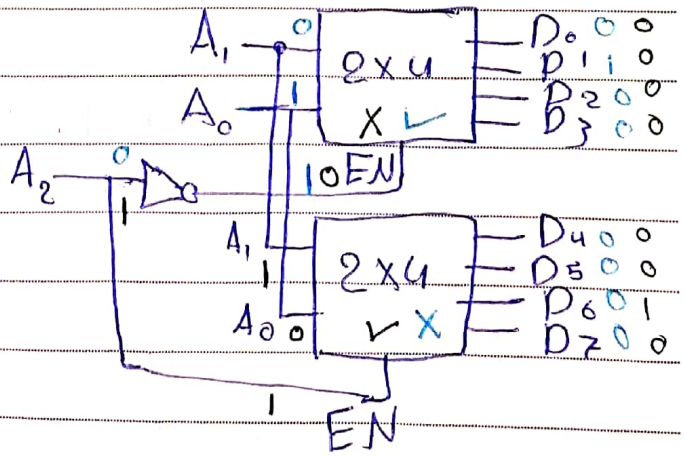
Ex: Constract a 3x8 decoder using 2x4 decoders with enable inputs.

3 x 8 I have 2 x 4

(1)

$A_2 A_1 A_0 \rightarrow 001$

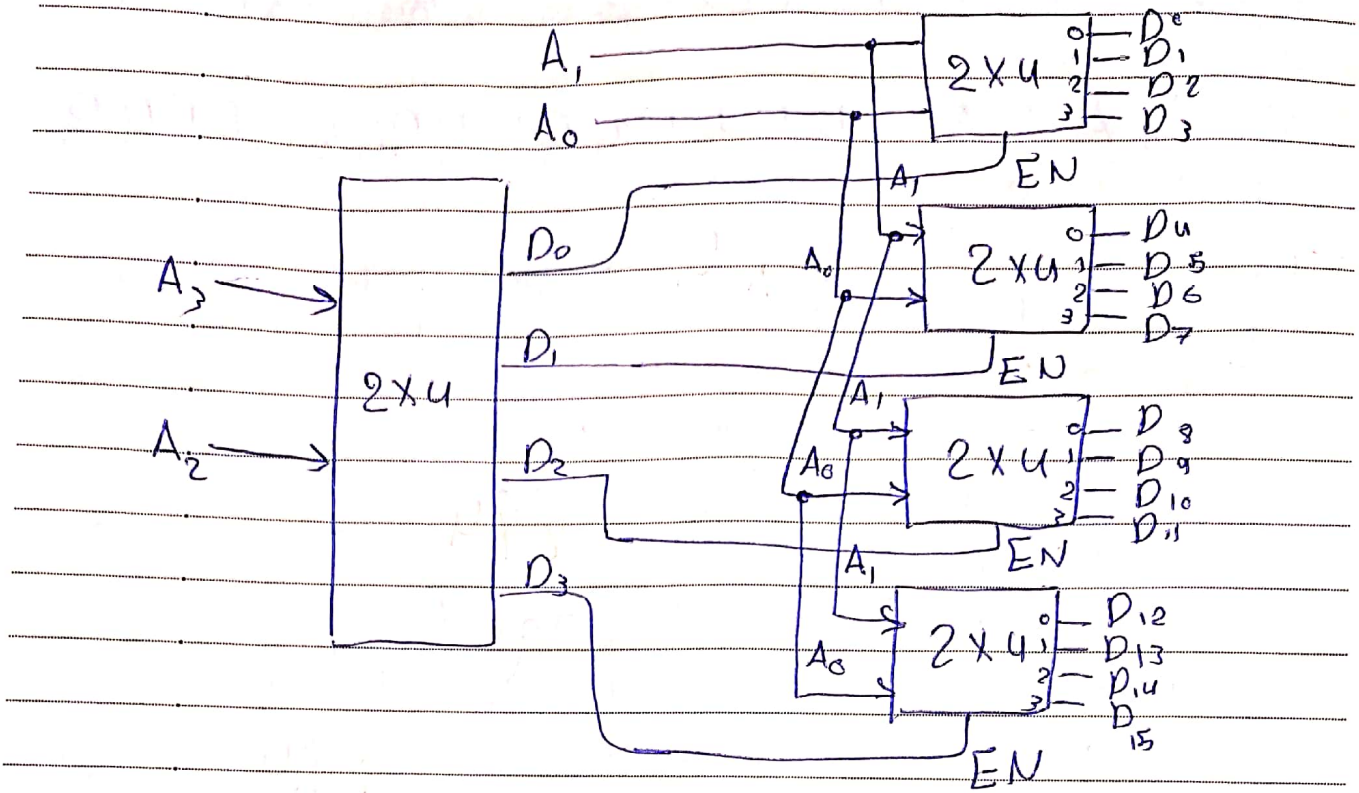
$A_2 A_1 A_0 \rightarrow 110$



Ex: Construct a 4×16 decoder using 2×4 decoders

A_3 A_2 A_1 A_0	D_0 D_1 D_2 D_3	D_4 D_5 D_6 D_7	D_8 D_9 D_{10} D_{11}	D_{12} D_{13} D_{14} D_{15}
EN 0 0 0 0	1 0 0 0			
EN 0 0 0 1	0 1 0 0	∅	∅	∅
EN 0 0 1 0	0 0 1 0			
EN 0 0 1 1	0 0 0 1			
0 1 0 0	↓ 00	1 0 0 0		
0 1 0 1	∅	0 1 0 0	∅	∅
0 1 1 0		0 0 1 0		
0 1 1 1		0 0 0 1		
1 0 0 0		↓ 01	1 0 0 0	
1 0 0 1	∅	∅	0 1 0 0	∅
1 0 1 0			0 0 1 0	
1 0 1 1			0 0 0 1	
1 1 0 0			↓ 10	1 0 0 0
1 1 0 1	∅	∅	∅	0 1 0 0
1 1 1 0				0 0 1 0
1 1 1 1				0 0 0 1

11



$A_3A_2 = 00 \rightarrow D_0 = 1$ & on ① ② ③ off

$D_1, D_2, D_3 = 0$

$A_3A_2 = 11 \rightarrow D_3 = 1$

$D_0, D_1, D_0 = 0$

* Using Decoder to implement a Boolean function

→ Decoder output are minterms.

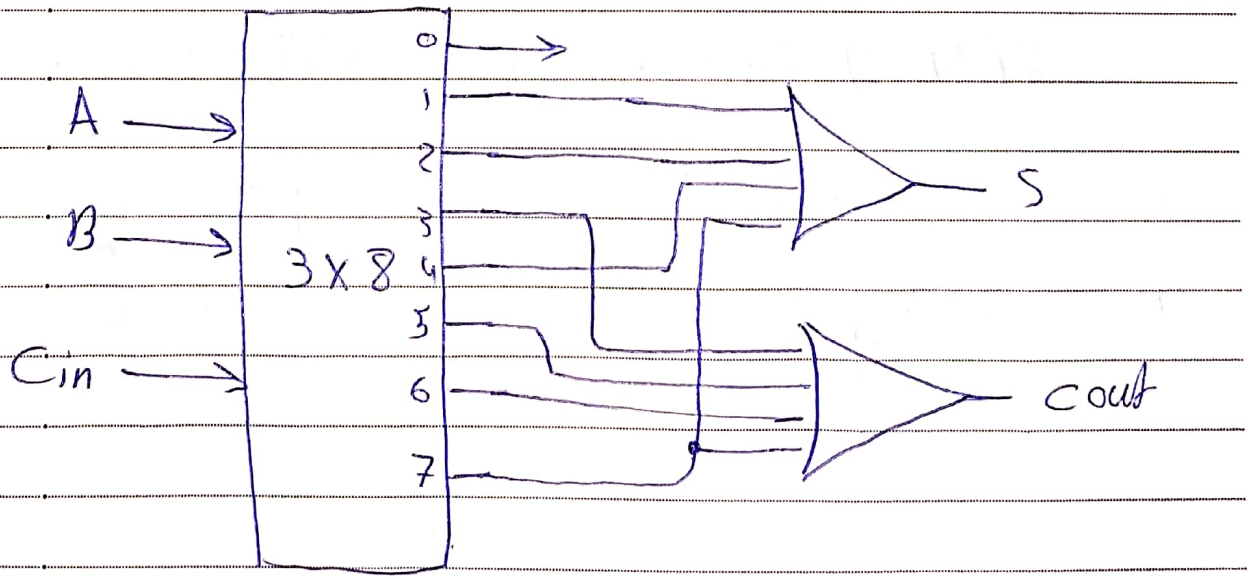
→ Any Boolean function in a sum of minterms

Ex: Implement a full adder using decoder

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$S = \Sigma(1,2,4,7)$

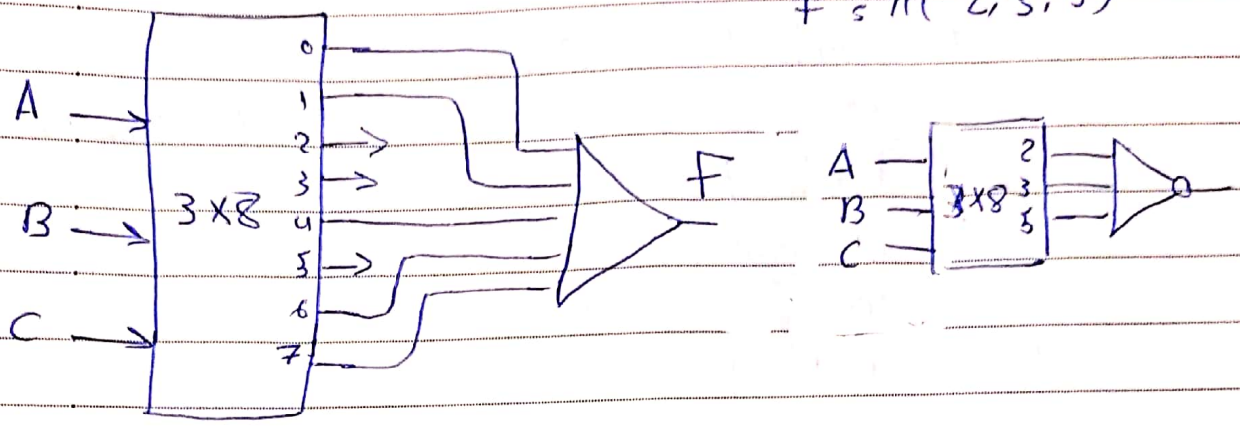
$Cout = \Sigma(3,5,6,7)$



Ex: Implement the Boolean function
$F(ABCD) = \Sigma(0,1,4,6,7)$
using 3 x 8 Decoder

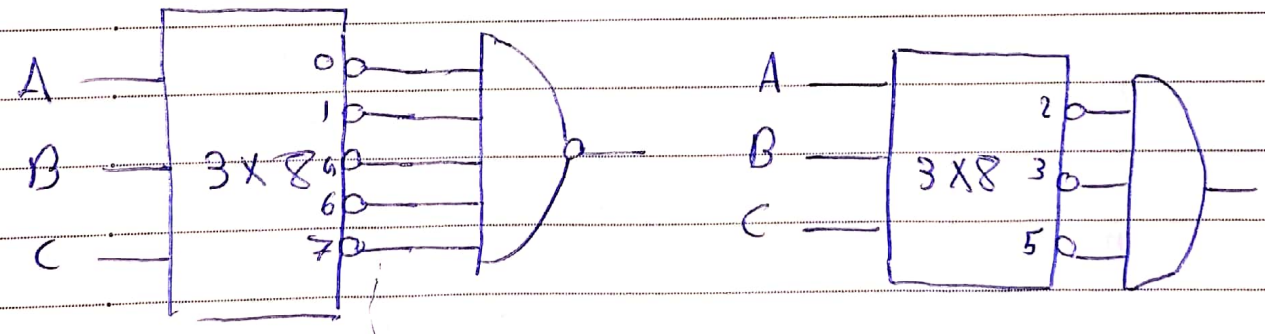
① Active high

$F = \pi(2,3,5)$



② Active low

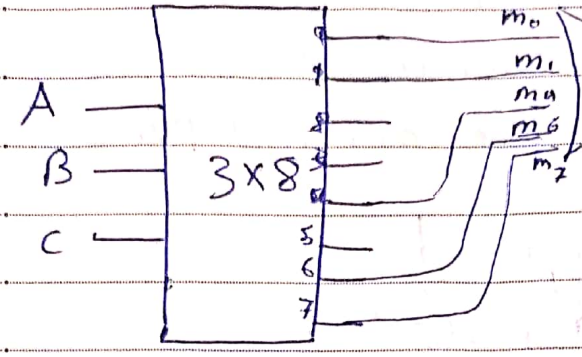
$F(ABC) = (0,1,4,6,7)$ $F = \pi(2,3,5)$



Ex: Implement $F(A,B,C) = \Sigma(0,1,4,6,7)$ using decoder

* Σ + Active high → OR
* Π + Active high → NOR

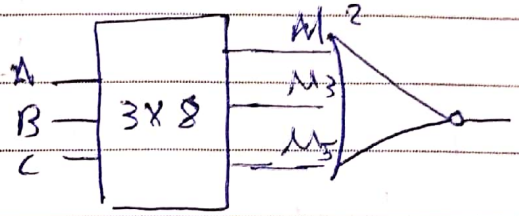
Active high decoder



$m_0 + m_1 + m_4 + m_6 + m_7$

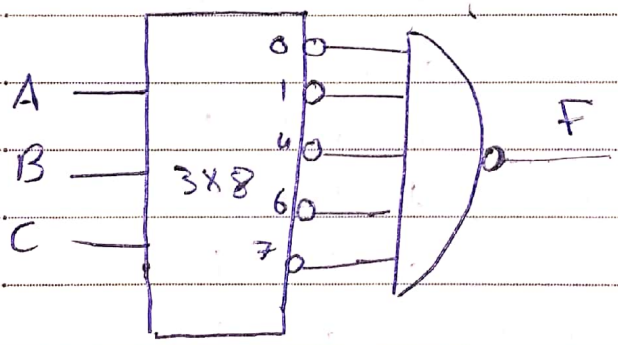
$= \Sigma(0,1,4,6,7)$

$F = \Pi(2,3,5)$ →



Active low decoder

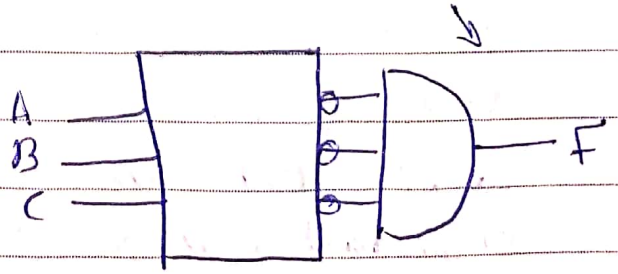
* Σ + Active low → Nand



$F = \Sigma(0,1,4,6,7)$

* Π + Active low → And

$F = \Pi(2,3,5)$



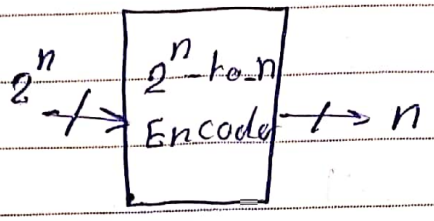
* Encoders:-

→ Performs the inverse operation of a decoder

→ # inputs = 2^n , # outputs = n

→ only one input is allowed to be active

→ output lines generate the binary code corresponding to the active input



Ex: octal to binary encoder 8X3

Inputs								outputs			
Invalid input ← E_7	E_6	E_5	E_4	E_3	E_2	E_1	E_0	A_2	A_1	A_0	V
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	0	1	0	0	0	1	1
0	0	0	0	0	1	0	0	0	1	0	1
0	0	0	0	1	0	0	0	0	1	1	1
0	0	0	1	0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0	1	0	1	1
0	1	0	0	0	0	0	0	1	1	0	1
1	0	0	0	0	0	0	0	1	1	1	1

* two Problems

→ All outputs are 0's in 2 cases:

① all inputs 0 ② E_0 = 1

solution: add one more output (V) if inputs is valid

→ If two inputs are active at the same time

$A_2A_1A_0$

$E_5 = 1$ 101 → 111 → 7

$E_2 = 1$ 010 ✗

Solution:-

use Priority Encoder → Produces output corresponding to the input with higher priority

$E_5 = 1$ $A_2A_1A_0$ * نأخذ الرقم الأكبر

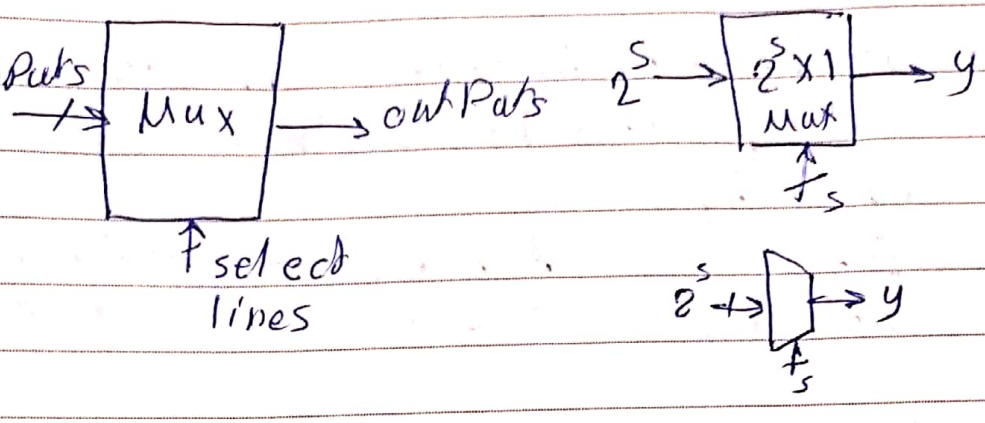
$E_2 = 1$ x 101

Ex: 4-to-2 Priority Encoder

E_3	E_2	E_1	E_0	A_1	A_0	V
0	0	0	0	x	x	0
0	0	0	1	0	0	1
0	0	1	0	0	1	1
0	0	1	1	0	1	1
0	1	0	0	1	0	1
0	1	0	1	1	0	1
0	1	1	0	1	0	1
0	1	1	1	1	0	1
1	0	0	0	1	1	1
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

* Multiplexers (data selector)

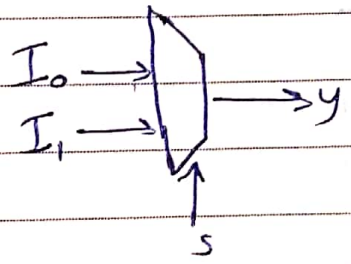
- A multiplexer is a combinational circuit that selects on of the 2^s input lines to be passed to a single output line



Ex: 2 × 1 Mux

① → # select s l

2 × 1

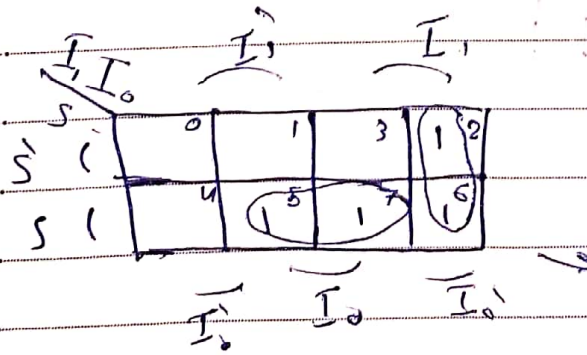


$S = 0 \rightarrow y = I_0$

$S = 1 \rightarrow y = I_1$

Ol: $I_s l_i$ *

I_1	I_0	S	y	
0	0	0	0	0
0	0	1	0	1
0	1	0	1	2
0	1	1	0	3
1	0	0	0	4
1	0	1	1	5
1	1	0	1	6
1	1	1	1	7

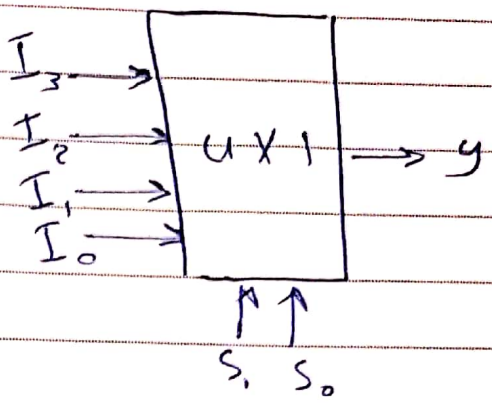


$y = S I_0 + I_1 I_0$

Ex: 4x1 Mux

$4 \times 1 = 2^{(2)} \times 1$ → select

S_1 S_0	y
0 0	I_0
0 1	I_1
1 0	I_2
1 1	I_3



* Implementation of logic circuits using Mux

- If we have an n-variable logic function we can implement using Mux.

① using a mux with n-select lines
② using a mux with less than n-select lines

Ex: Implement

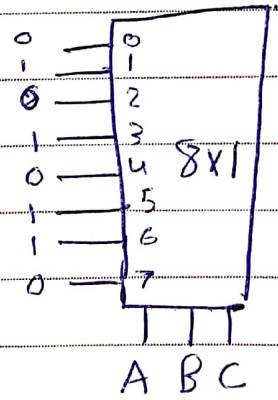
$$F(A,B,C) = \Sigma(1,3,5,6)$$

using

① 8x1 Mux

② ↓

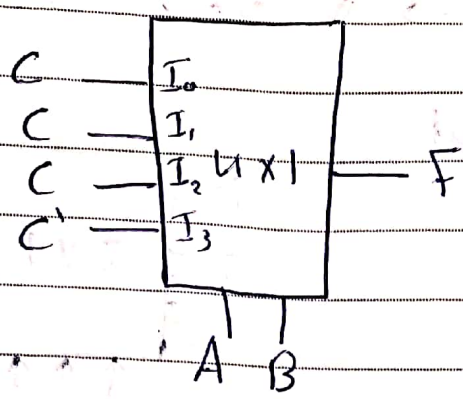
2 x 1 → #select ≤ 3 ≤ #inputs



A B C	F
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	1
1 1 0	1
1 1 1	0

② use a 4x1 Mux $F(ABC) = \Sigma(1,3,5,6)$

$4 \times 1 = 2^2 \times 1 \rightarrow$ # select = 2

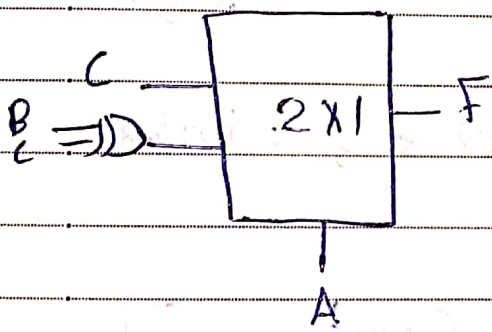


select msB select

A	B	C	F	
0	0	0	0	I_0
0	0	1	1	
0	1	0	0	I_1
0	1	1	1	
1	0	0	0	I_2
1	0	1	1	
1	1	0	1	I_3
1	1	1	0	

③ use 2x1 Mux $F(ABC) = \Sigma(1,3,5,6)$

$2^1 \times 1 \rightarrow$ # select = 1



select

A	B	C	F	
0	0	0	0	I_0
0	0	1	1	
0	1	0	0	
0	1	1	1	
1	0	0	0	I_1
1	0	1	1	
1	1	0	1	
1	1	1	0	

$I_0 = B'C + BC$

$= C(B' + B)$

$= C$

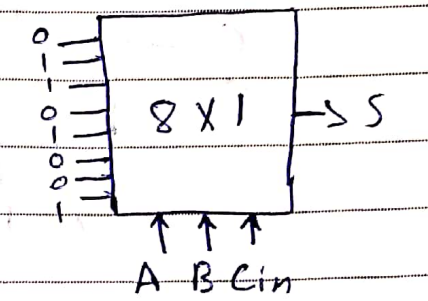
$I_1 = B \oplus C$

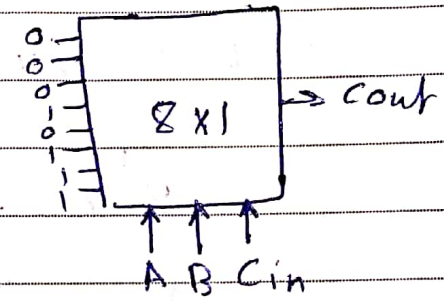
Ex: Implement FA using

① 8 x 1 Mux

8 x 1 → 2^3 x 1 → # select = 3 = inputs

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

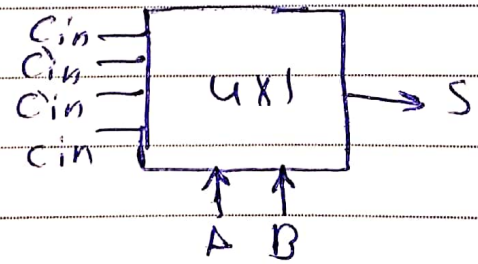


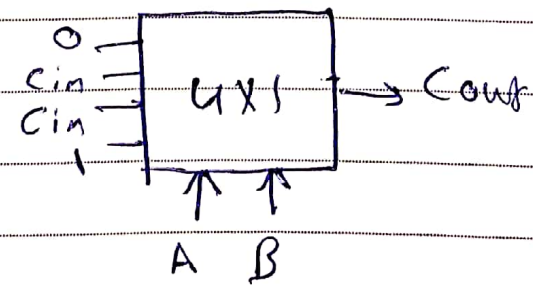


② 4 x 1 Mux

4 x 1 → 2^2 x 1 → select < 3 = inputs

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	Cin	0
0	1	0	Cin	0
0	1	1	Cin	1
1	0	0	Cin	0
1	0	1	Cin	1
1	1	0	Cin	1
1	1	1	Cin	1



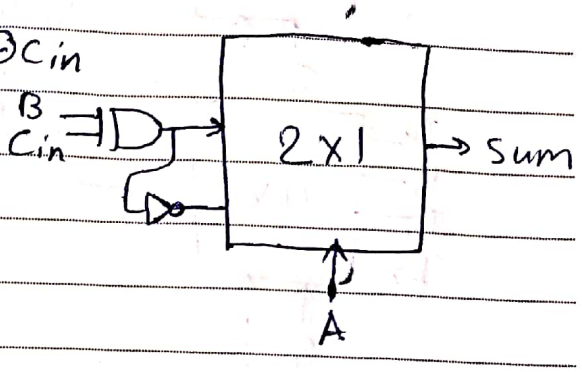


③ 2 x 1 Mux

select — 2 x 1 → 2' x 1 → #select = 1 (sum → msB)

A	B	Cin	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

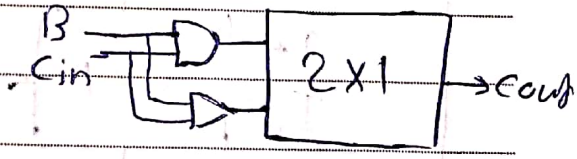
→ $B \oplus Cin$



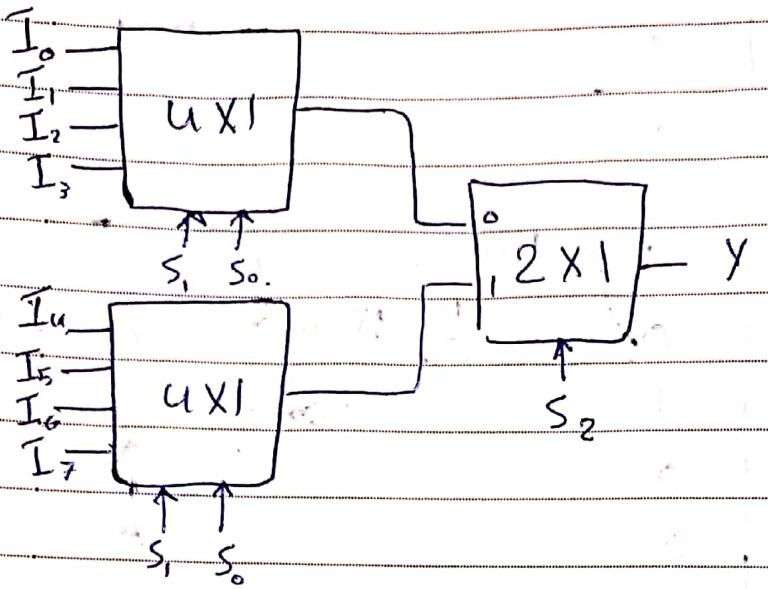
A	B	Cin	Cout
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Cout = B · Cin

Cout = B + Cin

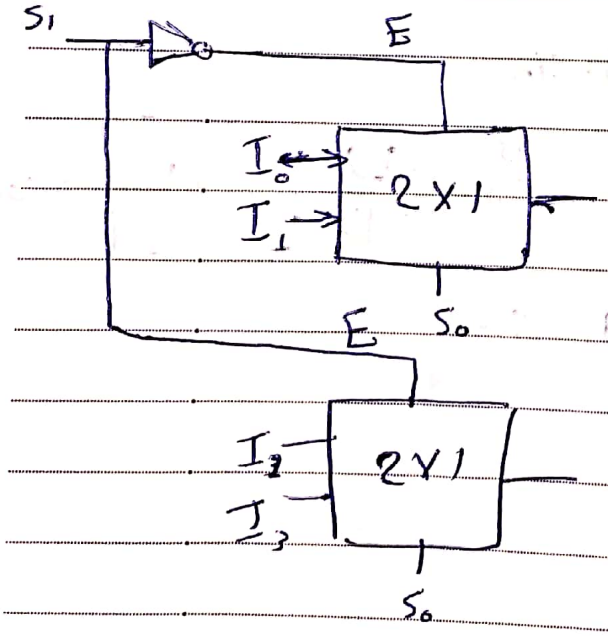


Ex: Construct an 8x1 Mux using 4x1 and 2x1 Muxs.



S_2 S_1 S_0	Y
0 0 0	I_0
0 0 1	I_1
0 1 0	I_2
0 1 1	I_3
1 0 0	I_4
1 0 1	I_5
1 1 0	I_6
1 1 1	I_7

Ex: Construct an 4x1 Mux using 2x1 Mux with enable.

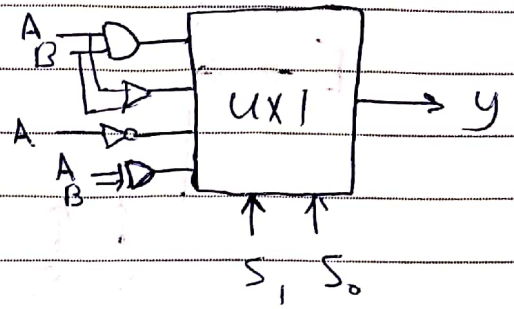


enable

E S_1 S_0	Y
0 0	I_0
0 1	I_1
1 0	I_2
1 1	I_3

Ex: Design a circuit to represent the following function using 4x1 Mux and external gates

S_1	S_0	output = y
0	0	A OR B
0	1	A and B
1	0	A`
1	1	A XOR B



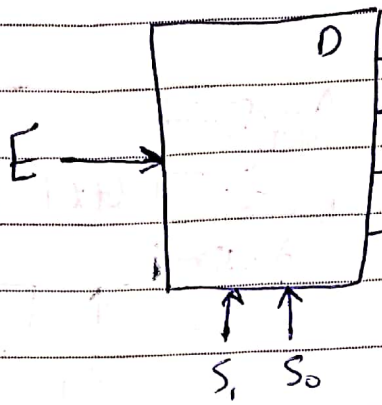
* Demultiplexer

→ Performes inverse of multiplexing operation
→ It has one input line (E) at transmits it to one of 2^n possible output lines
→ The selection of the specific output is controlled by the input on n-select lines

1x2	Mux →	# s = 1
1x4		# s = 2
1x8		# s = 3
1x16		# s = 4

Ex: 1x4 Demux

select lines = 2



$D_0 = E \cdot \bar{S_1} \bar{S_0}$ 0 0

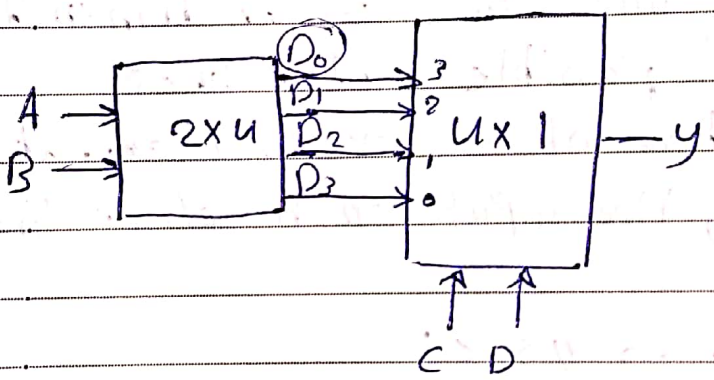
$D_1 = E \cdot \bar{S_1} S_0$ 0 1

$D_2 = E \cdot S_1 \bar{S_0}$ 1 0

$D_3 = E \cdot S_1 S_0$ 1 1

Examples

[1] Fill in the T.T



$D_0 D_1 D_2 D_3$

AB → 00 → 1 0 0 0

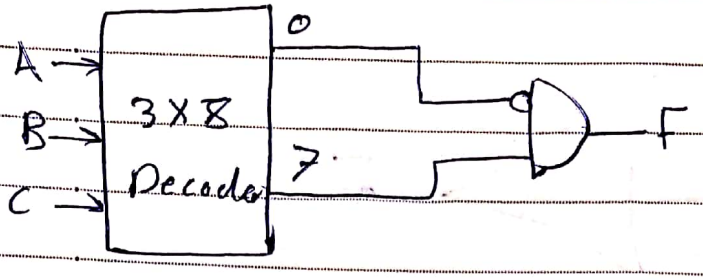
AB → 01 → 0 1 0 0

AB → 10 → 0 0 1 0

AB → 11 → 0 0 0 1

A B C D	y
0 0 0 0	0
0 0 0 1	0
0 0 1 0	0
0 0 1 1	1
0 1 0 0	0
0 1 0 1	0
0 1 1 0	1
0 1 1 1	0
1 0 0 0	0
1 0 0 1	1
1 0 1 0	0
1 0 1 1	0
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	0

Ex [2] Find F(A,B,C) in SOP



* each output of decoder is minterm

$F = m_0 \cdot m_7$

$= (A'B'C')' \cdot (ABC)$

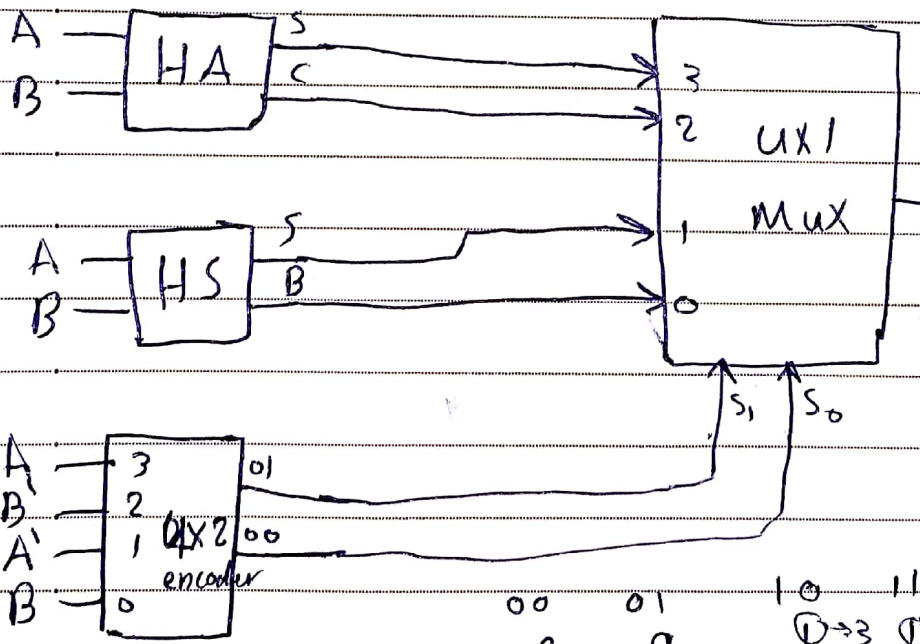
$= (A+B+C) \cdot (ABC)$

$= AABC + ABBC + ABCC$

$= ABC + ABC + ABC$

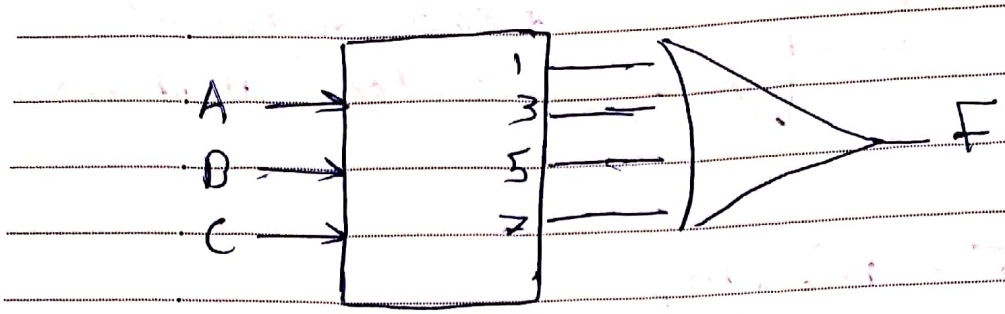
$= ABC$

Ex: fill the table

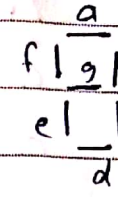


A	B	y
0	0	0
0	1	1
1	0	1
1	1	0

Ex. Build $F(A,B,C) = \Sigma(1,3,5,7)$ using decoder

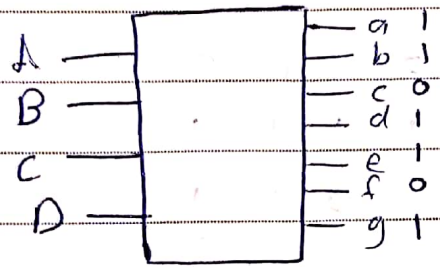


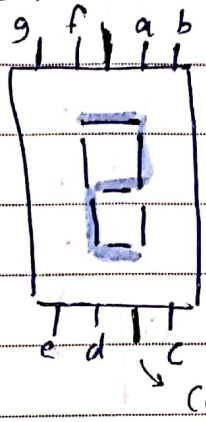
* BCD to 7 segment Display Decoder :-



- The BCD-7 segment decoder
  input : BCD code (4-bits)
  output : 7-outputs (a, b, c, d, e, f, g)

outputs drive the 7-segment display devices to produce decimal read out





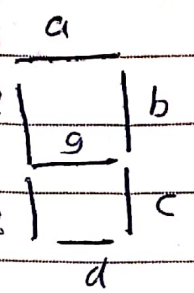
common

CA → 5 volt

CC → 0 volt

- Two types of 7-segment.
  ① Common Anode (Active low)
  ② common Cathode (Active high)

A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1
1	0	1	0	dont care						
:										
1	1	1	1							



$a = \Sigma(0, 2, 3, 5, 6, 7, 8, 9)$

$b = \Sigma(0, 1, 2, 3, 4, 7, 8, 9)$

$c = \Sigma(0, 1, 3, 4, 5, 6, 7, 8, 9)$

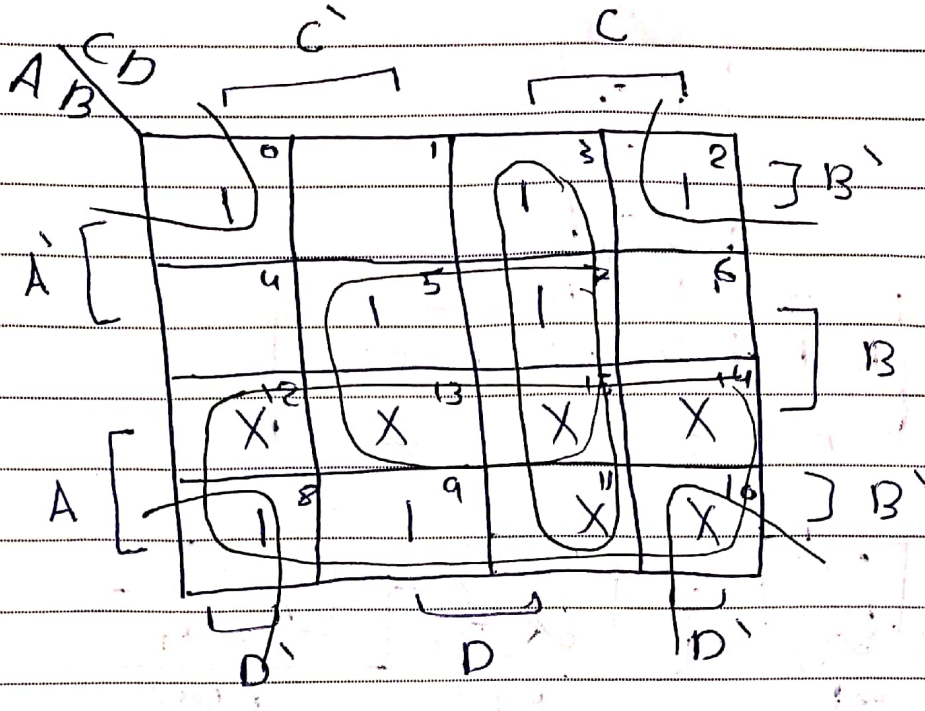
$d = \Sigma(0, 2, 3, 5, 6, 8, 9)$

$e = \Sigma(0, 2, 6, 8)$

$f = \Sigma(0, 4, 5, 6, 8, 9)$

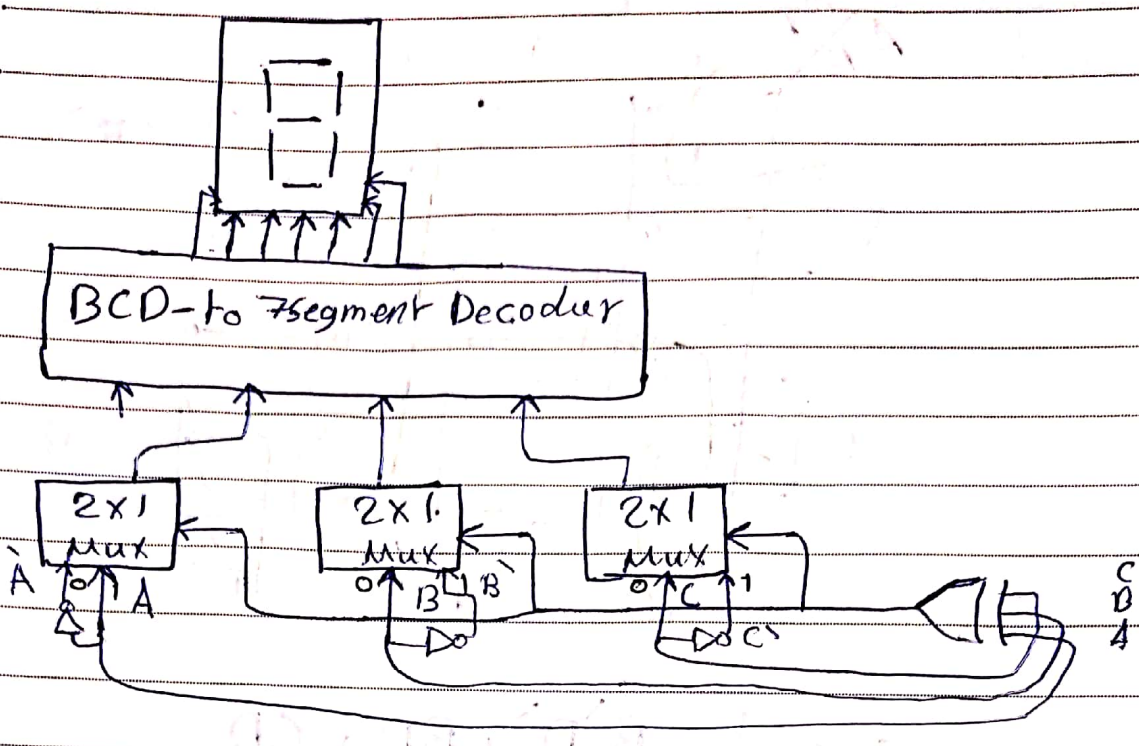
$g = \Sigma(2, 3, 4, 5, 6, 8, 9)$

$$a = \Sigma(0,2,3,5,7,8,9) + \Sigma_d(10,11,12,13,14,15)$$



$$a = A + B`D` + BD + CD$$

Ex: what is the output shown on the 7-segment display

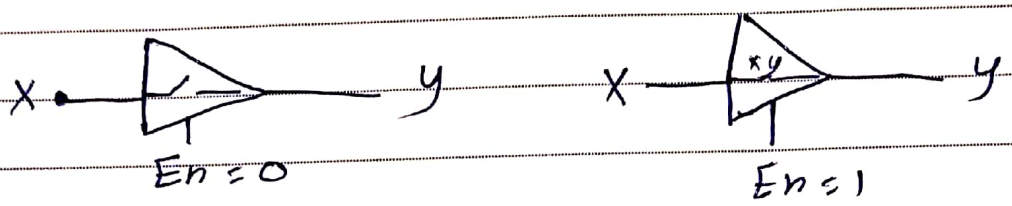


A B C	S	BCD	Display
0 0 0	0	0 0 0 0	0
0 0 1	1	0 1 1 0	6
0 1 0	1	0 1 0 1	5
0 1 1	0	0 0 1 1	3
1 0 0	1	0 0 1 1	3
1 0 1	0	0 1 0 1	5
1 1 0	0	0 1 1 0	6
1 1 1	1	0 0 0 0	0

* Buffer

- A basic logic gate that passes input unchanged to it output
- A Buffer has one input and one output. its output is always equal to its input
- Buffers are used to increas Propogation delay of circuits.

* Tri-state buffer



open circuit
y = high Impedence
open circuit (Z)

y = X

X	y
0	0
1	1

tri-state

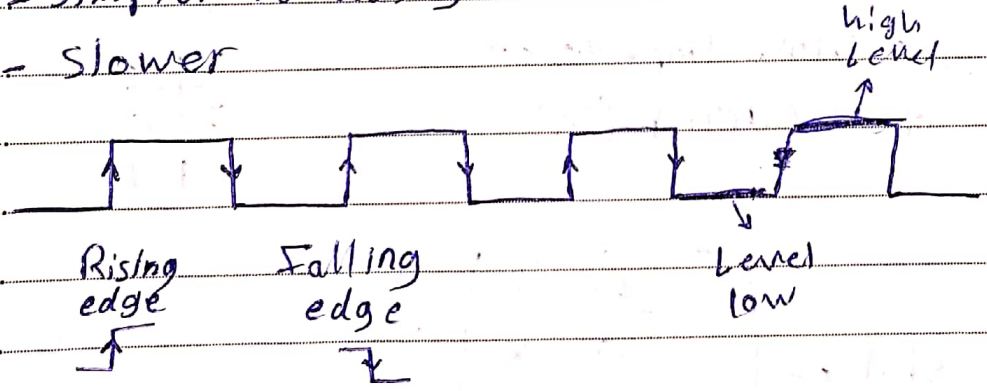
En	X	y
0	X	Z (open circuit)
1	0	0
1	1	1

* Sequential logic circuits

- sequential circuits include memory elements that store binary information.
- outputs in sequential circuits.
  ① Present state (strong element)
  ② Inputs.

* Two types of sequential circuits

[1] Synchronous sequential circuit
- The state is allowed to change at specifice time
- There is a clock (CLK) input
- simpler to design
- slower



[2] Asynchrounous
- The stored value is allowed to change at any time
- There is no clock
- complex to design
- faster

* Basic Memory Elements

[1] Latches:

- operate with signal Level (level sensitive)
- output can change many times in clock cycle

[2] Flip Flops (FFs)

- controled by clock transition (edge sensitive)
- output changes only once in clock cycle

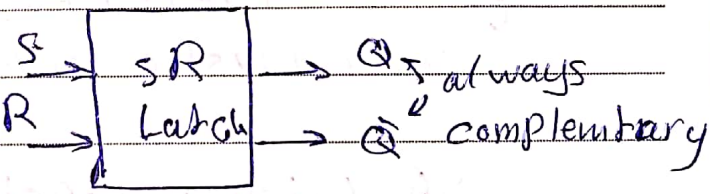
* Latches

[1] S-R Latch (set/Rest latch)

→ has two inputs

① set input

② Rest input



→ has three modes

① set mode → lunch output is set

$Q^+ = 1$
$R = 0 \quad S = 1$

② Rest mode → lunch output is Reset

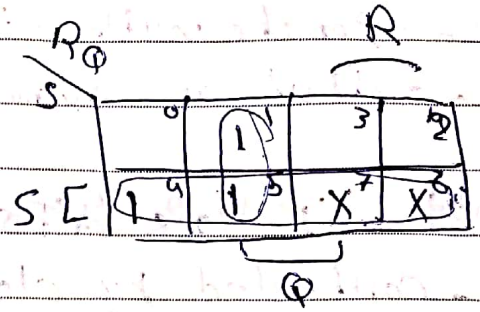
$S = 0 \quad R = 1$
$Q^+ = 0$

③ store mode → lunch output doesn't change

$Q^+ = Q$
$R = 0 \quad S = 0$

State table

	S	R	Q	Q^+
$Q^+ = Q$ store	0	0	0	0
	0	0	1	1
reset $Q^+ = 0$	0	1	0	0
	0	1	1	0
set $Q^+ = 1$	1	0	0	1
	1	0	1	1
un	1	1	0	X
	1	1	1	X

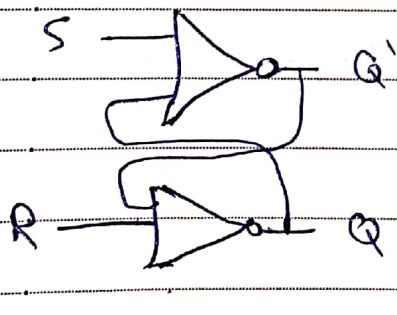


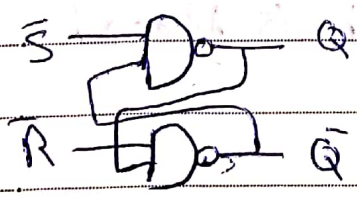
$Q^+ = S + R'Q$

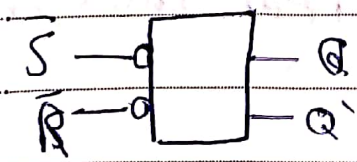
$Q^+ = \Sigma(1,4,5) + \Sigma d(6,7)$

Charecterstic equation

- SR Latch is constructed from two cross coupled nand/noR gates

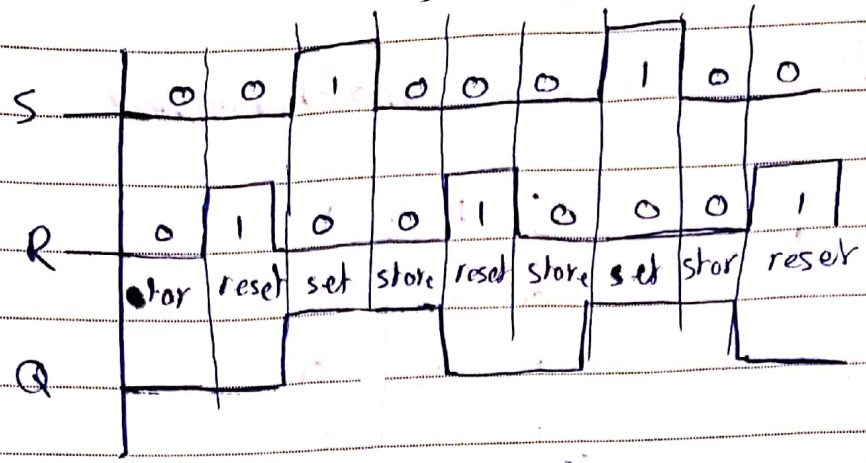






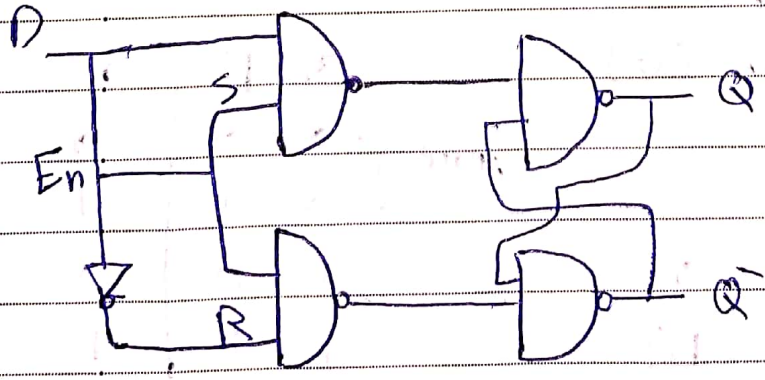
	$\bar{S}$	$\bar{R}$	Q	Q^+
SR || 0	0	0	X
un	0	0	1	X
SR	0	1	0	1
10 set	0	1	1	1
SR	1	0	0	0
01 reset	1	0	1	0
SR	1	1	0	0
00 state	1	1	1	1

Ex: (timing diagram) SR Latch



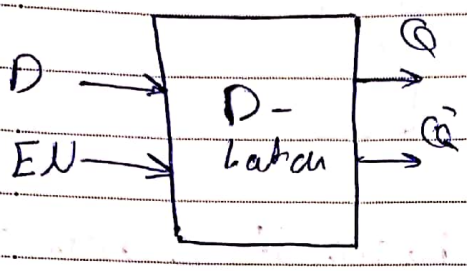
- The operation of the basic SR-latch can be modified by providing an additional input signal that controls when the latch operates

	En	S	R	Q^+
	0	X	X	Q
store ←	1	0	0	Q^+
reset ←	1	0	1	0
set ←	1	1	0	1
X ←	1	1	1	X



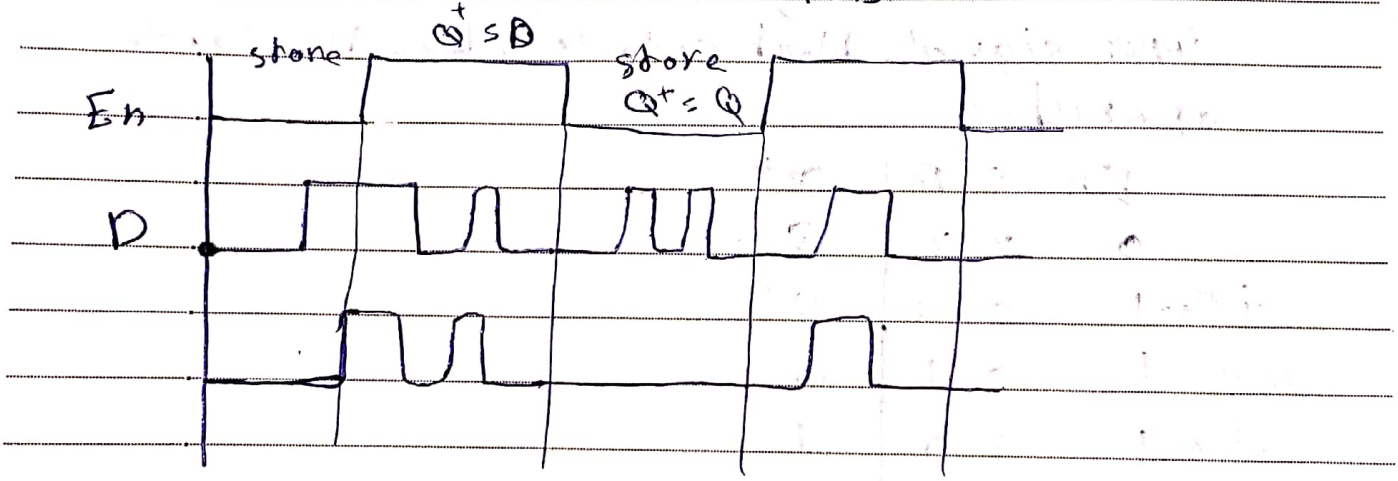
[2] D-Latch (Data Latch)

- To solve the condition of undetermined state



En	D	Q^+
0	x	Q
1	0	0
1	1	1

$Q^+ = D$



[3] Gated D-Latch (Transparent Latch)

- has two inputs
  1. Gate input (G)
  2. Data input

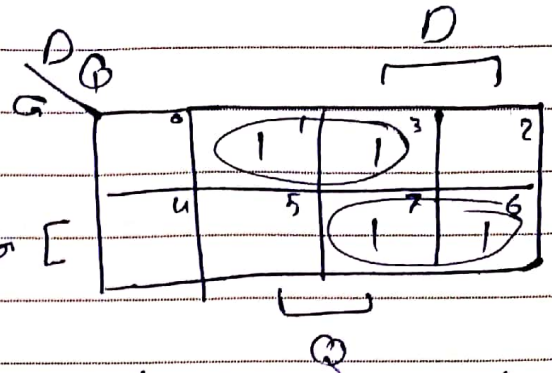
G	D	Q^+
0	0	0
0	1	1
1	0	0
1	1	1

store

$G = 0$ $(Q^+ = Q)$

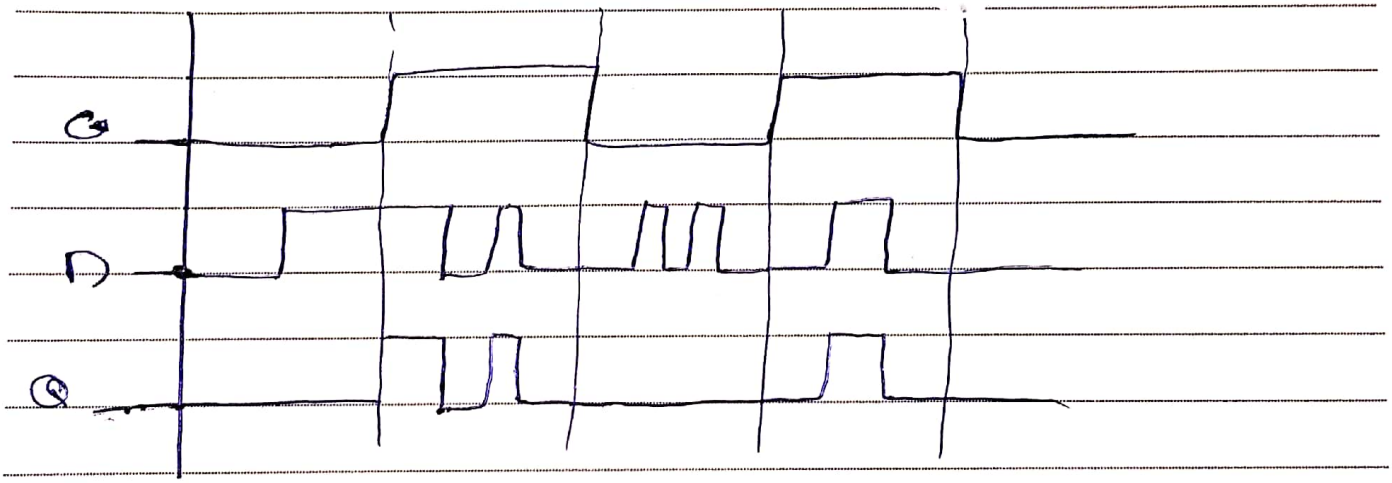
$G = 1$ $(Q^+ = D)$

G	D	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



$Q^+ = \Sigma(1,3,6,7)$

$= G'Q + GD$



22/8/

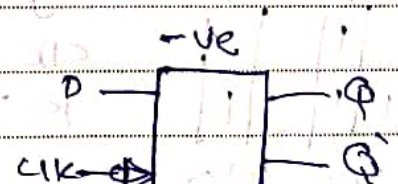
Flip Flops

- Edge sensitive memory elements
- Responce may be on either ① Rising edge ↑
  ② Falling edge ↓

- Operations Performed by ① Set ($Q^+=1$) ③ Store ($Q^+=Q$)
  ② Reset ($Q^+=0$) ④ Toggle ($Q^+=\bar{Q}$)

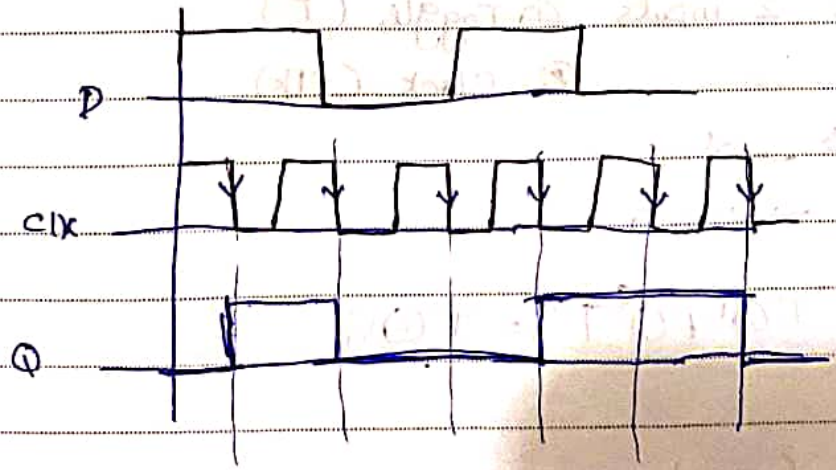
* Types of Flip Flops

1 - D - Flip Flop Has two inputs ① Data (D)
- The D-Flip Flop is edge triggred ② clk denoted by arrow
- positive edge triggred (responce rising edge)
- Negative " " (" falling ")



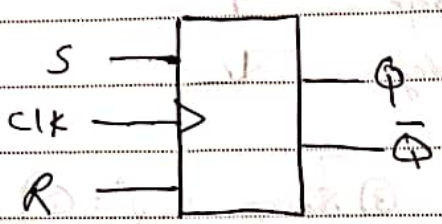
D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

$Q^+ = D$



Types of F.Fs

② SR - Flip Flop - Has 3 inputs ① Set (S - i/p)
② Reset (R - i/p)
③ Clock (Clk)



S	R	Q	Q^+	
0	0	0	0	$Q^+ = Q$
0	0	1	1	
0	1	0	0	$Q^+ = 0$
0	1	1	0	
1	0	0	1	$Q^+ = 1$
1	0	1	1	
1	1	0	x	undet.
1	1	1	x	

SR / Q	0	1
00		1
01	0	0
11	x	x
10	1	1

$Q^+ = S + R'Q$

Has 3 modes
① Set ($S=1, R=0$)
② Reset ($S=0, R=1$)
③ Store ($S=0, R=0$)

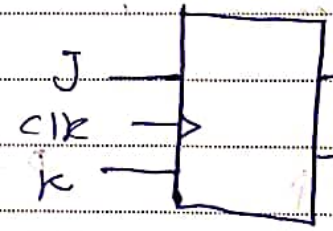
③ J-K-F.Fs - Has 3 inputs ① J → Set
② K → Reset
③ Clk

has 4 modes
1 - Set JK = 10 → $Q^+ = 1$
2 - Reset JK = 01 → $Q^+ = 0$
3 - Store JK = 00 → $Q^+ = Q$
4 - Toggle JK = 11 → $Q^+ = \bar{Q}$

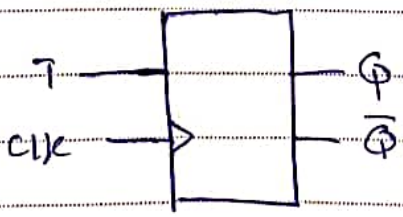
J	K	Q	Q^+	
0	0	0	0	$Q^+ = Q$
0	0	1	1	
0	1	0	0	$Q^+ = 0$
0	1	1	0	
1	0	0	1	$Q^+ = 1$
1	0	1	1	
1	1	0	1	$Q^+ = \bar{Q}$
1	1	1	0	

JK / Q	0	1
00	0	1
01	0	0
11	1	0
10	1	1

$Q^+ = J\bar{Q} + K'Q$



④ T-F.F. (Toggle F.F) Has 2 inputs ① Toggle (T)
② Clock (Clk)



$T = 0 \rightarrow Q^+ = Q$
$T = 1 \rightarrow Q^+ = \bar{Q}$

$Q^+ = T\bar{Q} + Q\bar{T} = T \oplus Q$

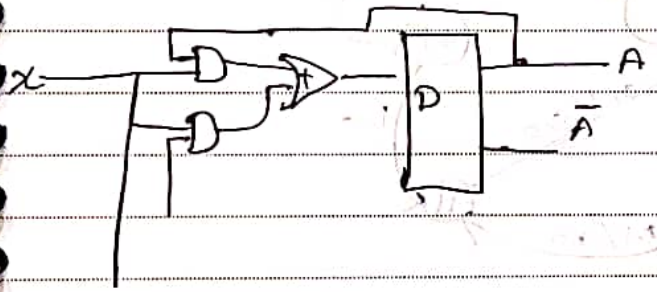
Has 2 modes
① store $Q^+ = Q$ ($T=0$)
② Toggle $Q^+ = \bar{Q}$ ($T=1$)

Analysis of sequential circuits

- Obtaining a table or diagram for the time sequence of i/ps & o/ps

- The behaviour of the seq. circuit can be described
  ① algebraically (transition equations)
  ② State table ③ State diagram.

Ex Analyze the sequantial circuit



① CCS Eqn (D) → $Q^+ = D$

$Q_A^+ = D_A \Leftrightarrow A(t+1) = D_A$

$B(t+1) = B_A$

② $D_A = Ax + Bx$

$A(t+1) = A(t)x + B(t)x$ ——— ①

③ $D_B = A'x$

$B(t+1) = A'(t)x$ ——— ②

④ $y(t) = [A(t) + B(t)] \cdot x'(t)$

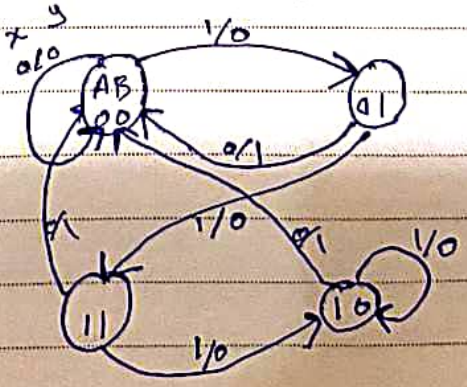
$= A(t)x'(t) + B(t)x'(t)$

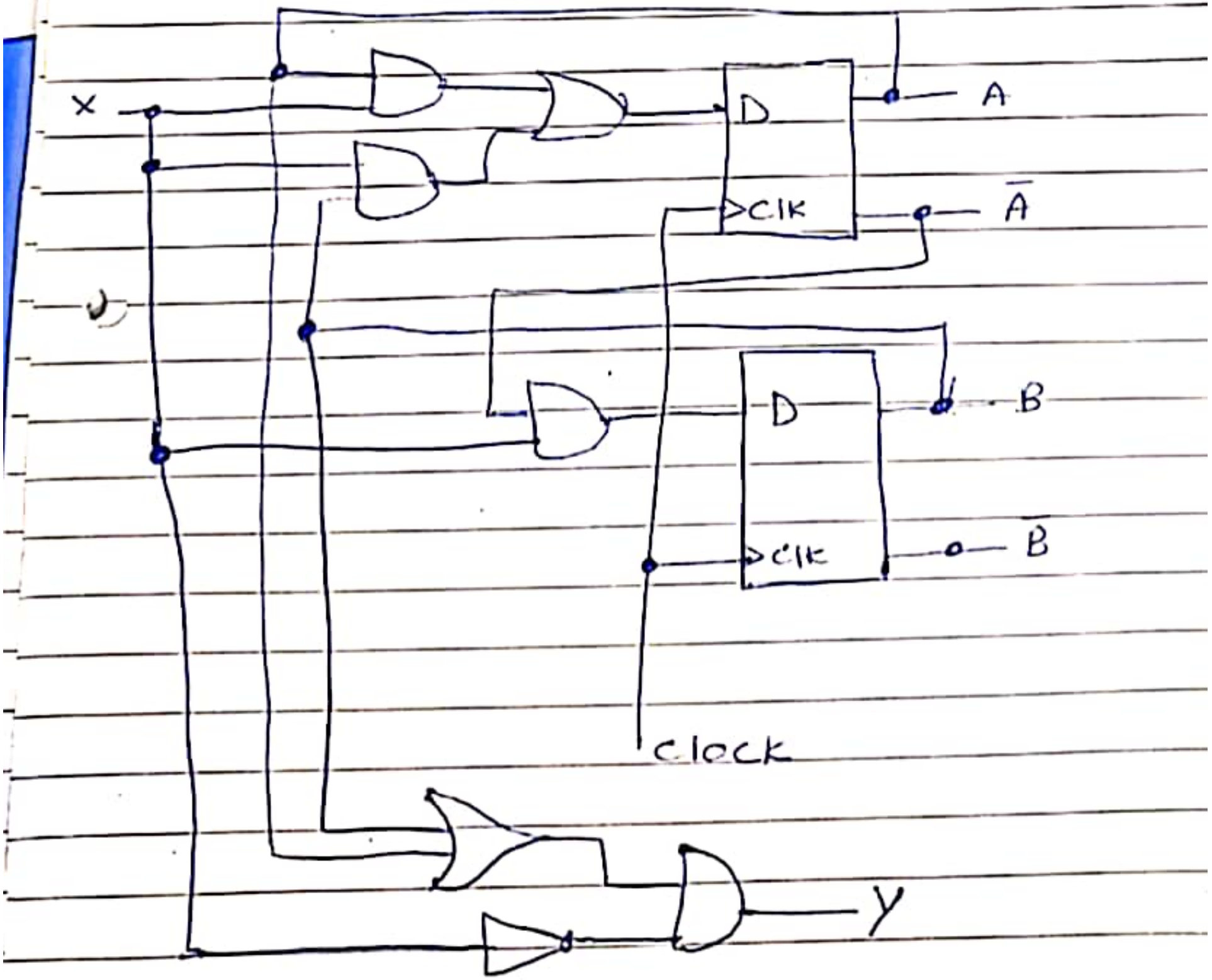
② State table

Present		i/p	Next		o/p
A(t)	B(t)	x	A(t+1)	B(t+1)	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

machines ① mealy y α x [o/p dep on i/p]
② moor

——— state diagram




X
D
A
>CIK
$\bar{A}$
D
B
>CIK
$\bar{B}$
clock
Y

EX2 Derive the state table & State diagram

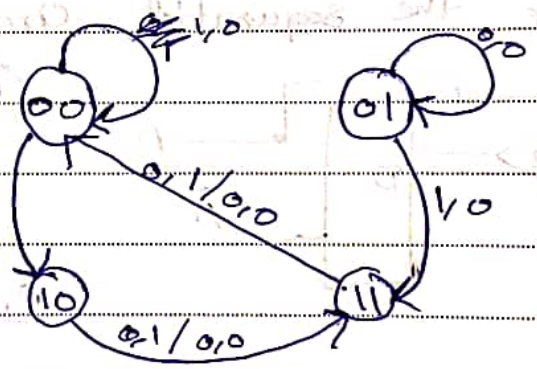
D- F.F → $Q^+ = D$

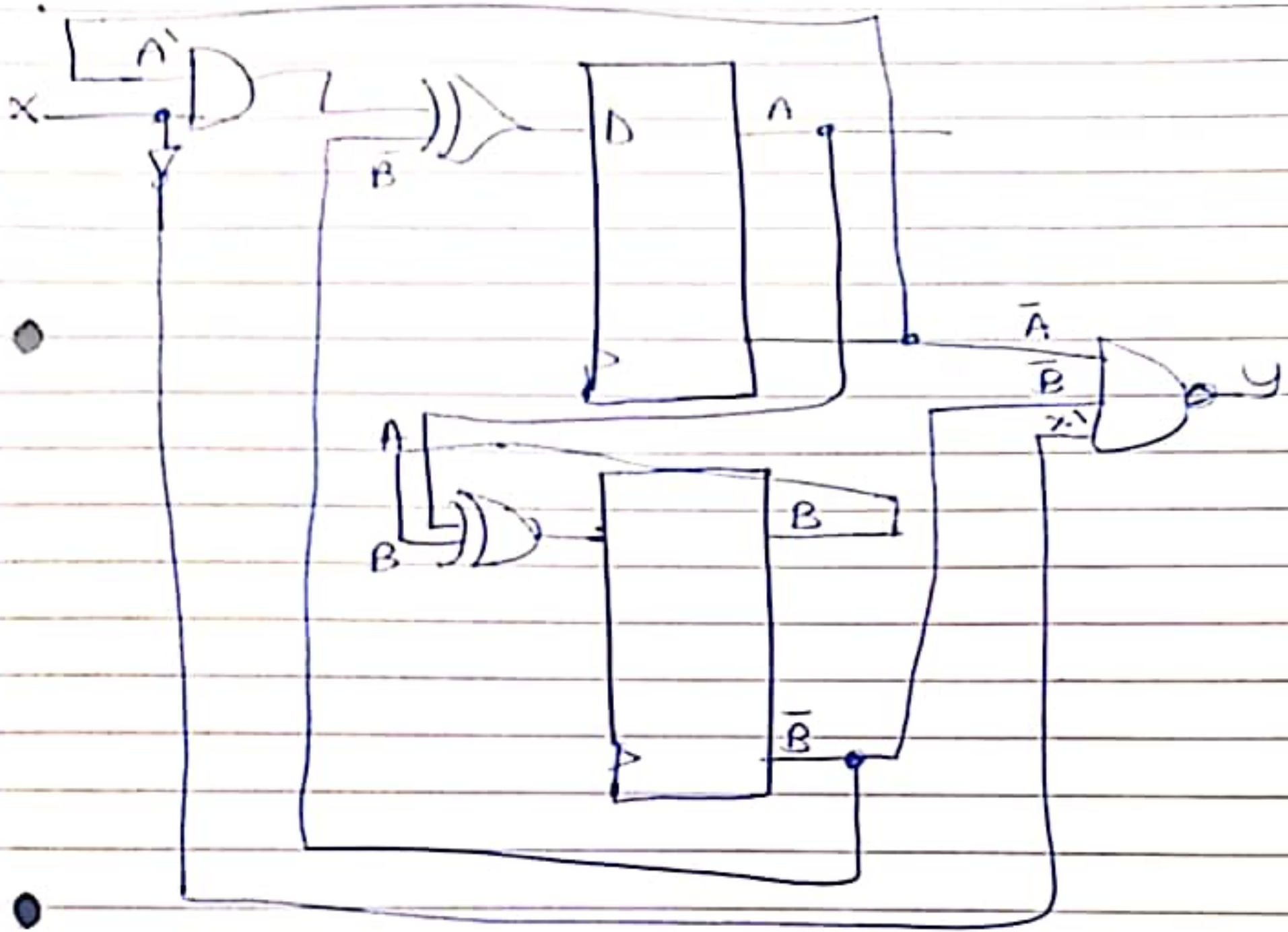
$A^+ = D_A = A'x \oplus B'$

$B^+ = D_B = A \oplus B$

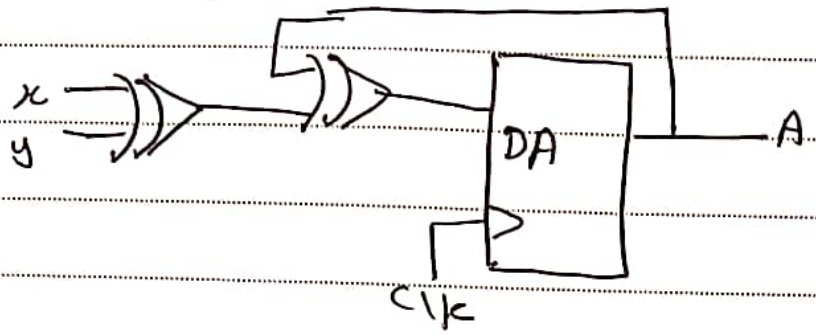
$y = A'B'x'$

A	B	x	A^+	B^+	y
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	1	1	0
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	0	0	0
1	1	1	0	0	0




X
Y
B
D
A
$\bar{A}$
$\bar{B}$
Y
B
B
$\bar{B}$

Ex3 Analyze

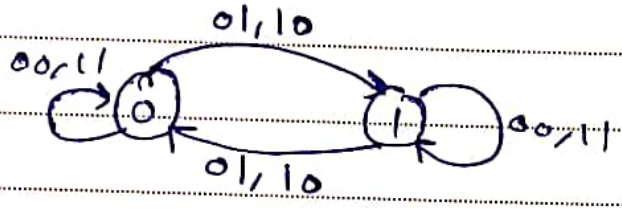


$Q^+ = D$

$A^+ = DA$

$A^+ = x \oplus y \oplus A$

A	x	y	A+
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



Ex: Analysis with JK - F.F

① ccs equation

$JK \rightarrow Q^+ = JQ' + K'Q$

$A^+ = J_A A' + K_A' A$, $B^+ = J_B B' + K_B' B'$

$J_A = B$, $K_A = x'B$ $J_B = \bar{x}$, $K_B = A \oplus x$

Sub. J_A, K_A in A^+ Sub J_B, K_B in B^+

$\rightarrow A^+ = BA' + (x + B')A$ $\rightarrow B^+ = x'B' + (A + x')(A' + x)B$

$= A'B + Ax + AB'$ $= B'x' + ABx + A'Bx'$

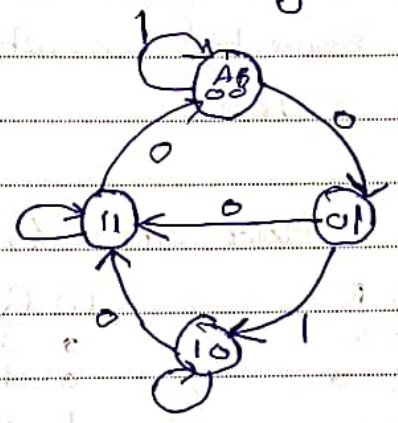
② State table

A	B	x	A^+	B^+
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

states → 4

$AKB = 2^2 =$

③ State diagram



Ex T - F.F

① CCS → $Q^+ = T \oplus Q$ $B^+ = T_B \oplus B$

$A^+ = T_A \oplus A$ $T_B = x$

$T_A = Bx$ $B^+ = x \oplus B$

$\leftarrow A^+ = Bx \oplus A$ $y = AB$

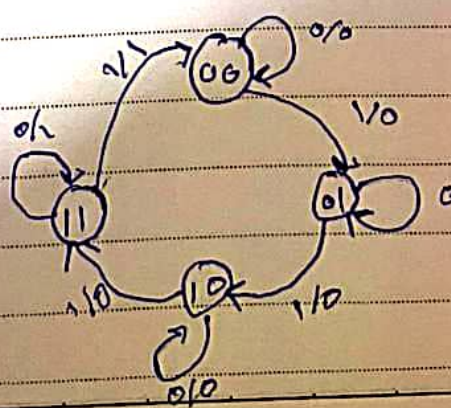
Bx

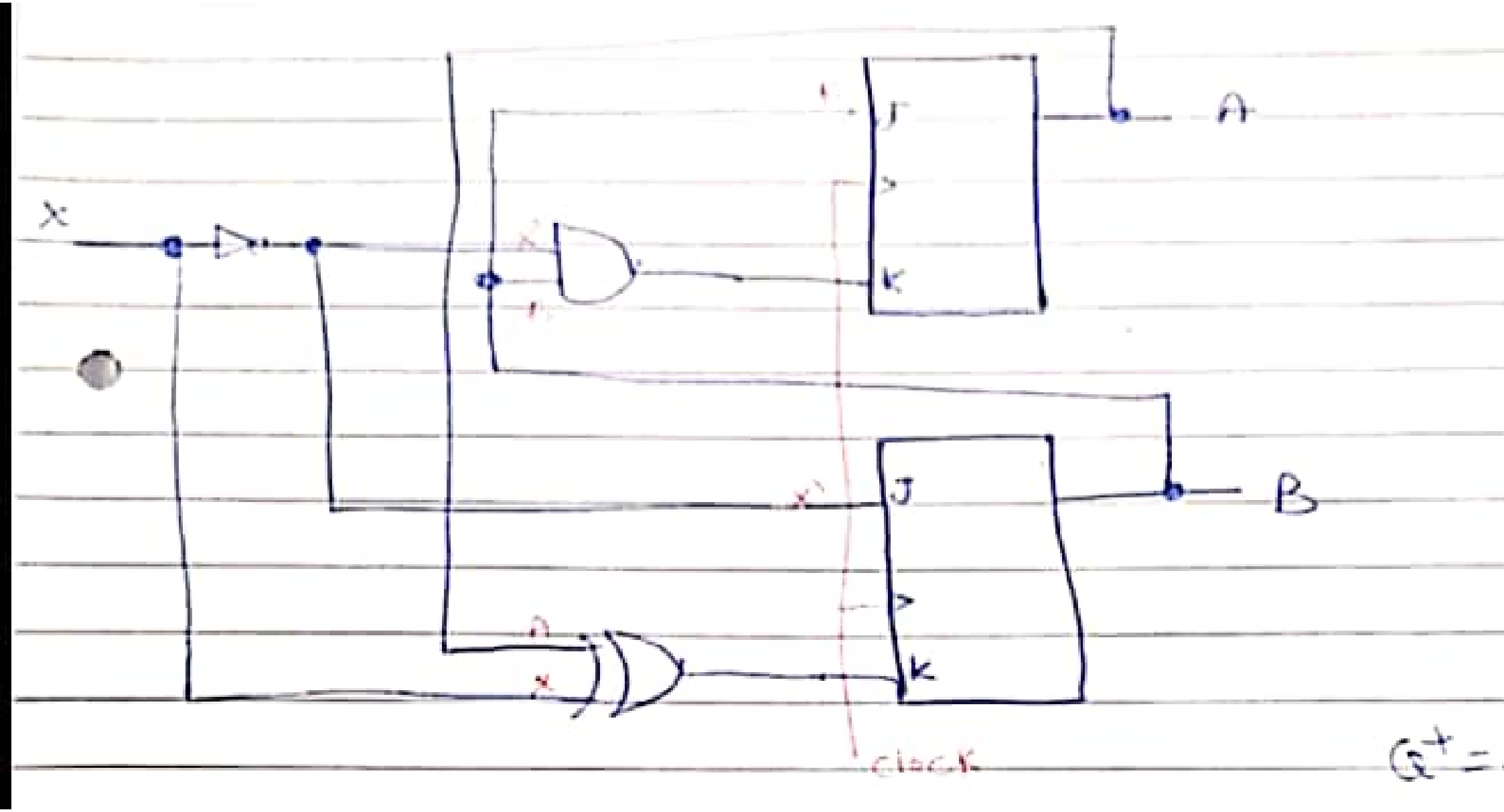
① = A مكتوبة

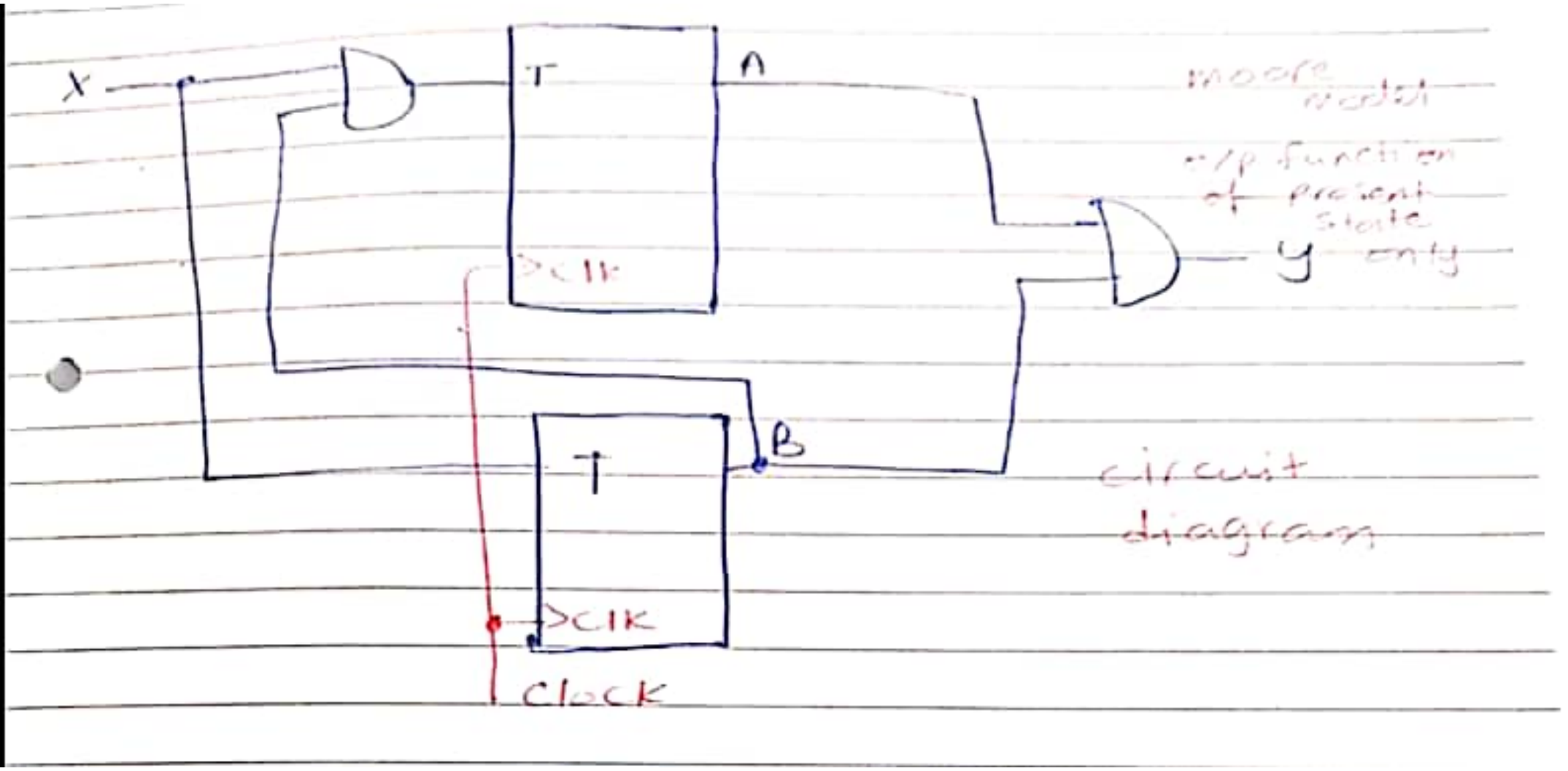
② State table

A	B	x	A^+	B^+	y
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

③ State diagram



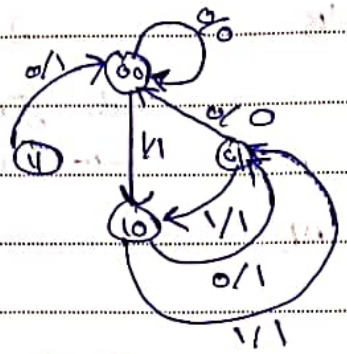

x
J
K
A
J
K
B
clock


X
T
A
>CLK
B
T
>CLK
Clock
moore model
o/p function of present state only
circuit diagram

Ex SR

① CCS equation ⇒ Q^+

A	B	x	A^+	B^+	y
0	0	0	0	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	1	0	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	0	0	0
1	1	1	0	0	1



$S_A = A'x$ $S_B = (A'+B)'$

$R_A = (A'x)'$ $R_B = A'+B$

$S_R \rightarrow Q^+ = S + R'Q$

$A^+ = S_A + R_A'Q$

$= A'x + A'x\,A$

$A^+ = A'x$

$B^+ = S_B + R_B'B$

$= AB' + AB'B$

$B^+ = AB'$

$y = x + AB'$

mealy model

* Design of sequential circuits

* Excitation Tables

- Excitation tables are derived from characteristic eqs

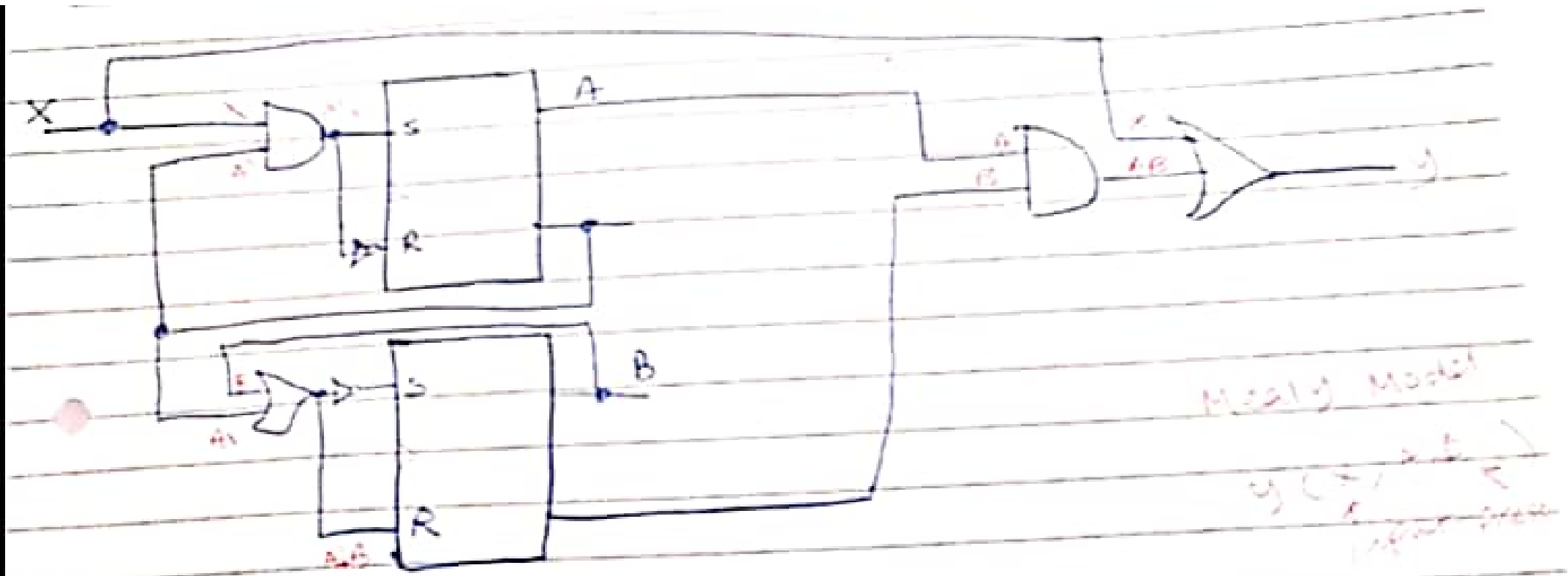
Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

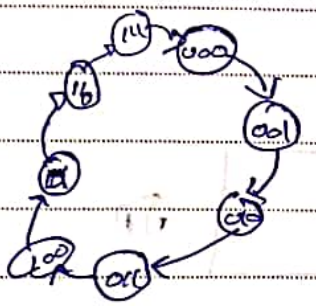
ALADIB


X
S
R
A
B
S
R

Ex Design a 3 bit up Counter using T-F-F 000 → 111

A	B	C	A^+	B^+	C^+	T_A	T_B	T_C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

T →	Q	Q^+	T
	0	0	0
	0	1	1
	1	0	1
	1	1	0

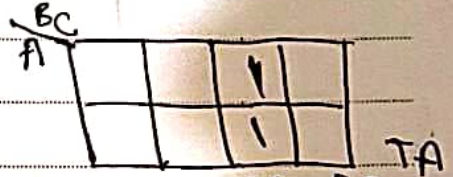


$T_C = 1$

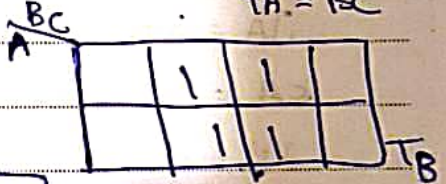
$T_A = \Sigma(3,7) \rightarrow BC$

$T_B = \Sigma(1,3,5,7) \rightarrow C$

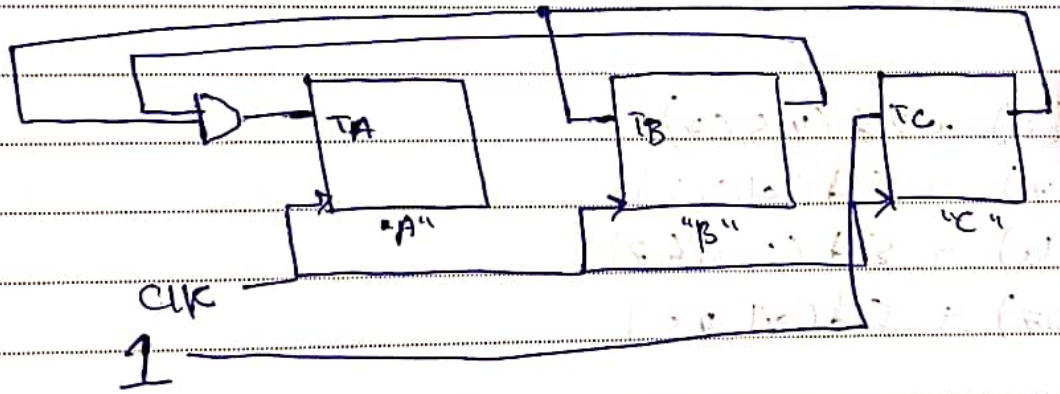
Kmap



$T_A = BC$



$T_B = C$



EX Design a 3-bit up counter using JK

A	B	C	A^+	B^+	C^+	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Q	Q^+	JK
0	0	0 X
0	1	1 X
1	0	X 1
1	1	X 0

$J_A = BC$, $J_C, K_C = 1$

$K_A = BC$

$J_B = C$

$K_B = C$

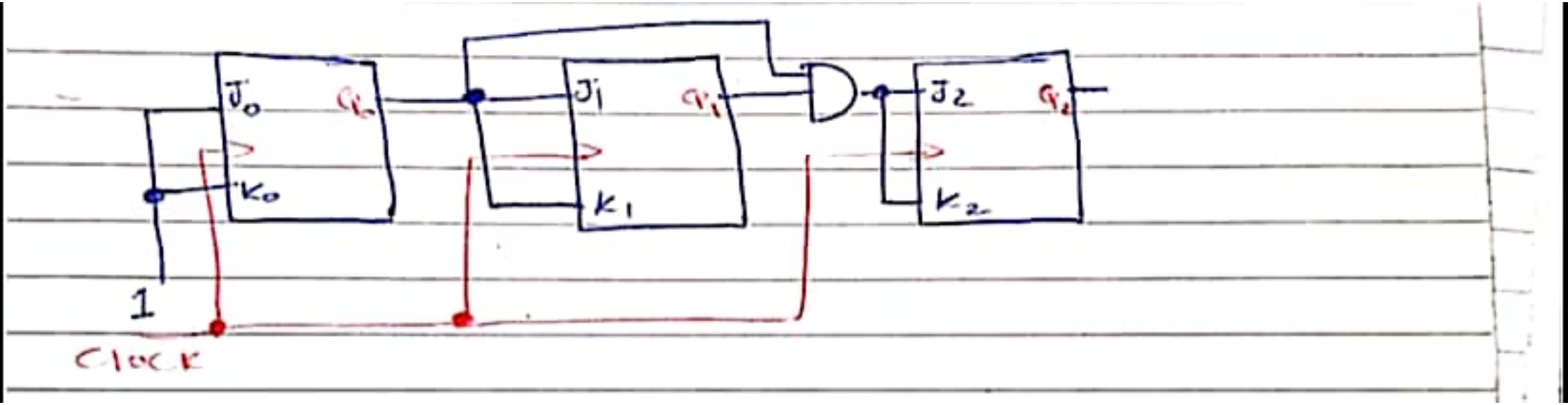
J_A

$J_A = \Sigma(3) + \Sigma d(4,5,6,7)$

$K_A = \Sigma(7) + \Sigma d(0,1,2,3)$

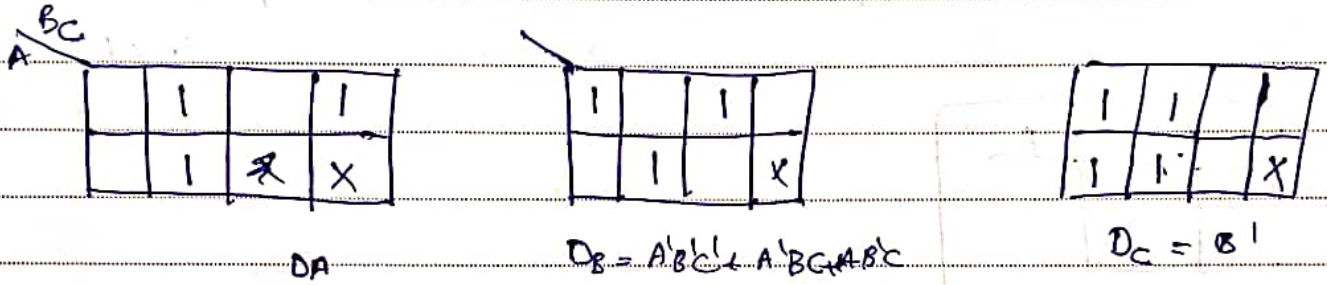
$J_B = \Sigma(1,5) + \Sigma d(2,3,6,7)$

$K_B = \Sigma(3,7) + \Sigma d(0,1,4,5)$


J_0
Q_0
K_0
J_1
Q_1
K_1
J_2
Q_2
K_2
1
Clock

EX Design a counter that goes through the sequence 0,3,2,4,1,5,7 & repeat (use D)

A	B	C	A^+	B^+	C^+	D_A	D_B	D_C
0	0	0	1	1	1	1	1	1
0	1	1	1	0	0	1	0	0
1	0	0	0	0	1	0	0	1
1	1	1	x	x	x	x	x	x
			0	0	0	0	0	0

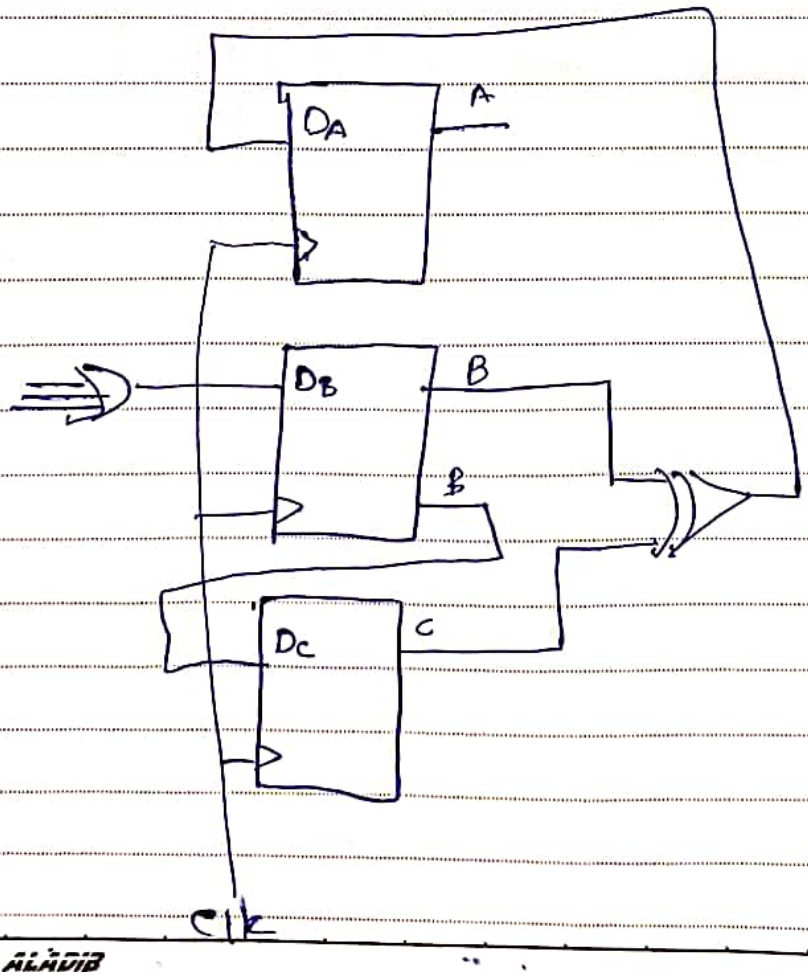


D_A

$D_B = A'B'C' + A'BC + AB'C$

$D_C = B'$

$D_A = B'C + C'B = C \oplus B$



Ex Design a 3 bit down odd counter use T-F-F

↑ 7, 5, 3, 1 → 7, x, 5, x, 3, x, 1, x

A	B	C	A^+	B^+	C^+	T_A	T_B	T_C
0	0	0	x	x	x	x	x	x
0	0	1	0	1	1	0	1	0
0	1	0	x	x	x	x	x	x
0	1	1	1	0	1	1	1	0
1	0	0	x	x	x	x	x	x
1	0	1	1	1	1	0	1	0
1	1	0	x	x	x	x	x	x
1	1	1	0	0	1	1	1	0

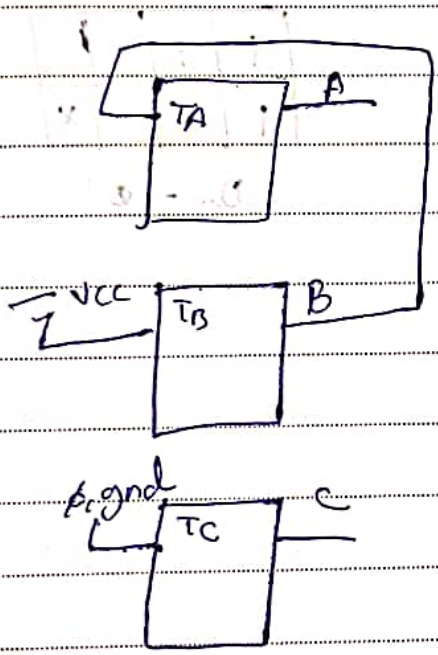
$T_C = 0$

$T_B = 1$

$T_A = \Sigma(3,7) + \Sigma d(0,2,4,6)$

x		1	x
x		1	x

$T_A = B$



EX Design a circuit for the following state table

use JK

Present A	Present B	input x	next A^+	next B^+	output y	J_A	K_A	J_B	K_B
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	1	0	X	1	X
0	1	0	0	1	0	0	X	X	0
0	1	1	1	0	1	1	X	X	1
1	0	0	1	0	0	X	0	0	X
1	0	1	1	1	1	X	0	1	X
1	1	0	1	1	0	X	0	X	0
1	1	1	0	0	1	X	1	X	1

QQ^+	JK
0 0	0 X
0 1	1 X
1 0	X 1
1 1	X 0

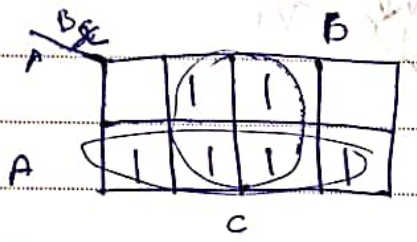
$y = \sum(1,3,4,5,6,7)$

$J_A = \sum(3) + \sum d(4,5,6,7)$

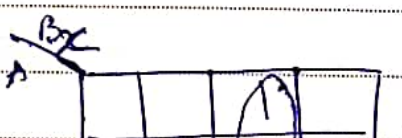
$K_A = \sum(2) + \sum d(0,4,2,3)$

$J_B = \sum(1,5) + \sum d(2,3,6,7)$

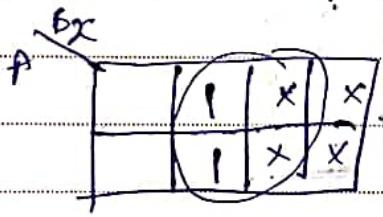
$K_B = \sum(3,7) + \sum d(0,1,4,5)$



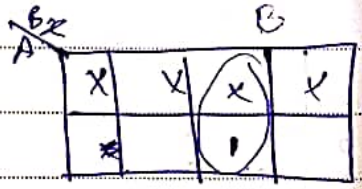
$y = A + x$



$J_A = Bx$

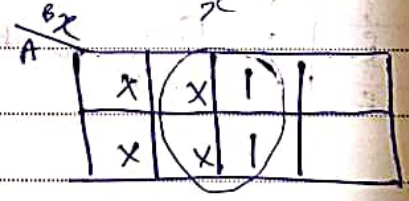


$J_B = x$

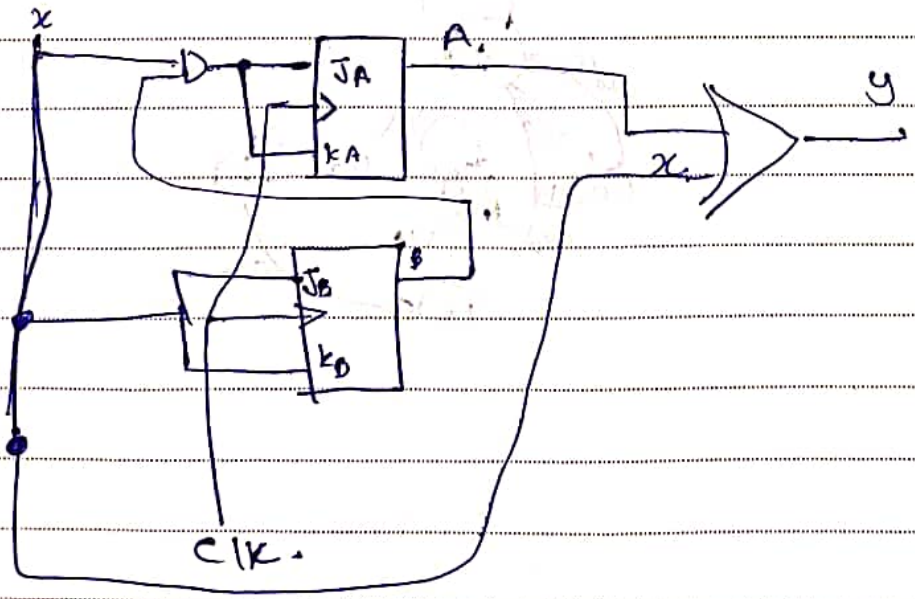


$= Bx$

K_A



$K_B = x$



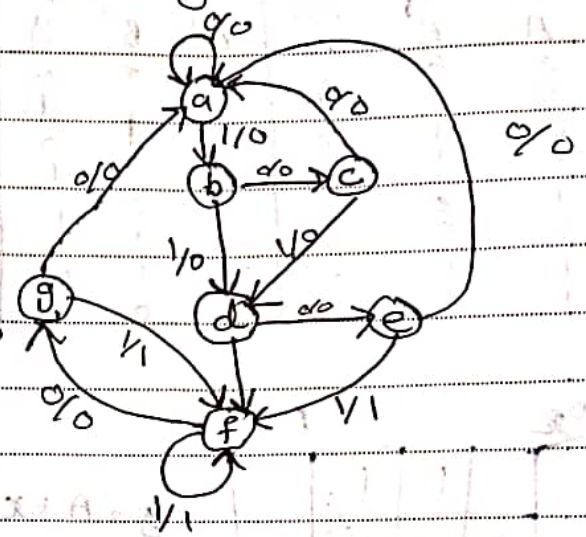
ALADDIN
صنع الحضارة

* State Reduction

EX Reduce the Following State diagram

Present	next x=0	next x=1	output x=0	output x=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
→ g	a	f	0	1

Cancel



∴ g = e

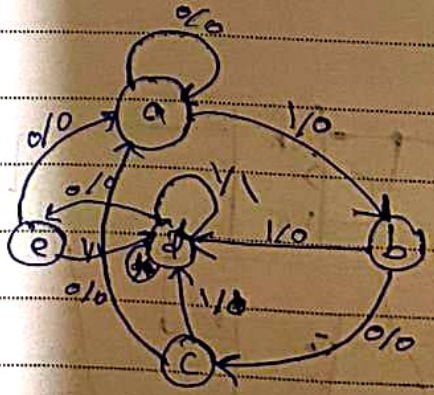
replace g → e

a	a	b	0 0
b	c	d	0 0
c	a	d	0 0
d	e	f	0 1
e	a	f	0 0
f	e	f	0 0

=

P → d

→ a	a	b	0 0
b	c	d	0 0
c	a	d	0 0
d	e	d	0 1
e	a	d	0 1



* Sequence detector detector

- A sequential circuit that takes an input string of bits and generates an output "1" when ever the target sequence has been detected.

EX Design a circuit to detect 110 from given input stream

x 0 1 1 0 1 1 0 1 0 1 1 0 .

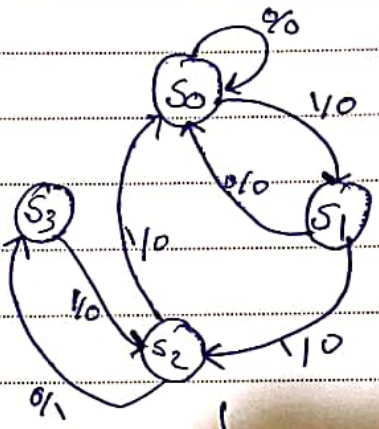
0 0 0 1 0 0 1 0 0 0 0 1

$S_0 = \phi$ (Reset)

$S_1 = 1$

$S_2 = 11$

$S_3 = 110$



↳ State table

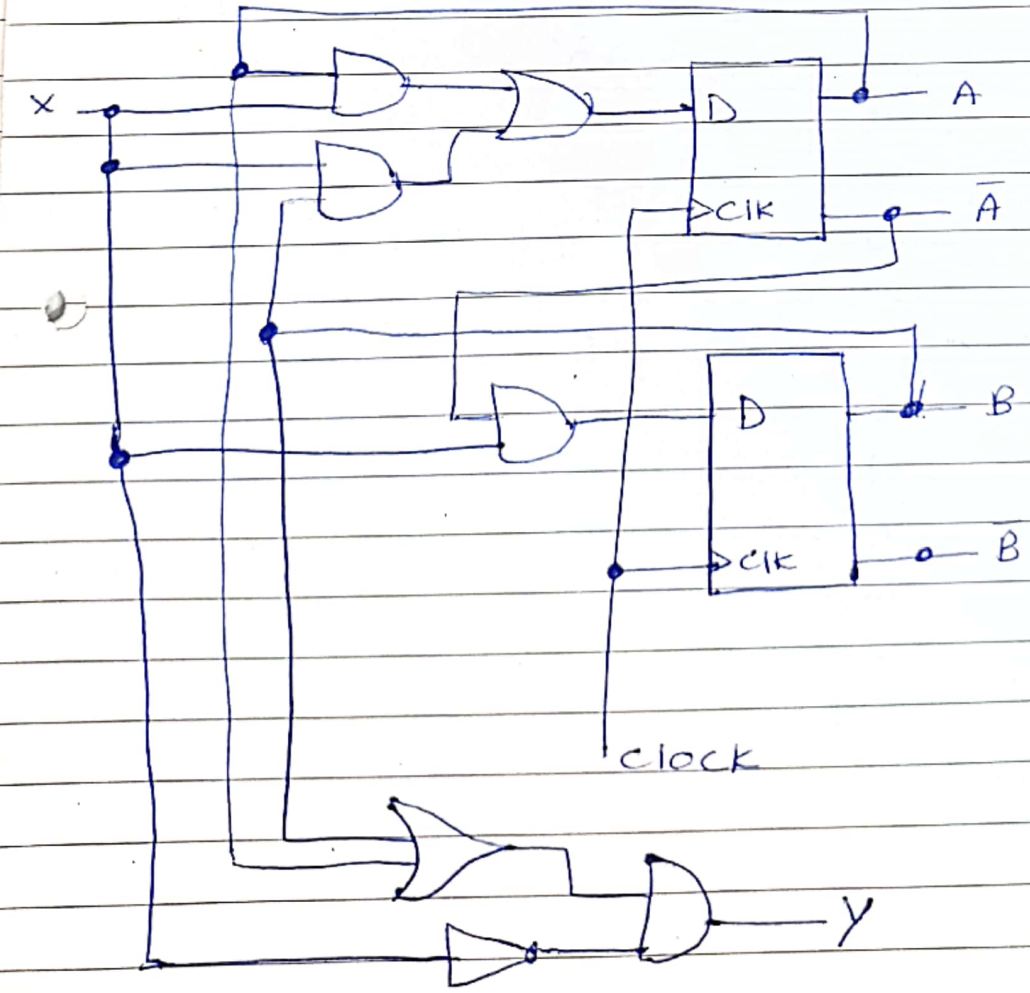
excitation table

Analysis of clocked sequential Circuits:

- Obtaining a table or diagram for the time sequence of inputs, outputs & internal states.

- The Behaviour of the circuit can be described algebraically by means of state equations (transition equation) ⇒ specifies the next state as a function of the present state & inputs.

ex. Fig 5.15 P210



$A(t+1) = A(t)\,x(t) + B(t)\,x(t)$

$B(t+1) = A'(t)\,x(t)$

$y(t) = [A(t) + B(t)]\,x'(t)$

State Equations

$Q^+ = D$

Mealy model

1) output y is a function of both input x & the present state of A and B

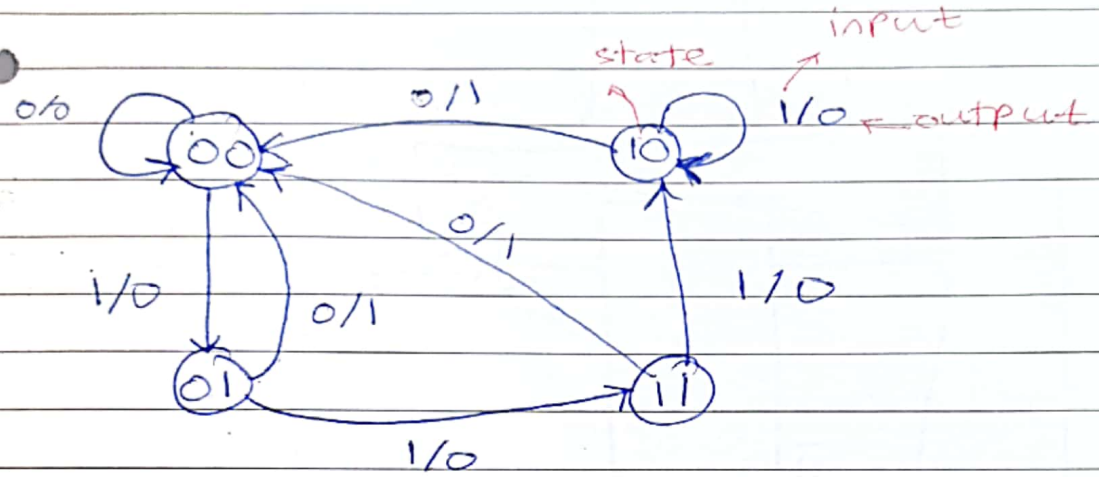
State table.

Present state A	Present state B	input x	Next state A	Next state B	Output Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

OR

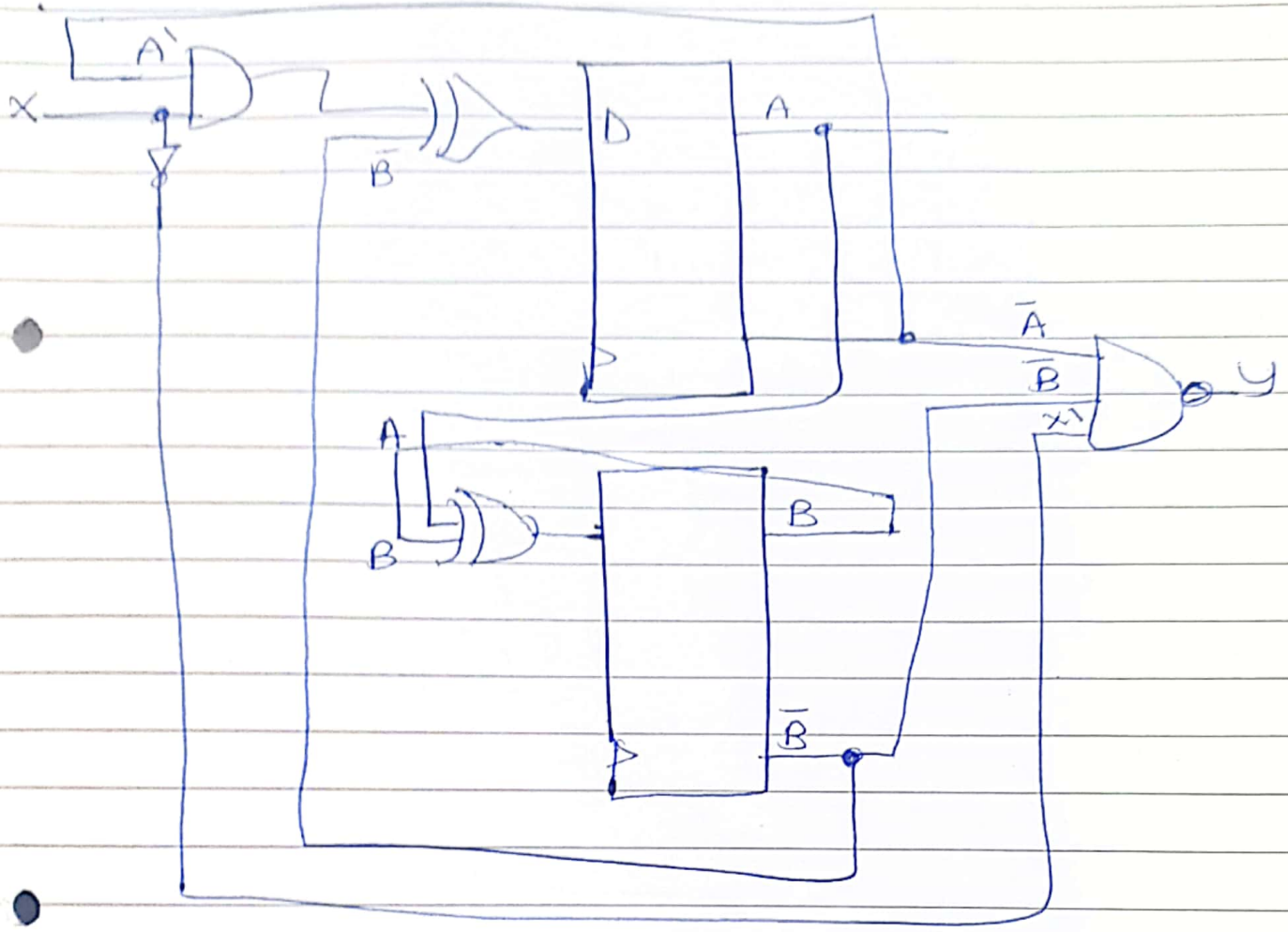
CamScanner

Present state A B	Next state x=0 A B	Next state x=1 A B	output x=0 Y	output x=1 Y
0 0	0 0	0 1	0	0
0 1	0 0	1 1	1	0
1 0	0 0	1 0	1	0
1 1	0 0	1 0	1	0



state diagram

Derive the state table & the State diagram



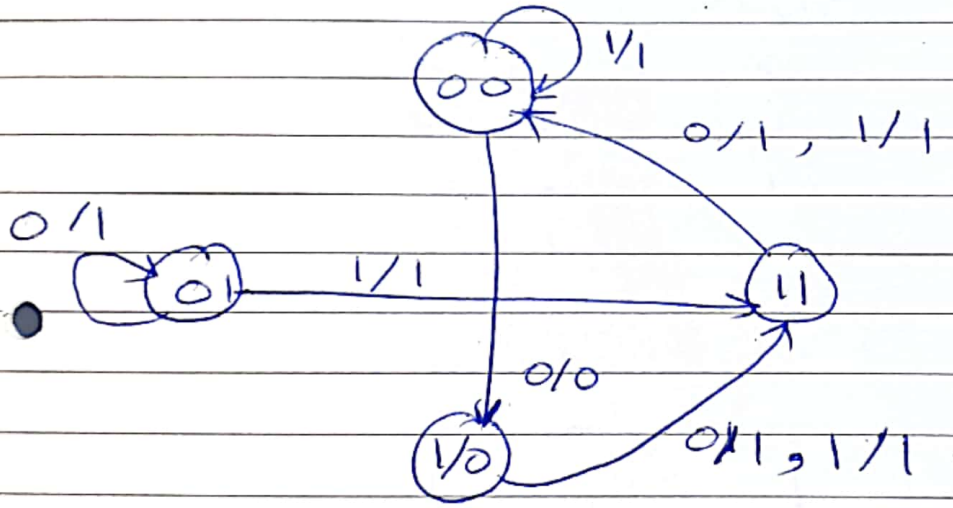
$A(t+1) = A'(t)\, x \oplus B'(t)$

$B(t+1) = A(t) \oplus B(t)$

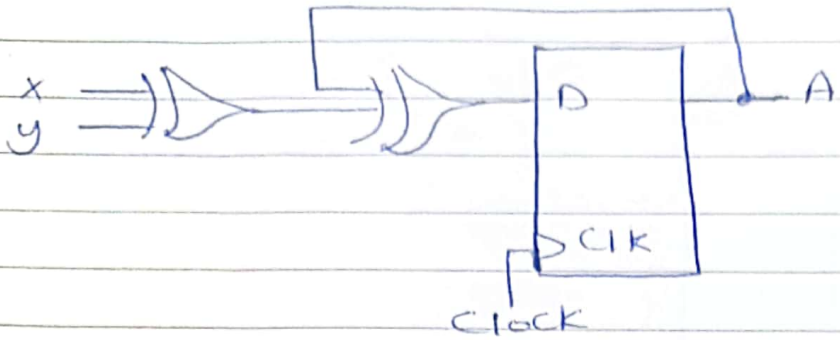
$y(t) = (A'(t) \cdot B'(t) \cdot x')' = A(t) + B(t) + x$

CamScanner

Present state		Input	Next state		o/p
A	B	x	A	B	y
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1



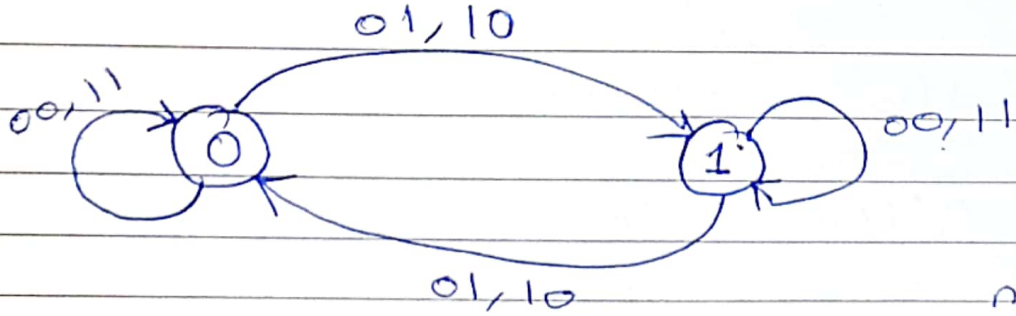
ex.



$A(t+1) = A(t) \oplus x \oplus y$

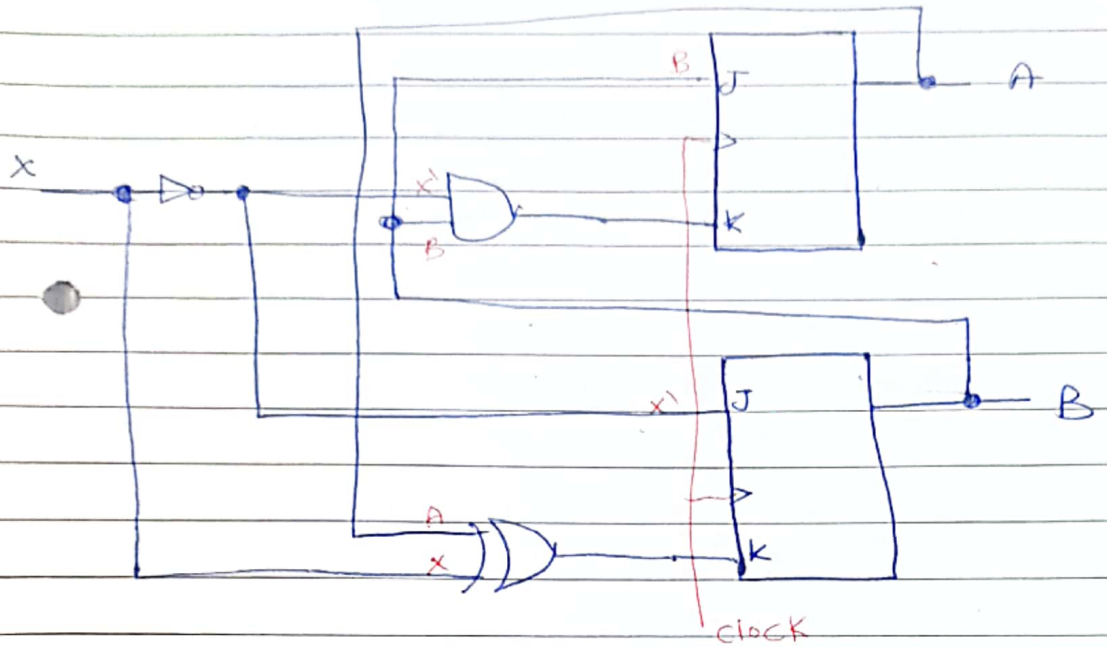
Moore Model output is a function of Present state only

Present state	Inputs x	Inputs y	Next state
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



no need for slash because there is no output (y) from the comb. circuit.

Ex. Analysis with JK Flip Flops



$Q^+ = JQ' + K'Q$

$$A(t+1) = J_A A' + K_A' A$$
$$B(t+1) = J_B B' + K_B' B$$

$J_A = B$ $\quad K_A = Bx'$

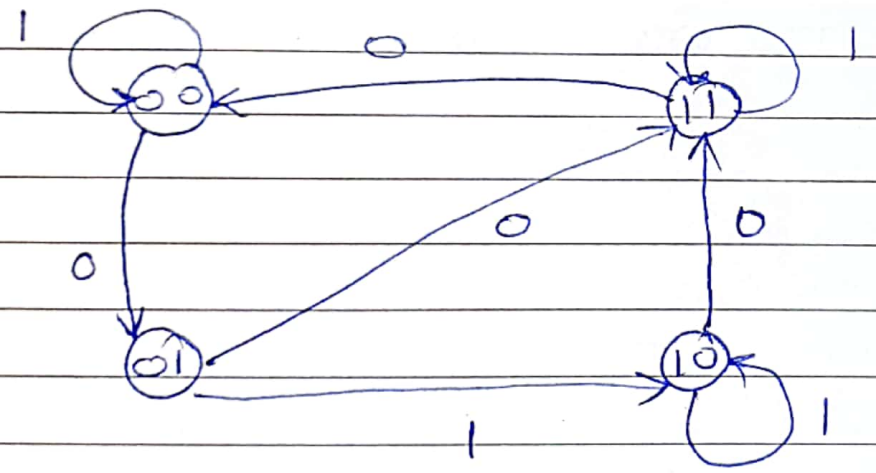
$J_B = x'$ $\quad K_B = A'x + Ax' = A \oplus x$

substitute J_A & K_A in $A(t+1)$

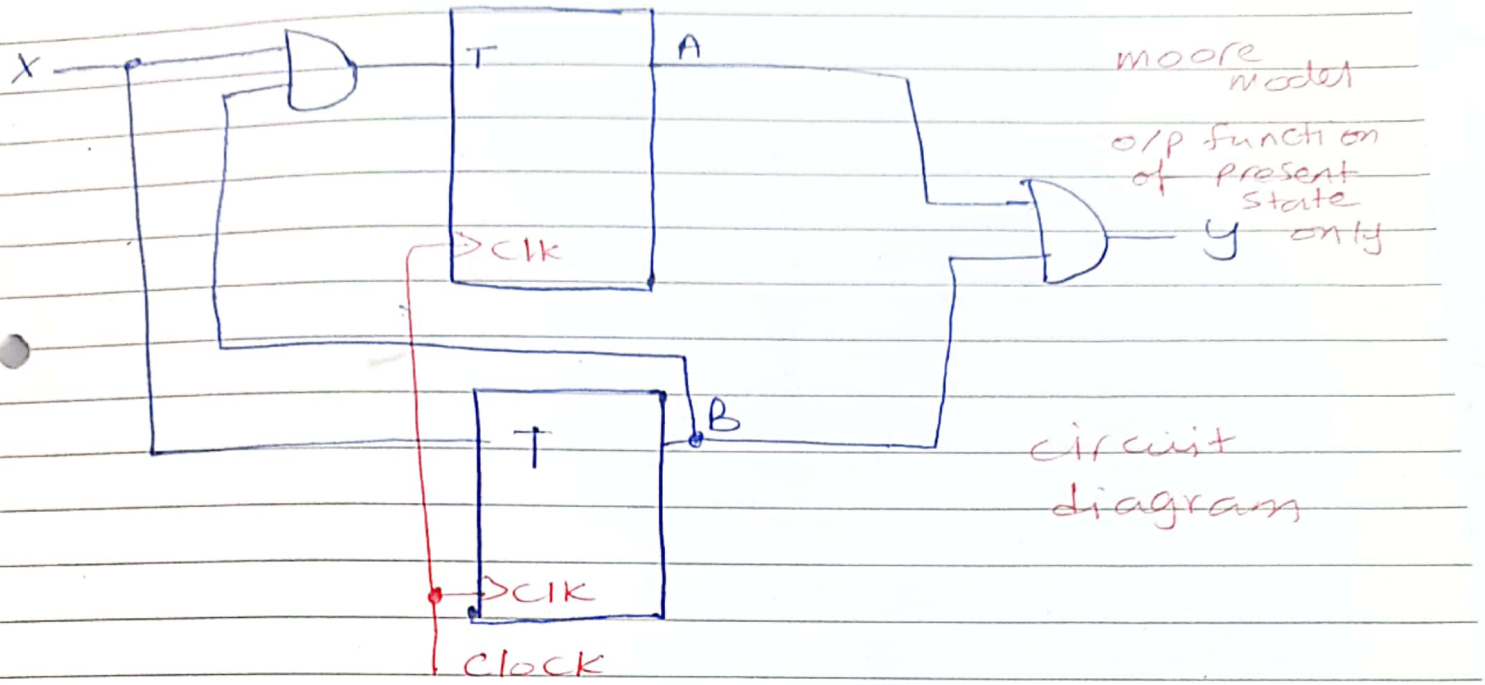
$$\Rightarrow A(t+1) = BA' + (Bx')' A$$
$$= BA' + (B' + x) A$$
$$= A'B + AB' + Ax$$

$$B(t+1) = x'B' + (A \oplus x)' B$$
$$= B'x' + ABx + A'Bx'$$

Present state A	B	Input X	Next state A	B
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1



ex. Analysis with T Flip-Flops



$Q(t+1) = T \oplus Q = T'Q + TQ'$

$T_A = Bx$

$T_B = x$

$y = AB$

$$\begin{aligned} A(t+1) &= T_A \oplus A \\ &= T'A + TA' \\ &= (Bx)'A + BxA' \\ &= (B' + x')A + BxA' \\ &= AB' + Ax' + A'Bx \end{aligned}$$

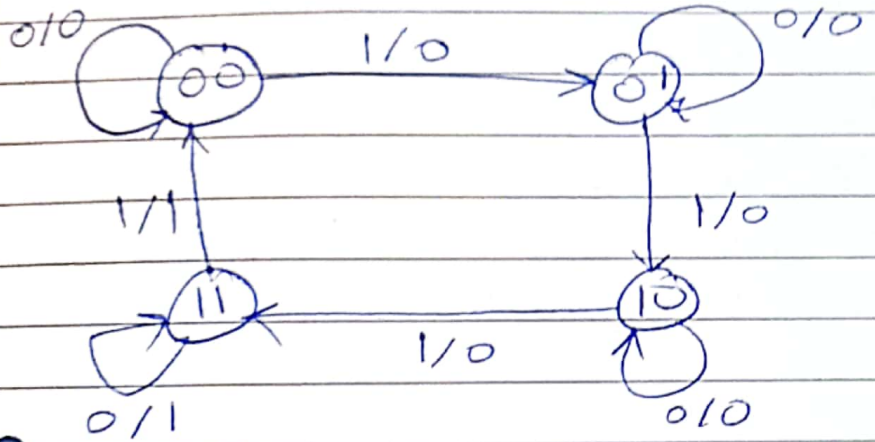
$B(t+1) = x'B + xB'$

state Equations

CamScanner

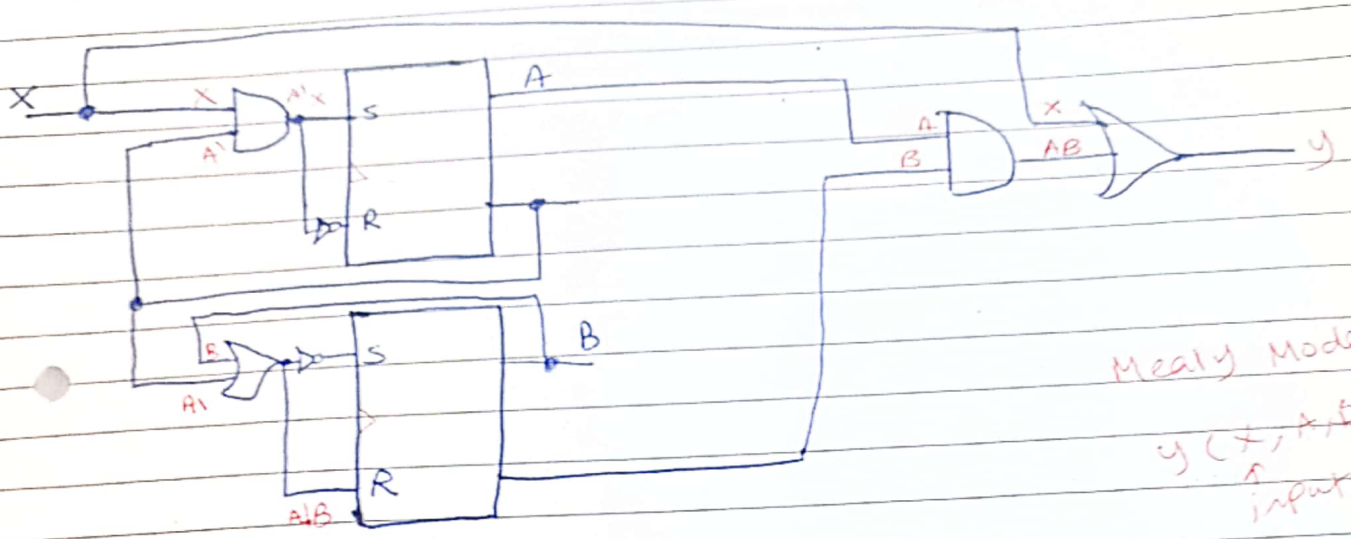
State table

Present state A	Present state B	Input x	Next state A	Next state B	Output y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1



state diagram

ex. SR



Mealy Model

$y(x, A, B)$

↑ input present

$S_A = A'x$, $R_A = (A'x)' = A + x'$

$S_B = (A' + B)' = AB'$, $R_B = A' + B$

$y(t) = x + AB'$

$Q+ = S + R'Q$

$A(t+1) = S_A + (R_A' \cdot A) = A'x + A'xA = A'x$

$B(t+1) = S_B + R_B' \cdot A = AB' + AB'A = AB'$

Present A B	input x	next A B	output y
0 0	0	0 0	0
0 0	1	1 0	1
0 1	0	0 0	0
0 1	1	1 0	1
Φ 0	0	0 1	1
Φ 0	1	0 1	
Φ 1	0	0 0	1
1 1	1	(11) 0 0	1

CamScanner

* Flip Flops:-

- The D-Flip Flop is the most economical & efficent Flip Flop.

- Other types of Flip Flops (JK, T, SR) can be constructed from the D-flip-flop.

- Three operations can be performed with a flip-flop

  ① set to "1".
  ② Reset to "0".
  ③ Complement Q

1. D-flip Flop

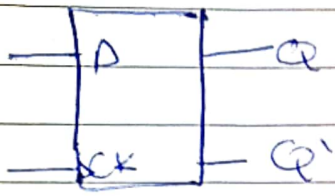
- Has two inputs ① clock (CLK) denoted by a small arrow head, ② Data (D).

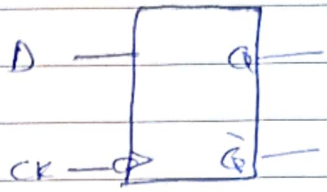
- The output of the D Flip-Flop changes in response to the clock input only, not in response to a change in the D-input.

- The D Flip-Flop is edge-triggered not level sensitive.

- positive (rising) edge triggered "0" → "1"
- negative (falling) edge triggered "1" → "0"

CamScanner





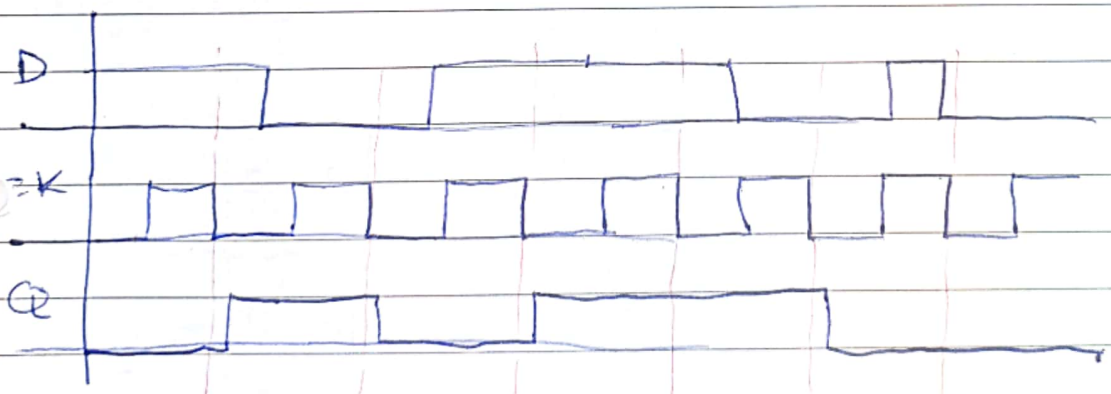
rising edge triggered

Falling edge trigger

D	Q	Q+
0	0	0
0	1	0
1	0	1
1	1	1

$Q^+ = D$ CC eqn.

Falling edge.



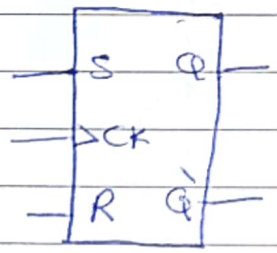
SR Flip-Flop:-

- Has three inputs
  1. Clock (CK)
  2. Set
  3. Reset.

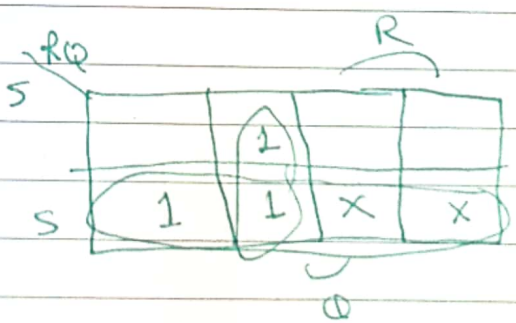
- Similar to SR Latch
  S = "1" sets the FF (Q^+ = "1")
  R = "1" Resets the FF (Q^+ = "0")

- Like the D-FF, the Q only changes in response to an active clock edge

- positive edge-triggered OR negative-edge triggered.



S	R	Q	Q^+	
0	0	0	0	
0	0	1	1	store $Q^+ = Q$
0	1	0	0	
0	1	1	0	Reset $Q^+ = 0$
1	0	0	1	
1	0	1	1	set $Q^+ = 1$
1	1	0	X	not allowed
1	1	1	X	don't care (3)



Char. eqn.

$Q^+ = S + R'Q$

$Q(t+1) = S + R'Q(t)$

JK-FlipFlop:

- Has 3 inputs J, K and clk.
  (J: Set, K: Reset)

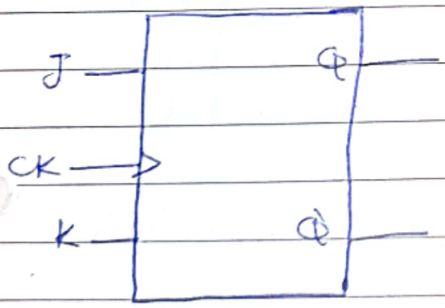
- performs three operations:

① set JK = 10

② Reset JK = 01

③ complement JK = 11

④ store JK = 00



J	K	Q	Q+	
0	0	0	0	store
0	0	1	1	$Q^+ = Q$
0	1	0	0	Reset
0	1	1	0	$Q^+ = 0$
1	0	0	1	Set
1	0	1	1	$Q^+ = 1$
1	1	0	1	complement
1	1	1	0	toggle
				$Q^+ = Q'$

char. equation

$$Q^+ = JQ^* + K'Q$$

$$Q(t+1) = JQ'(t) + K'Q(t)$$

4) T- Flip Flop.

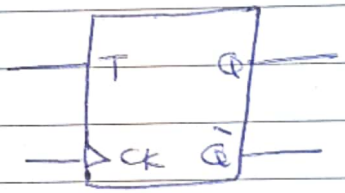
- Toggle (T) Flip-Flop has two inputs CK (Clock) & Toggle (T)

- The T-input controls the state change

T=0, State does not change $Q+ = Q$
T=1, state changes on active trigger
$Q+ = Q'$

- Often used in the design of counters.

T	Q	Q+
0	0	0
0	1	1
1	0	1
1	1	0



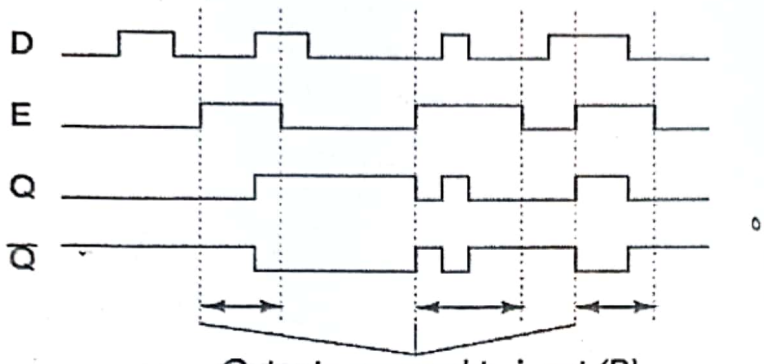
char. equation

$$Q+ = TQ' + T'Q$$

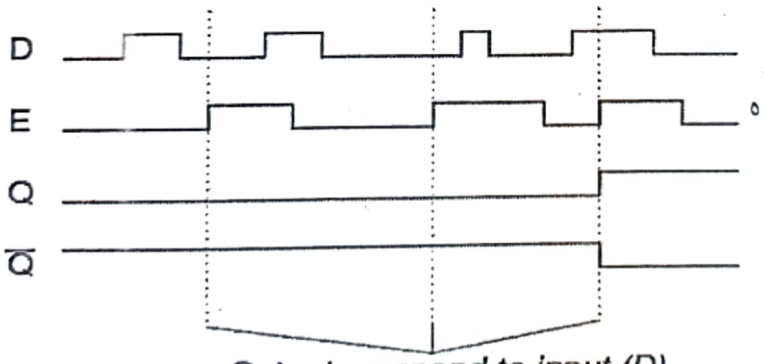
$$= T \oplus Q$$

CamScanner

Regular D-latch response

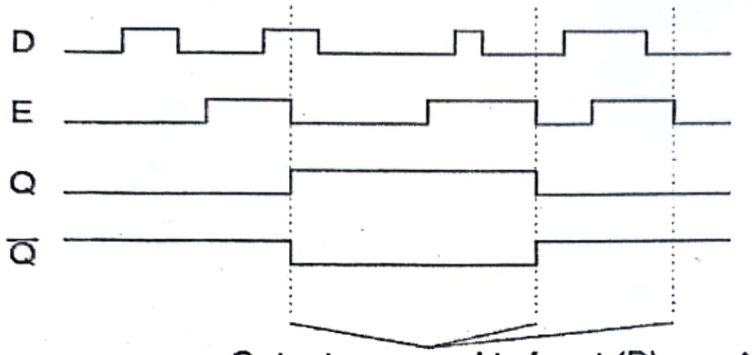


Positive edge-triggered D-latch response



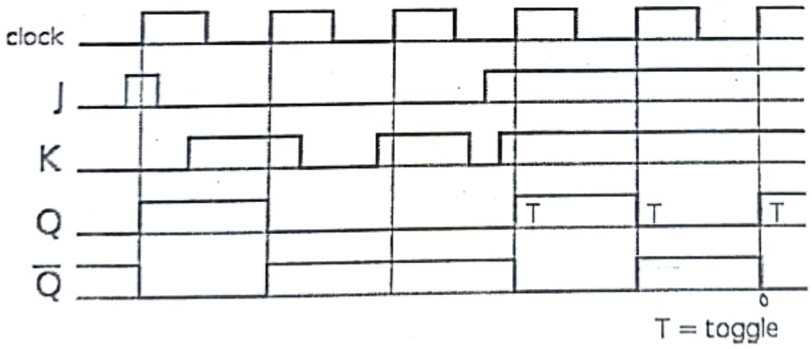
CamScanner

Negative edge-triggered D-latch response



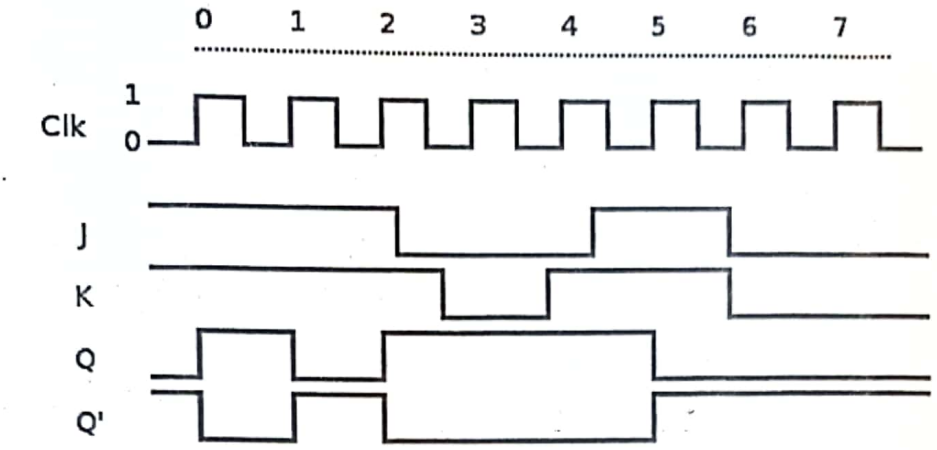
Outputs respond to input (D) only when enable signal transitions from high to low

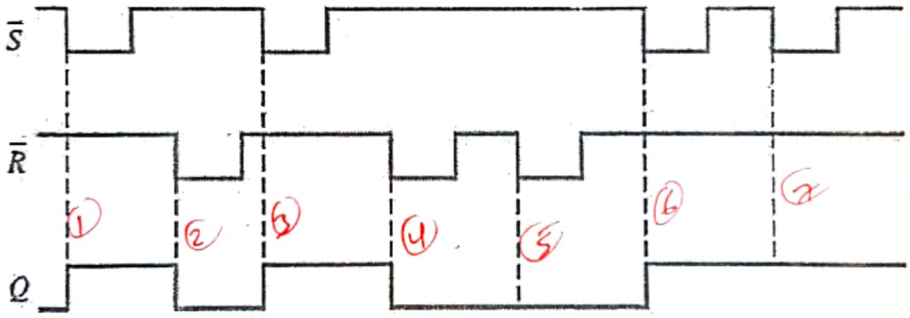
+ve edge

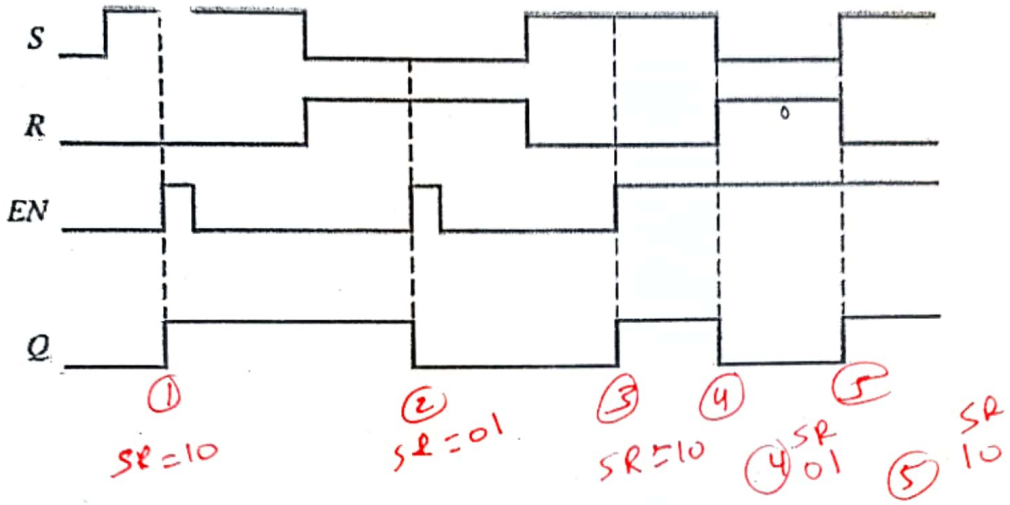


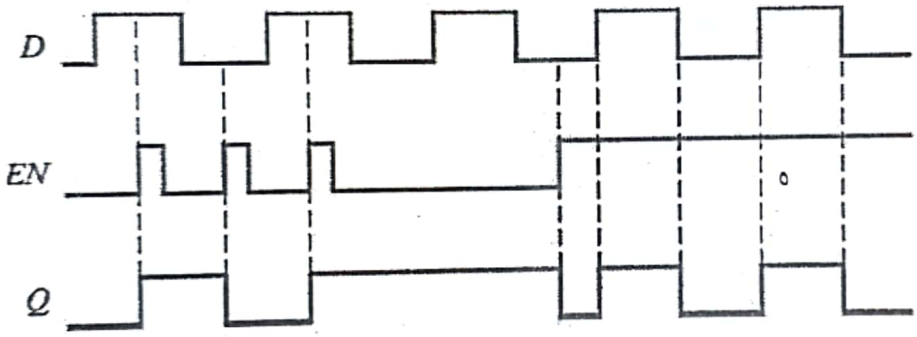
T = toggle

CamScanner


0 1 2 3 4 5 6 7
Clk
1
0
J
K
Q
Q'

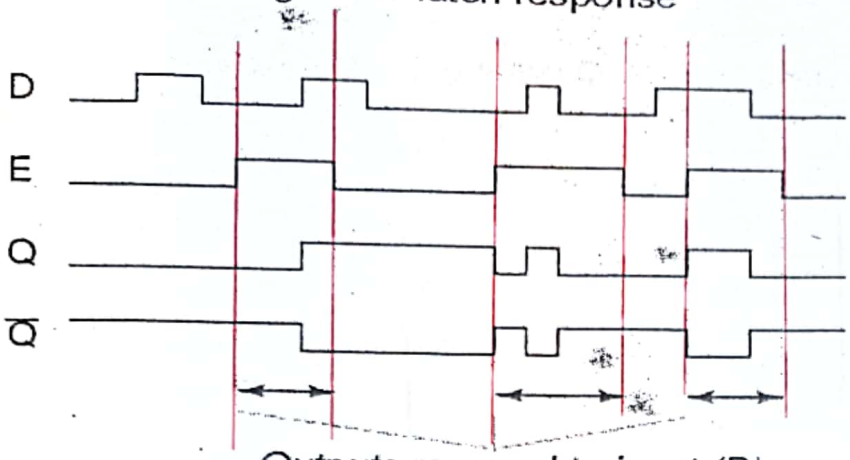

$\overline{S}$
$\overline{R}$
Q
1
2
3
4
5
6
7





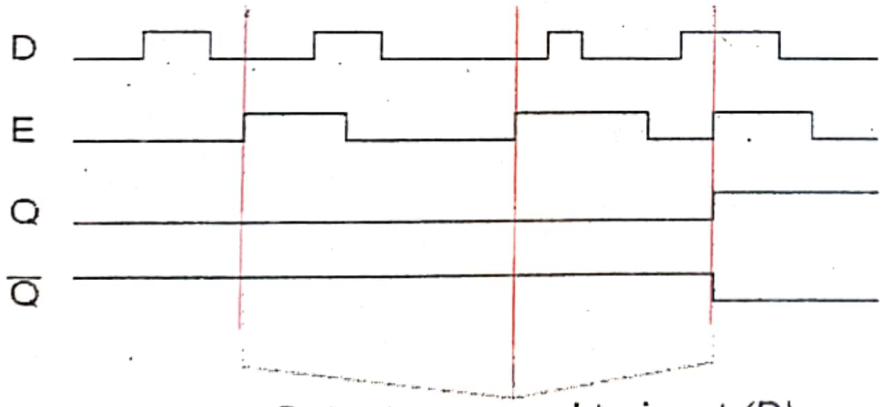
CamScanner

Regular D-latch response



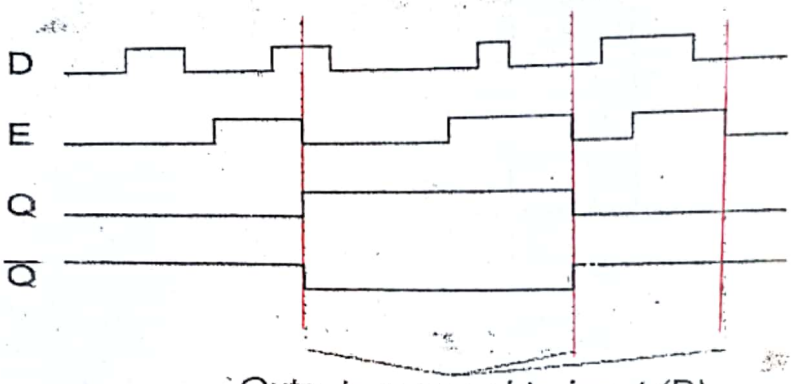
Outputs respond to input (D) during these time periods EN=1

Positive edge-triggered D-latch response

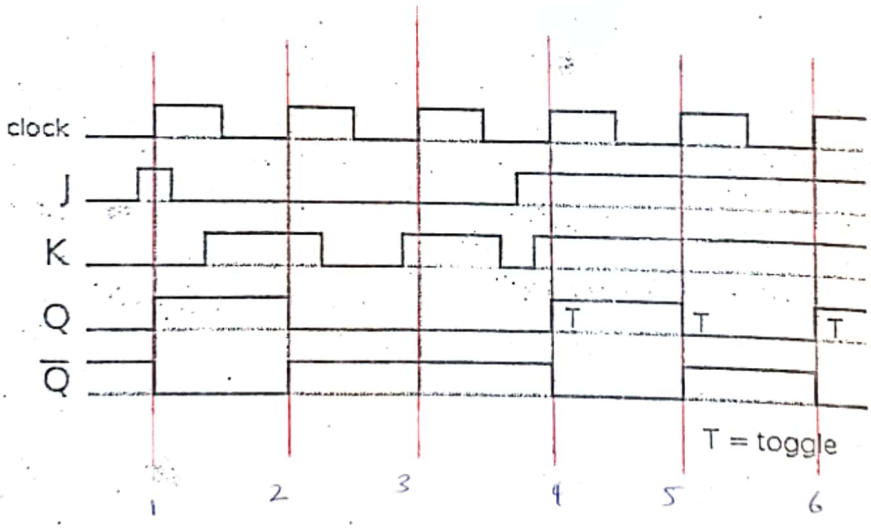


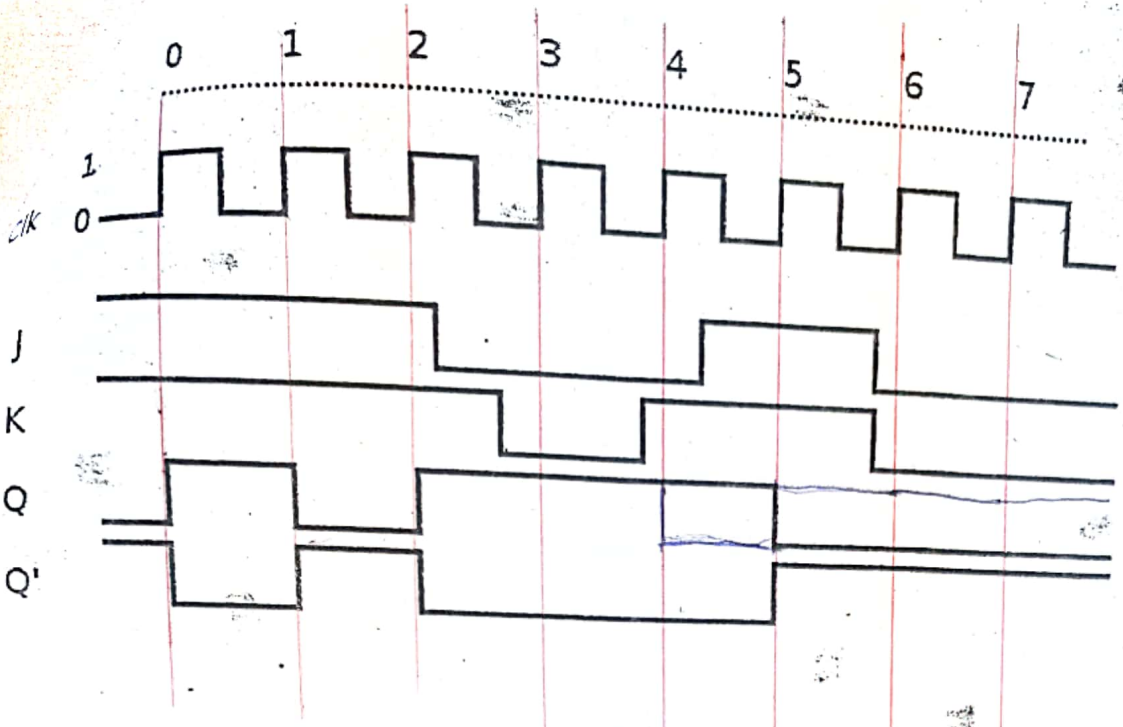
Outputs respond to input (D) only when enable signal transitions from low to high

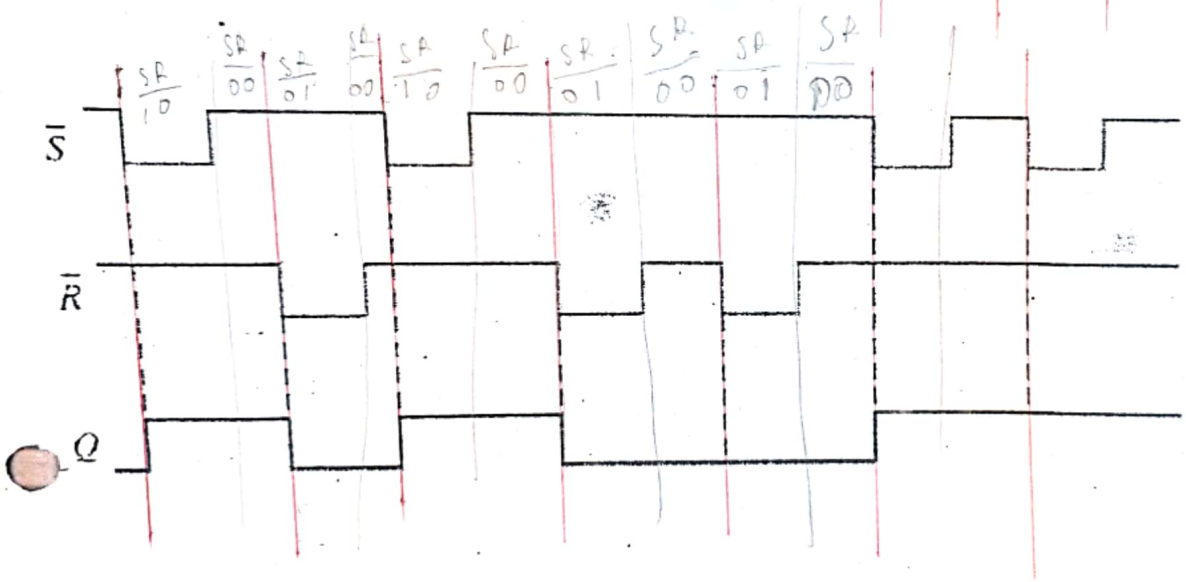
Negative edge-triggered D-latch response



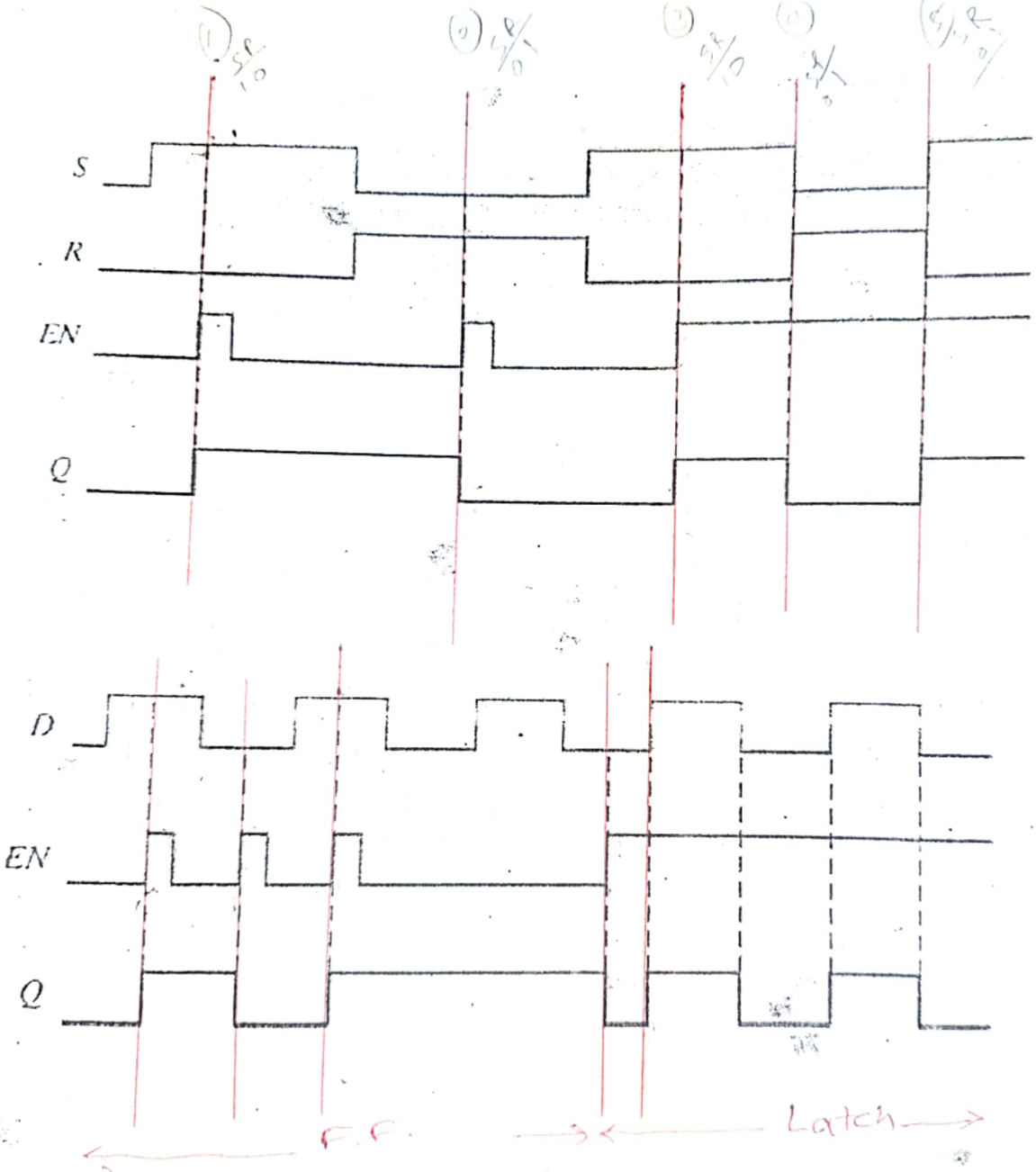
Outputs respond to input (D) only when enable signal transitions from high to low




0
1
2
3
4
5
6
7
1
0
clk
J
K
Q
Q'


SR 10
SR 00
SR 01
SR 00
SR 10
SR 00
SR 01
SR 00
SR 01
SR 00
$\overline{S}$
$\overline{R}$
Q

CamScanner


S
R
EN
Q
D
EN
Q
F.F.
Latch

moore $y(A,B)$

mealy mach $y(x,A,B)$

* Design of sequential Circuits:-

* Excitation Tables.

Q	Q+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S	R	Q	Q+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

SR-FF

$Q^+ = S + R'Q$

Q	Q+	D
0	0	0
0	1	1
1	0	0
1	1	1

D	Q	Q+
0	0	0
0	1	0
1	0	1
1	1	1

D-FF

Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

J	K	Q	Q+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

JK-FF

$Q^+ = JQ' + K'Q$

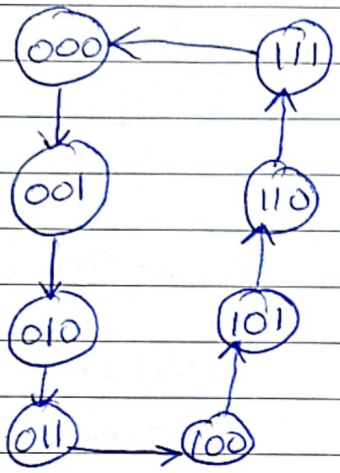
Q	Q+	T
0	0	0
0	1	1
1	0	1
1	1	0

T	Q	Q+
0	0	0
0	1	1
1	0	1
1	1	0

T-FF

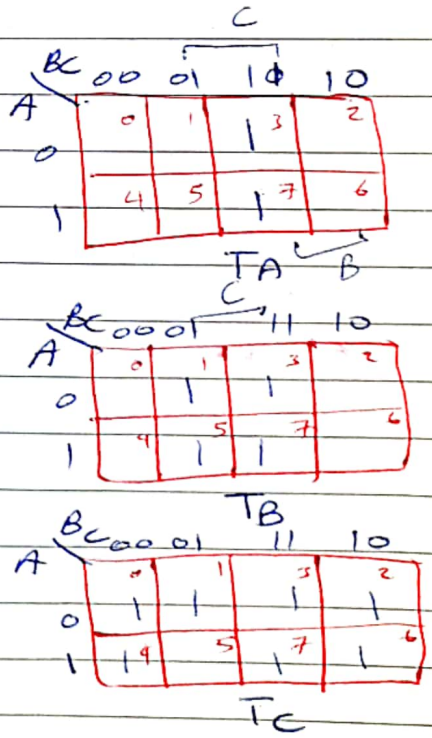
CamScanner

ex. Design a 3 bit up counter using T-FF



Q	Q_{t+}	T
0	0	0
0	1	1
1	0	1
1	1	0

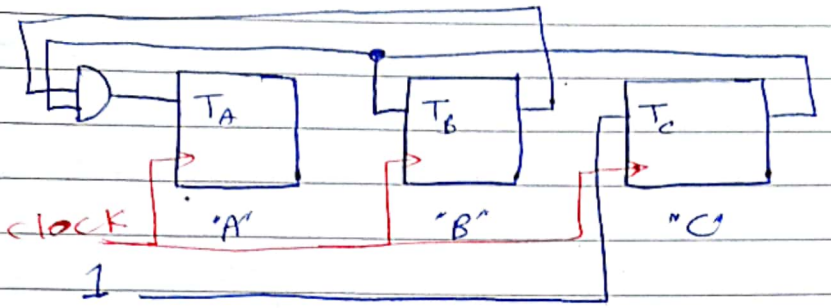
	Current state			next state			FF inputs		
	A	B	C	A	B	C	T_A	T_B	T_C
0	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	1
2	0	1	0	0	1	1	0	0	1
3	0	1	1	1	0	0	1	1	1
4	1	0	0	1	0	1	0	0	1
5	1	0	1	1	1	0	0	1	1
6	1	1	0	1	1	1	0	0	1
7	1	1	1	0	0	0	1	1	1



$T_A = BC$

$T_B = C$

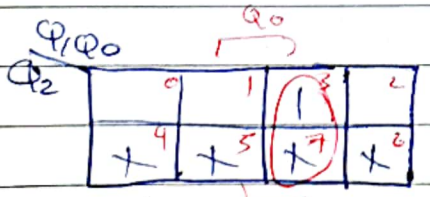
$T_C = 1$



(2) Design a 3-bit up counter using JK FF.

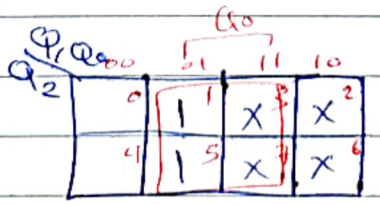
Present	next	F.F. inputs		
$Q_2Q_1Q_0$	$Q_2Q_1Q_0$	J_2K_2	J_1K_1	J_0K_0
0 0 0	0 0 1	0 X	0 X	1 X
1 0 0 1	0 1 0	0 X	1 X	X 1
2 0 1 0	0 1 1	0 X	X 0	1 X
3 0 1 1	1 0 0	1 X	X 1	X 1
4 1 0 0	1 0 1	X 0	0 X	1 X
5 1 0 1	1 1 0	X 0	1 X	X 1
6 1 1 0	1 1 1	X 0	X 0	1 X
7 1 1 1	0 0 0	X 1	X 1	X 1

Q_nQ_{n+1}	JK
0 0	0 X
0 1	1 X
1 0	X 1
1 1	X 0



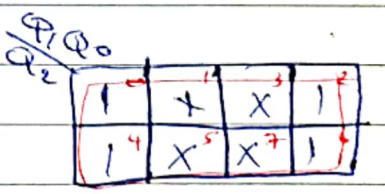
J_2

$J_2 = Q_0Q_1$



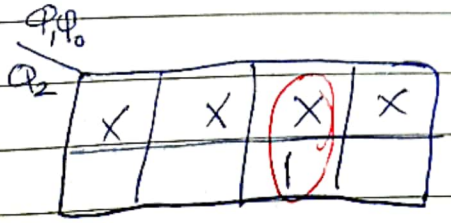
J_1

$J_1 = Q_0$



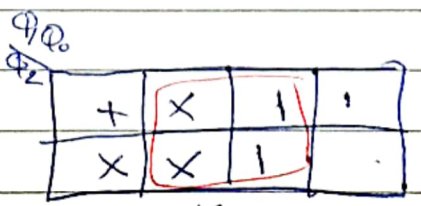
J_0

$J_0 = 1$



K_2

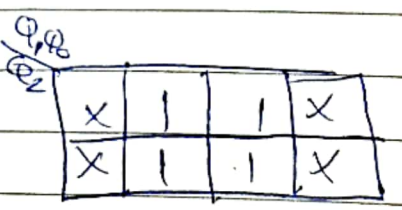
$K_2 = Q_0Q_1$



K_1

$K_1 = Q_0$

(3)



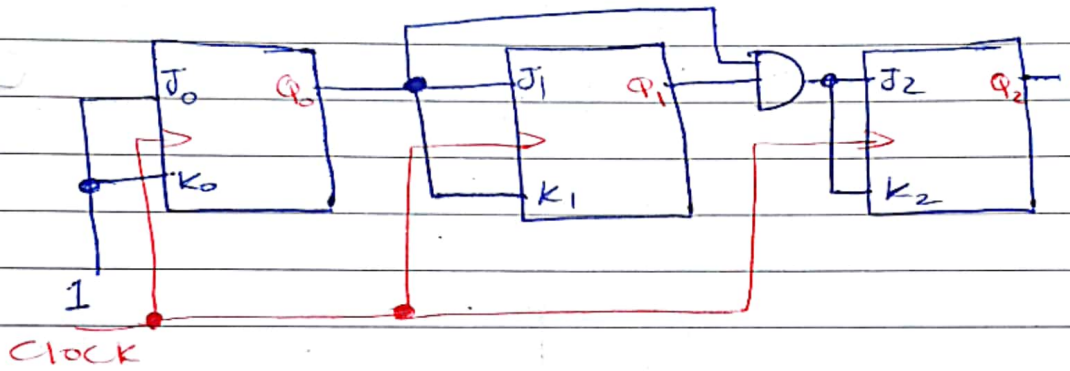
K_0

$K_0 = 1$

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

$J_2 = K_2 = Q_1 Q_0$

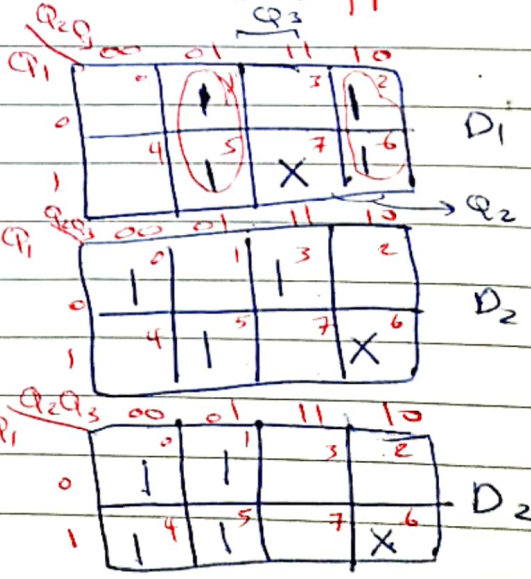


ex (3) Design a counter that goes through the sequence

0, 3, 2, 4, 1, 5, 7 and repeat.

Present $Q_1 Q_2 Q_3$	next $Q_1 Q_2 Q_3$	$D_1 D_2 D_3$
0 0 0 0	0 1 1	0 1 1
1 0 0 1	1 0 1	1 0 1
2 0 1 0	1 0 0	1 0 0
3 0 1 1	0 1 0	0 1 0
4 1 0 0	0 0 1	0 0 1
5 1 0 1	1 1 1	1 1 1
6 1 1 0	X X X	X X X
7 1 1 1	0 0 0	0 0 0

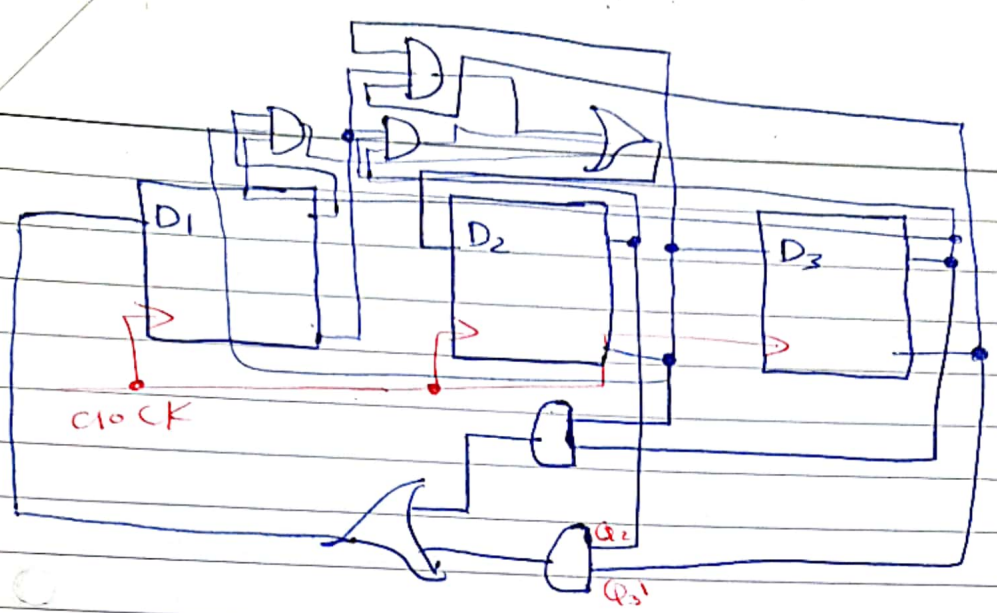
Q	Q+	D
0	0	0
0	1	1
1	0	0
1	1	1



$D_1 = Q_2' Q_3 + Q_2 Q_3'$

$D_2 = Q_1' Q_2' Q_3' + Q_1' Q_2 Q_3 + Q_1 Q_2' Q_3$ (4)

$D_3 = Q_2'$



JK

$J_1 = Q_2' Q_3 + Q_2 Q_3'$

$J_2 = Q_1' Q_3' + Q_1 Q_3$

$J_3 = Q_2'$

$K_1 = Q_3' + Q_2$

$K_2 = Q_1 + Q_3'$

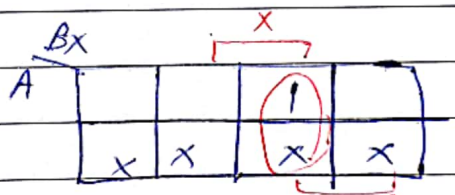
$K_3 = Q_2$

ex (4)

Design a sequential circuit that implement the following State table, use JK F.F.

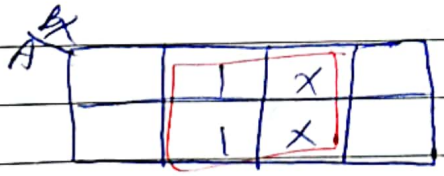
Present	input	next	output	F.F
AB	x	AB	y	$J_A K_A$ $J_B K_B$
0 0	0	0 0	0	0 x 0 x
0 0	1	0 1	1	0 x 1 x
0 1	0	0 1	0	0 x x 0
0 1	1	1 0	1	1 x x 1
1 0	0	1 0	1	x 0 0 x
1 0	1	1 1	1	x 0 1 x
1 1	0	1 1	1	x 0 x 0
1 1	1	0 0	1	x 1 x 1

Q	Q+	J K
0	0	0 x
0	1	1 x
1	0	x 1
1	1	x 0

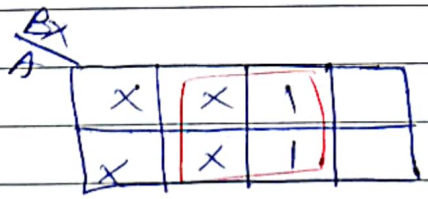


$J_A = xB$

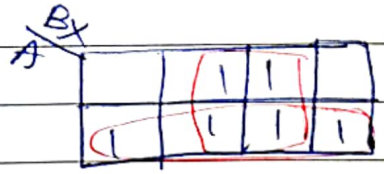
$K_A = xB$



$J_B = x$

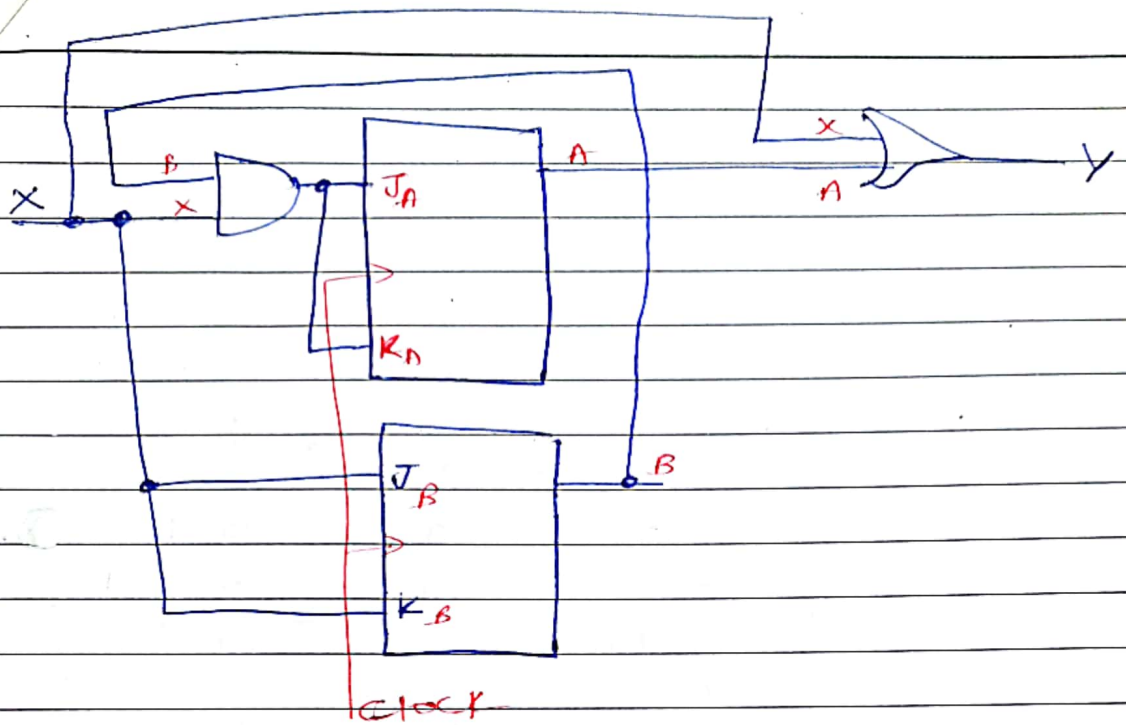


$K_B = x$



$y = x + A$

"mealy machine"


X
B
X
J_A
A
K_A
X
A
Y
J_B
B
K_B
Clock

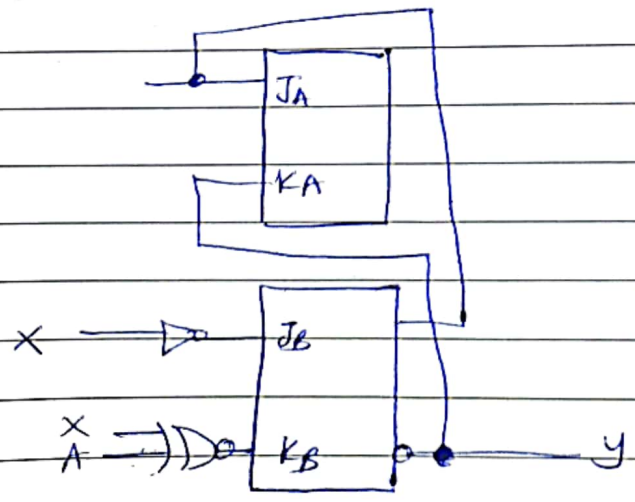
ex. Use JK FF to design

Present	input	next	output	FF.			
AB	x	AB	y	J_A	K_A	J_B	K_B
0 0	0	0 1	1	0	X	1	X
0 0	1	0 0	1	0	X	0	X
0 1	0	1 0	0	1	X	X	1
0 1	1	1 1	0	1	X	X	0
1 0	0	0 1	1	X	1	1	X
1 0	1	0 0	1	X	1	0	X
1 1	0	1 1	0	X	0	X	0
1 1	1	1 0	0	X	0	X	1

$J_A = B$, $K_A = \bar{B}$

$J_B = x'$, $K_B = A \oplus x$

$y = B'$ (Moore Machine)



ex (6)

Design an even counter

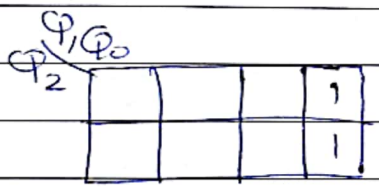
0 → 2 → 4 → 6 ↺

Using T FF

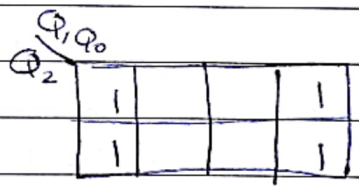
TFF excitation

Q Q+	T
0 0	0
0 1	1
1 0	1
1 1	0

Present	Next	FF Inputs
$Q_2 Q_1 Q_0$	$Q_2^+ Q_1^+ Q_0^+$	$T_2 T_1 T_0$
0 0 0	0 1 0	0 1 0
0 1 0	1 0 0	1 1 0
1 0 0	1 1 0	0 1 0
1 1 0	0 0 0	1 1 0



$T_2 = Q_1 Q_0'$



$T_1 = Q_0'$

$T_0 = \phi$

ex (3)

Design the following counter (with unused states) using SR FF.

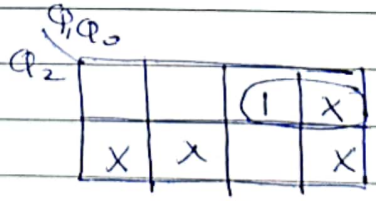
$0 \to 1 \to 3 \to 4 \to 6 \to 7$

(2, 5) unused states (x for simplification)

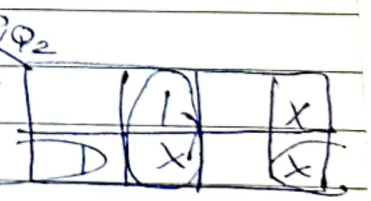
Present	next	FF-inputs		
$Q_2Q_1Q_0$	$Q_2^+Q_1^+Q_0^+$	S_2R_2	S_1R_1	S_0R_0
0 0 0	0 0 1	0 x	0 x	1 0
0 0 1	0 1 1	0 x	1 0	x 0
0 1 0	x x x	x x	x x	x x
0 1 1	1 0 0	1 0	0 1	0 1
1 0 0	1 1 0	x 0	1 0	0 x
1 0 1	x x x	x x	x x	x x
1 1 0	1 1 1	x 0	x 0	1 0
1 1 1	0 0 0	0 1	0 1	0 1

SR excitation table

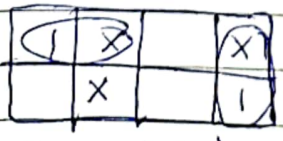
QQ^+	SR
0 0	0 x
0 1	1 0
1 0	0 1
1 1	x 0



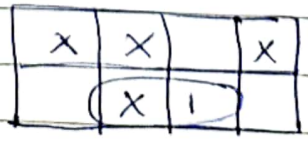
$S_2 = \bar{Q}_2 Q_1$



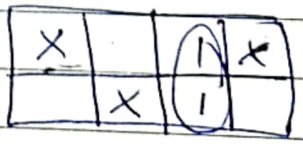
$S_1 = Q_1' Q_0 + Q_2 Q_0'$



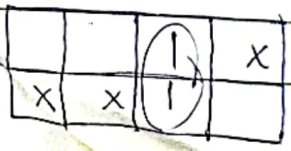
$S_0 = Q_2' Q_1' + Q_1 Q_0$



$R_2 = Q_2 Q_0$



$R_1 = Q_1 Q_0$



$R_0 = Q_1 Q_0$